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CONTRIBUTORS

John Fortin, Priscilla Bloomfield, Joseph Mahaz, and Laith Alfaqih; Airport Cooperative Research Program; Transportation Research Board; National Academies of Sciences, Engineering, and Medicine

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AIRPORT COOPERATIVE RESEARCH PROGRAM

ACRP RESEARCH REPORT 155

Guidebook for Advanced Computerized Maintenance Management System Integration at Airports

John Fortin
Priscilla Bloomfield
CH2M
Boston, MA

Joseph Mahaz
ELECTRONIC DATA, INC.
St. Petersburg, FL

AND

Laith Alfaqih
STANTEC
Cincinnati, OH

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AIRPORT COOPERATIVE RESEARCH PROGRAM

Airports are vital national resources. They serve a key role in transportation of people and goods and in regional, national, and international commerce. They are where the nation's aviation system connects with other modes of transportation and where federal responsibility for managing and regulating air traffic operations intersects with the role of state and local governments that own and operate most airports. Research is necessary to solve common operating problems, to adapt appropriate new technologies from other industries, and to introduce innovations into the airport industry. The Airport Cooperative Research Program (ACRP) serves as one of the principal means by which the airport industry can develop innovative near-term solutions to meet demands placed on it.

The need for ACRP was identified in *TRB Special Report 272: Airport Research Needs: Cooperative Solutions* in 2003, based on a study sponsored by the Federal Aviation Administration (FAA). ACRP carries out applied research on problems that are shared by airport operating agencies and not being adequately addressed by existing federal research programs. ACRP is modeled after the successful National Cooperative Highway Research Program (NCHRP) and Transit Cooperative Research Program (TCRP). ACRP undertakes research and other technical activities in various airport subject areas, including design, construction, legal, maintenance, operations, safety, policy, planning, human resources, and administration. ACRP provides a forum where airport operators can cooperatively address common operational problems.

ACRP was authorized in December 2003 as part of the Vision 100—Century of Aviation Reauthorization Act. The primary participants in the ACRP are (1) an independent governing board, the ACRP Oversight Committee (AOC), appointed by the Secretary of the U.S. Department of Transportation with representation from airport operating agencies, other stakeholders, and relevant industry organizations such as the Airports Council International-North America (ACI-NA), the American Association of Airport Executives (AAAE), the National Association of State Aviation Officials (NASAO), Airlines for America (A4A), and the Airport Consultants Council (ACC) as vital links to the airport community; (2) TRB as program manager and secretariat for the governing board; and (3) the FAA as program sponsor. In October 2005, the FAA executed a contract with the National Academy of Sciences formally initiating the program.

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Once selected, each ACRP project is assigned to an expert panel appointed by TRB. Panels include experienced practitioners and research specialists; heavy emphasis is placed on including airport professionals, the intended users of the research products. The panels prepare project statements (requests for proposals), select contractors, and provide technical guidance and counsel throughout the life of the project. The process for developing research problem statements and selecting research agencies has been used by TRB in managing cooperative research programs since 1962. As in other TRB activities, ACRP project panels serve voluntarily without compensation.

Primary emphasis is placed on disseminating ACRP results to the intended users of the research: airport operating agencies, service providers, and academic institutions. ACRP produces a series of research reports for use by airport operators, local agencies, the FAA, and other interested parties; industry associations may arrange for workshops, training aids, field visits, webinars, and other activities to ensure that results are implemented by airport industry practitioners.

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COOPERATIVE RESEARCH PROGRAMS

CRP STAFF FOR ACRP RESEARCH REPORT 155

Christopher J. Hedges, *Director, Cooperative Research Programs*
Lori L. Sundstrom, *Deputy Director, Cooperative Research Programs*
Michael R. Salamone, *Manager, Airport Cooperative Research Program*
Marci A. Greenberger, *Senior Program Officer*
Brittany Summerlin-Azeez, *Program Coordinator*
Eileen P. Delaney, *Director of Publications*
Natalie Barnes, *Associate Director of Publications*
Sharon Lamberton, *Editor*

ACRP PROJECT 09-14 PANEL
Field of Maintenance

Sarah J. Demory, *City of Boise, Boise, ID* (Chair)
Majed Khater, *Clark County (NV) Department of Aviation, Las Vegas, NV*
Casey Martin, *Jacobs, Houston, TX*
Bob Montgomery, *Southwest Airlines, Dallas, TX*
Robert C. Wilson, *Kent County (MI) Department of Aeronautics, Grand Rapids, MI*
Thomas Mai, *FAA Liaison*
Raymond "Ray" Zee, *FAA Liaison*
Paul J. Eubanks, *Airports Council International—North America Liaison*
Christine Gerencher, *TRB Liaison*

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- **Case Studies—**Port Authority of New York and New Jersey, Salt Lake City International Airport, Seattle-Tacoma International Airport, Toronto Pearson International Airport, General Mitchell International Airport, and Gwinnett County Department of Water Resources;
- **Focus Group—**Atlanta Hartsfield-Jackson International Airport (Aaron Davis), Northwest Arkansas Regional Airport (Charles Butcher), Toronto Pearson International Airport, Los Angeles World Airports (Christine Salvaggio), Columbus Regional Airport (Casey Denny), General Mitchell International Airport (Timothy Pearson), and Denver International Airport (Stephanie Sarmiento and Betty Trujillo);

(*continued on page viii*)

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FOREWORD

By Marci A. Greenberger
Staff Officer
Transportation Research Board

A Computerized Maintenance Management System (CMMS) can be used to manage a variety of assets across a number of different airport systems. The robust features and varied functionalities of the CMMS software available today offer users opportunities and challenges. Given the breadth, complexity, and integration of the systems involved, implementing a new or replacement CMMS can be overwhelming. Phasing systems into the CMMS may result in a more successful implementation, and a business case is almost always the first step to help determine which systems should be the first to be managed through the CMMS.

Successful integration of a CMMS can result in better decision making because the system provides timely and systemic information, but many industries have found implementation of a CMMS to be a challenging endeavor. This guidebook is designed to assist airport staff in determining which systems are best managed in the CMMS and in what order of implementation.

As part of ACRP Project 09-14, CH2M was selected to develop guidance on the steps necessary to implement a CMMS, factors for consideration in prioritizing which systems should be included in the CMMS using a phased approach, and the steps for integrating CMMS data into performance management and business decision making. Their research included gathering information and data from airports and organizations from other industries through case studies, focus groups, and surveys.

This guidebook will assist airport staff in overcoming some of the challenges associated with implementing a new or replacement CMMS allowing for improved integration with their asset management program and decision-making processes.

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- **Survey**—Addison Airport, Atlanta Hartsfield-Jackson International Airport, Boston Logan International Airport, Charlotte Douglas International Airport, Chicago O'Hare International Airport, John Glenn Columbus International Airport, Denver International Airport, Edmonton International Airport, Eugene Airport, General Mitchell International Airport, Los Angeles World Airports, Louis Armstrong New Orleans International Airport, Memphis International Airport, Metropolitan Nashville Airport Authority, Minden-Tahoe Airport, Northwest Arkansas Regional Airport Authority, Orlando International Airport, Phoenix Sky Harbor International Airport, Port Authority of New York and New Jersey, Reagan National Airport, Reid-Hillview and San Martin Airport, Salt Lake City International Airport, Seattle-Tacoma International Airport, and Tri-Cities Regional Airport; and
- **CH2M Team**—Jim Oldach, Joy Swenson, Craig Omundsen, and Rose Keelor.

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CHAPTER 1

Introduction

Project Background

Airports are made up of a complex set of interconnected systems that can rival those of a small city. Conducting timely, systematic maintenance activities on those systems with adequate resources is a challenging venture, but doing so is necessary to optimize airport operations and avoid negative impacts to airlines, the traveling public, and tenants. Succeeding at such work often depends on the accuracy and quality of data residing in an airport's computerized maintenance management system (CMMS).

Most organizations that use a CMMS find it challenging to appropriately and fully deploy and use the CMMS functions, given the system's complexity and highly customizable nature. Numerous industry surveys about CMMS have elicited significant findings regarding organizations' success in implementing or using a CMMS to its full potential. Wireman (2016) cites these reported findings:

- A success rate less than 50% (Thomas Registry Survey 1987); 50% of CMMS implementation fails (plantservices.com 1999)
- The majority of organizations use only parts of a CMMS (Engineer's Digest 1992)
- Organizations only use 30% of [CMMS] capabilities (ARC Study 2003)
- Organizations only realize 50% of [the CMMS's] potential (Aberdeen Research 2007)
- Only 16% of the respondents indicated that their CMMS is integrated with other systems (reliabilityweb.com 2011)
- A survey in which 91% of respondents viewed work order management is the most important feature of a CMMS and 34% of respondents rated work order management as excellent in their existing CMMS (reliabilityweb.com 2011)

Research also shows that 40% to 80% of all CMMS implementations will fail to deliver the expected return on investment (ROI) (Wireman 2016). Turner (2015) has listed some of the pitfalls that can compromise the success of a CMMS implementation, such as an organization's current infrastructure and work culture, lack of staffing, or failure to understand the scope of the implementation.

In respect to achieving success in implementing and fully using a CMMS, airports are no different than other industries. As documented in *ACRP Web-Only Document 23: Guidance on Successful Computer Maintenance Management Systems (CMMS) Selection and Practices*, effectively implementing and using a CMMS in the airport environment requires overcoming a unique set of challenges, including the large, diverse set of assets and airport systems, and the complexity of features and capabilities of a CMMS.

The challenges of data transfer and sharing data among systems also are compounded by limited resources and the lack of supporting infrastructure. Many organizations facing these challenges have not identified appropriate practices to ensure that their CMMS investment

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is operations and maintenance (O&M) ready and able to produce valuable benefits, such as improved quality and relevant data outputs for informed decision making. Further, many CMMS users do not leverage the full capabilities of these systems.

Leveraging a CMMS as single source-of-truth data repository for maintenance and asset management strategy implementation and reporting offers a ripe opportunity for many airports to enhance overall operations, maintenance, and business decision making. Airport managers require guidance and resources to improve the design and implementation of their CMMS if they are to develop robust and reliable systems that can:

- Classify and group assets logically to achieve various levels tracking
- Define and measure levels of service (LOS)
- Produce high quality, reliable data
- Track key performance indicators (KPIs)
- Generate meaningful reports

These benefits empower executives and managers to improve business decisions surrounding accurate timing of renewal and replacement schedules, data-driven short- and long-term budget plans, total cost of ownership requirements, overall asset and systems performance, and customer experience optimization.

In addition to *ACRP Web-Only Document 23,* recent ACRP research that has advanced the progress of CMMS practices in the aviation industry include the following:

- *ACRP Report 69: Asset and Infrastructure Management for Airports—Primer and Guidebook,* which addresses the theory, practice, and technology used in asset and infrastructure management.
- ACRP Project 09-07, "Airport Maintenance Roles in Planning, Design, Construction, Commissioning, and Activation," which establishes critical milestones for maintenance input on infrastructure projects.

Equally important is previous research related to risk management and life cycle costing (LCC), as both topics are closely related to effective CMMS implementation programs. ACRP research and guidebooks in this area include:

- *ACRP Report 19: Developing an Airport Performance-Measurement System*
- *ACRP Report 19A: Resource Guide to Airport Performance Indicators*
- *ACRP Report 74: Application of Enterprise Risk Management at Airports*
- *ACRP Report 116: Guidebook for Successfully Assessing and Managing Risks for Airport Capital and Maintenance Projects*
- *ACRP Report 131: A Guidebook for Safety Risk Management at Airports*
- *ACRP Report 172: Guidebook for Considering Life-Cycle Costs in Airport Asset Procurement*

This previous research has advanced understanding and practices regarding CMMS utility in the airport environment; however, a gap remains in the literature related to advising airports on implementing and utilizing a CMMS successfully. Given that CMMSs are critical for many organizations and industries, it is advisable that advanced practices be identified, modified, and simply translated to be useful resources to airport managers and executives.

CMMS at Airports

Delivering the best investment value for airports' physical assets can be a complex task. It involves careful consideration of the trade-offs between performance, cost, risk, and the overall effects of the assets' life cycle stages. Expenditures involve inherently conflicting factors. Moreover, assets can be identified and managed at differing levels, ranging from discrete equipment

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items or components to complex functional systems, networks, sites, or diverse portfolios. As an example, discrete equipment items may have identifiable, individual life cycles that can be optimized, whereas asset systems may have an indefinite horizon of required usage.

Sustainability considerations also should be integral to optimized decision making. An organization with diverse asset systems, while each contributing to the overall goals of the organization, may also present widely different investment opportunities, performance challenges, and risks. An integrated CMMS is, therefore, essential in an airport's ability to understand short-term versus long-term benefits, expenditures versus performance levels, planned and unplanned availability, or capital costs versus operating costs. Successful implementation of a CMMS also can help an airport understand the diversity and complexity of assets and align their asset management strategy to the airport's overall objectives, priorities, and chosen risk profile.

During the past 25 years, organizations such as airports have looked for ways to better manage their assets and implement overall asset management strategies. A CMMS can help move them from a reactive maintenance philosophy to a more proactive philosophy. Many important factors contribute to an airport's success or failure in this effort.

Since September 11, 2001, an immense strain has been placed on the fiscal stability of airports across the country. Commercial service airports in the United States spend substantially more on security-related issues than they did prior to September 11, 2001. Some of the increase in spending has been budgeted, but additional money has come via cuts and reductions in other airport spending. In recent years, rising costs have forced airlines out of business, necessitating mergers and, in general, a reduction in the number of flights, which has also significantly impacted airport revenues. One area in which airport spending has been cut is airport asset management/facility maintenance. Many U.S. airports have either decreased or, in some cases, completely cut all new spending relating to airport maintenance. Maintenance departments are now expected to do more with fewer resources and, although their budgets have been significantly reduced, their workloads have increased.

Many U.S. airports have found it impossible to operate in the black, let alone invest in implementing modern technology like airport CMMS, which provides major efficiency enhancements over existing manual processes. Yet, by implementing the appropriate technology in the right way, other airports have found they can do more with fewer resources and still provide a higher quality of service to their customers while helping to stabilize rates and charges to the airlines.

Today, airports are experiencing significant growth, which puts additional strain on their maintenance resources as well as a significant load increase to their airside and landside assets. Unfortunately, the management style of maintenance departments is primarily reactive, and the ratio of corrective maintenance (CM) to preventive and/or predictive maintenance (PM) is significantly lopsided. Best maintenance practices would have an airport CM to PM ratio at 75% PM and 25% CM. Unfortunately, most airports are the reverse, 75% CM to 25% PM. National averages show that it costs an airport far more to perform unplanned or unscheduled CM than it does to perform planned and scheduled PM. The conundrum is that it takes money to save money and, until such time that an airport can invest in modern energy-efficient equipment, enterprise facilities, and/or CMMSs to improve their PM program, they cannot gain the desired efficiencies.

Most facility maintenance professionals constantly search for ways to trim expenses and closely scrutinize their airport's equipment, parts, labor, and energy costs. Fortunately, maintenance operations do not have to be the budget-depleting necessity they have long been considered. With properly implemented business processes and the new technology offered by today's

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CMMSs, an airport maintenance manager can move his or her maintenance department into the black. This can be accomplished because the right maintenance business processes, aligned with the right airport-specific CMMS, can make maintenance operations more efficient.

During the past few years, implementing a CMMS process and software has made the job of facilities service managers easier by significantly reducing the amount of paperwork associated with manual work order systems. This approach provides better coordination of maintenance tasks, which, in turn, improves inventory management and supports PM programs. According to case study research, airport maintenance departments look to CMMS functions to help justify staffing levels, identify appropriate replacement schedules, and better forecast pending failures, among other realized benefits.

Today, asset management programs provide more advantages, including creating a single source of information. Today's systems excel in tracking assets, parts, energy, and labor, allowing managers to make more informed assessments about repairs, purchases, and staffing levels. These programs also offer improved automation of work orders and can provide overall airport asset management, to include facilities, information technology (IT), and fleet asset management.

With the purchase and implementation of mobile solutions that are integrated with the airport CMMS program, maintenance technicians can fill out work orders and perform inspections with smart phones and tablets. These devices offer real-time data collection and reporting. Furthermore, by employing systems linked to bar coding, users can track inventory and assets electronically.

Implementing a CMMS program using today's available technology will further an airport's ROI by properly documenting historical equipment information, affording the airport the opportunity to mature into a true Reliability-Centered Maintenance (RCM) Program. This type of program relies on the historical data housed in today's modern systems to identify and predict equipment failures and recommend proactive processes. These programs will reduce the number of failures, extend the life of equipment, and improve equipment uptime, resulting in decreased costs. In general, the proper implementation, use, and maintenance of a CMMS program will also afford airports the ability to execute a world-class maintenance practice that meets the nationally accepted business process flow for enterprise facilities and asset management systems (AMSs) illustrated in Figure 1-1.

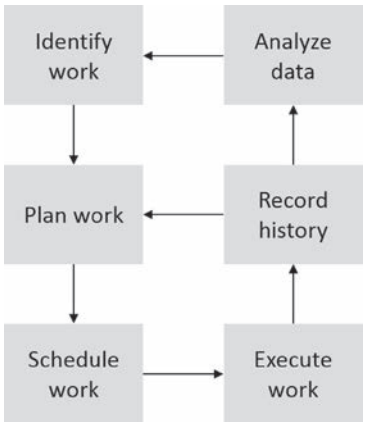


Figure 1-1. Process flow for enterprise facilities and AMSs.

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Guidebook Content and Organization

The research objectives of ACRP Project 09-14 included development of a guidebook to help airports (1) identify which airport system is likely to provide the highest value when implementing a CMMS; (2) describe the steps necessary to implement the identified systems into a CMMS; (3) provide the factors that an airport should consider when prioritizing the systems for inclusion into the CMMS in a phased approach; and (4) provide the steps for integrating a CMMS into business decision making. Accordingly, the research team developed and organized this guidebook on behalf of the airport management industry, based on the knowledge, expertise, opinions, and recommendations of airport personnel and other aviation professionals. The guidebook provides a comprehensive collection of survey results, case studies and best practices to facilitate the implementation of a robust and reliable CMMS program. The procedures suggested have been designed to be easily integrated within airports' existing and planned asset management programs.

The guidebook is intended as a resource to support implementation of a CMMS program by airport personnel. It has been organized to allow readers to find and focus on particular areas of information or interest within the guidebook, as needed.

- Chapter 1 provides a general overview of ACRP Project 09-14, "Advanced Computer Maintenance Management System (CMMS) Integration for Airports" and CMMS history at airports, along with a description of the purpose, content, and organization of this guidebook.
- Chapter 2 introduces, benchmarks, and documents the state of the aviation industry in implementing a CMMS.
- Chapter 3 introduces the roadmap for implementing a CMMS. It addresses how to design and implement a CMMS at the airport while considering how the new system will integrate with existing systems, and how it will function and be used by various stakeholder groups within the organization, especially maintenance and leadership.
- Chapter 4 presents a summary of five implementation case studies (four involving airports and one a non-airport organization). Detailed information about the case studies is presented in Appendix A.
- Chapter 5 provides guidance to help airport CMMS developers, integrators, and users integrate and manage the CMMS in relation to a performance management program, thereby improving metrics-based decision making.
- Chapter 6 introduces best practices in establishing asset taxonomy, hierarchy, and attributes.
- Chapter 7 addresses how implementing a CMMS will enable airports to meet requirements for AMS, as defined by the International Organization for Standardization (ISO) in ISO 55000, ISO 55001, and ISO 55002.

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CHAPTER 2

State of the Airport Industry in CMMS Implementation

This chapter summarizes the results of a survey conducted as part of ACRP Project 09-14 to benchmark and document the state of the airport industry in implementing a CMMS. The survey results also allowed the research team to identify airports to highlight in the case studies.

Approach

The research team contacted 88 airports of varied classifications. The airport classifications included:

- Commercial large hub primary airports
- Commercial medium hub primary airports
- Commercial small hub primary airports
- Commercial non-hub primary airports
- Non-hub non-primary airports
- Reliever non-primary airports
- General aviation (GA) non-primary airports

Of the 88 airports and airport authorities to which the survey was sent, 25 completed the survey, for a response rate of 28%.

Commercial large hub airports had the most participants (13 airports, or 52%). The rest of the participants were distributed among the other airport classifications as presented in Figure 2-1.

Results and Analysis

State of Maintenance and CMMS at Airports

At the beginning of the survey, the participants were asked if they had implemented a CMMS. The majority of the participants (18 respondents, or 72%) said they had implemented a CMMS or had started the implementation process. The rest of the airports commented that they have systems in place but not a CMMS. Comments from airports that responded "No" to this question included the following:

- "Work order system dates back over 15-20 years and requires the faxing of material and work orders to respective trades/stores. For airfield work, there is a proprietary GIS data base in use for lighting, pavement and safety area/signage maintenance." (Commercial large hub airport)

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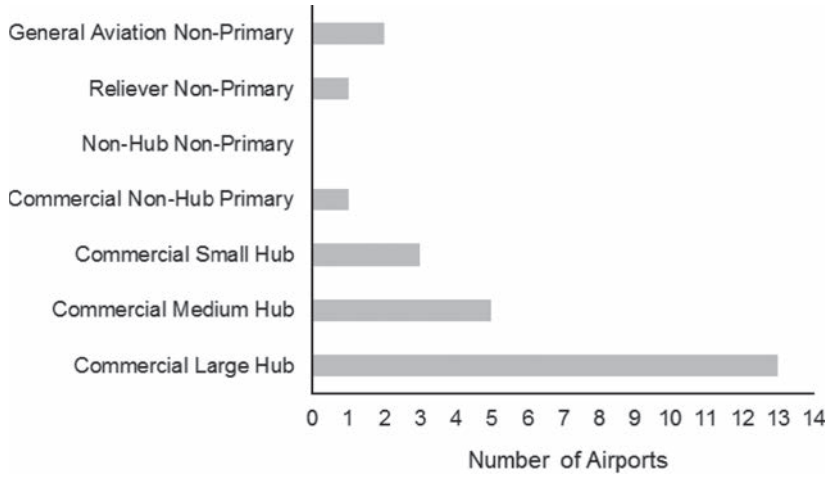


Figure 2-1. Distribution of participating airports by airport classification.

- "We currently have a software solution that provides limited maintenance management capabilities for buildings." (Commercial medium hub airport)
- "We procured a CMMS late in FY16 and are in the process of implementing it." (Reliever non-primary airport)

Large, medium, and small commercial hub airports had the most positive responses about implementing a CMMS. Survey results indicate that airports classified as hubs, which deal with large volumes of passengers and services to the public along with having a variety of assets and many resources, are more likely to implement or consider implementing a CMMS.

Planning and scheduling maintenance activities is critical to the success of a maintenance program. If airport maintenance activities are properly planned and scheduled, they provide airports with better overall management of resources, reduce asset failures, maximize/improve "wrench" time, and extend asset life. Ultimately, these benefits help airports reduce maintenance costs and stabilize rates and charges to the airlines and tenants.

Most of the airport respondents indicated that they have some level of proactive maintenance in place. Few airports have proactive or reactive programs only. The results show that bigger airports have the resources and the ability to operate their maintenance programs at proactive and somewhat proactive levels. The survey shows that having such systems in place requires commitment from airport leadership, training, and availability of personnel.

Figure 2-2 illustrates the distribution of the types of maintenance programs per airport classification. Previous surveys have indicated that about 60% of airports have an 80:20 ratio of reactive to proactive maintenance programs (Mahaz and Alfaqih 2017).

CMMS Implementation and Utilization

As at other organizations, implementing a CMMS at an airport takes time and requires committed resources. Implementation typically is a multi-phased process with peaks and valleys during each implementation phase. Airports vary widely in terms of their culture, organizational processes, and organizational structures. Each airport has a unique set of factors to consider when prioritizing implementation of systems such as a CMMS. These factors include:

- Performance needs and goals (i.e., the ability of the system to enhance performance)
- System criticality

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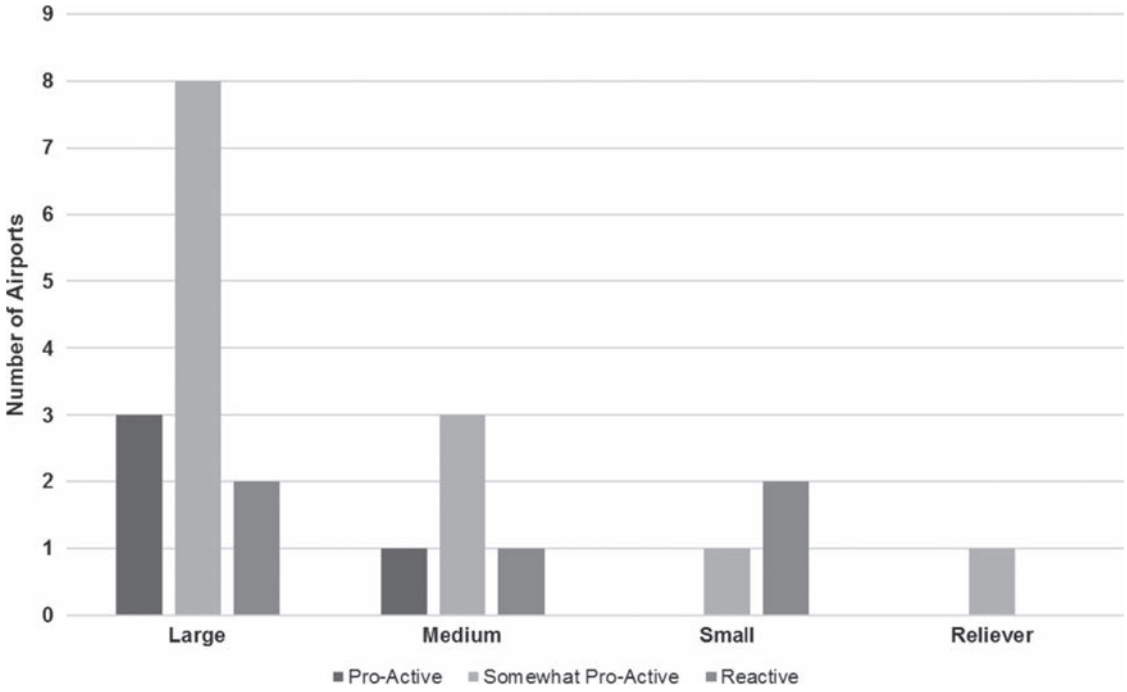


Figure 2-2. Maintenance programs at airports.

- Financial resources and constraints
- Regulatory compliance
- Leadership availability and commitment
- Piloting (e.g., information from implementation of the system at a similar organization, or opportunity to test the system)
- Internal IT resources
- Third-party IT resources

Figure 2-3 shows the ranking of these factors based on airports' survey responses. These responses show that the most important factor in implementing any system is whether doing so will help the airport become more efficient and *enhance performance.* Efficiency and enhanced performance directly impact the services airports provide to stakeholders, especially passengers. CMMS can help airports improve efficiency and enhance performance in a financially responsible manner while facilitating the airports' meeting their regulatory requirements.

At the majority of the airports surveyed, CMMS is mainly used to;

- Proactively manage PM
- Manage work requests

Airports are introducing mobile devices to support a CMMS. Mobile devices allow for more accurate collection and reporting of data from the field to expedite and improve the decision-making process for assets. Airports also use a CMMS to manage the inventory of material and spare parts, and they are starting to implement enterprise asset management (EAM) systems that allow linking and integrating a CMMS with other systems. Figure 2-4 summarizes airports' responses to survey questions about their use of CMMS.

At the majority of airports surveyed (68%), airport staff manage the CMMS internally. A few airports (18%) indicated that they contract with a solutions provider to manage and maintain the system. Other airports reported using a combination of internal system management with external software support.

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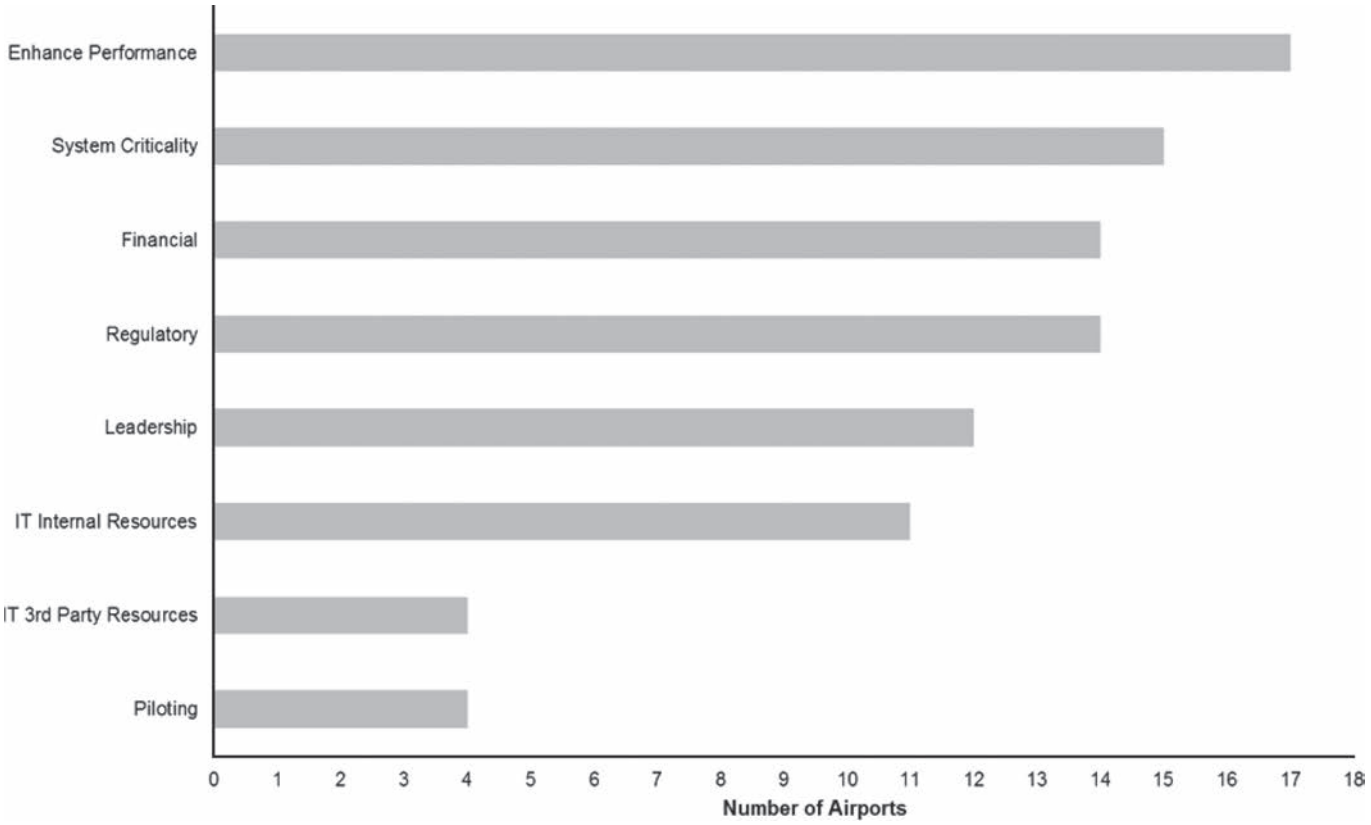


Figure 2-3. Factors to consider when prioritizing implementation (ranked).

When asked to rate the success of the CMMS implementation on a scale of 1 to 10 (where 1 is not very successful and 10 is very successful), the result was mid-range (5 or 6) for 40% of responding airports. Few airports (27%) gave an implementation success score of 8 or above. Figure 2-5 presents the CMMS implementation success rating scored by the airports.

Managing Assets Using a CMMS

The survey assumes that airports differ in terms of the types of assets they manage through a CMMS. Provided a list of 24 assets, airports were asked to select those they used as part of their CMMS. Figure 2-6 ranks the selected assets and Table 2-1 lists the top 10 asset systems selected by airports based on the survey results.

Figures 2-7, 2-8, and 2-9 present the comparisons between assets managed through a CMMS by airport classification (large, medium, small, and reliever non-primary). The percentage value (*x*-axis) represents the number of airport respondents in a certain classification that manage a particular type of asset through a CMMS. Five out of five (100%) medium airports manage heating, ventilation, and air conditioning (HVAC) through a CMMS, whereas only 11 of 13 (85%) of large airports and two out of three (66%) small airports manage HVAC through a CMMS.

The reliever non-primary airport selected and ranked only five assets, as follows: (1) runway lighting, (2) facilities, (3) hangars, (4) pavement, and (5) signage.

Figures 2-7 through 2-9 show that large and medium airports manage various types and sizes of assets using a CMMS. Although all airports focus on critical assets such as buildings, HVAC,

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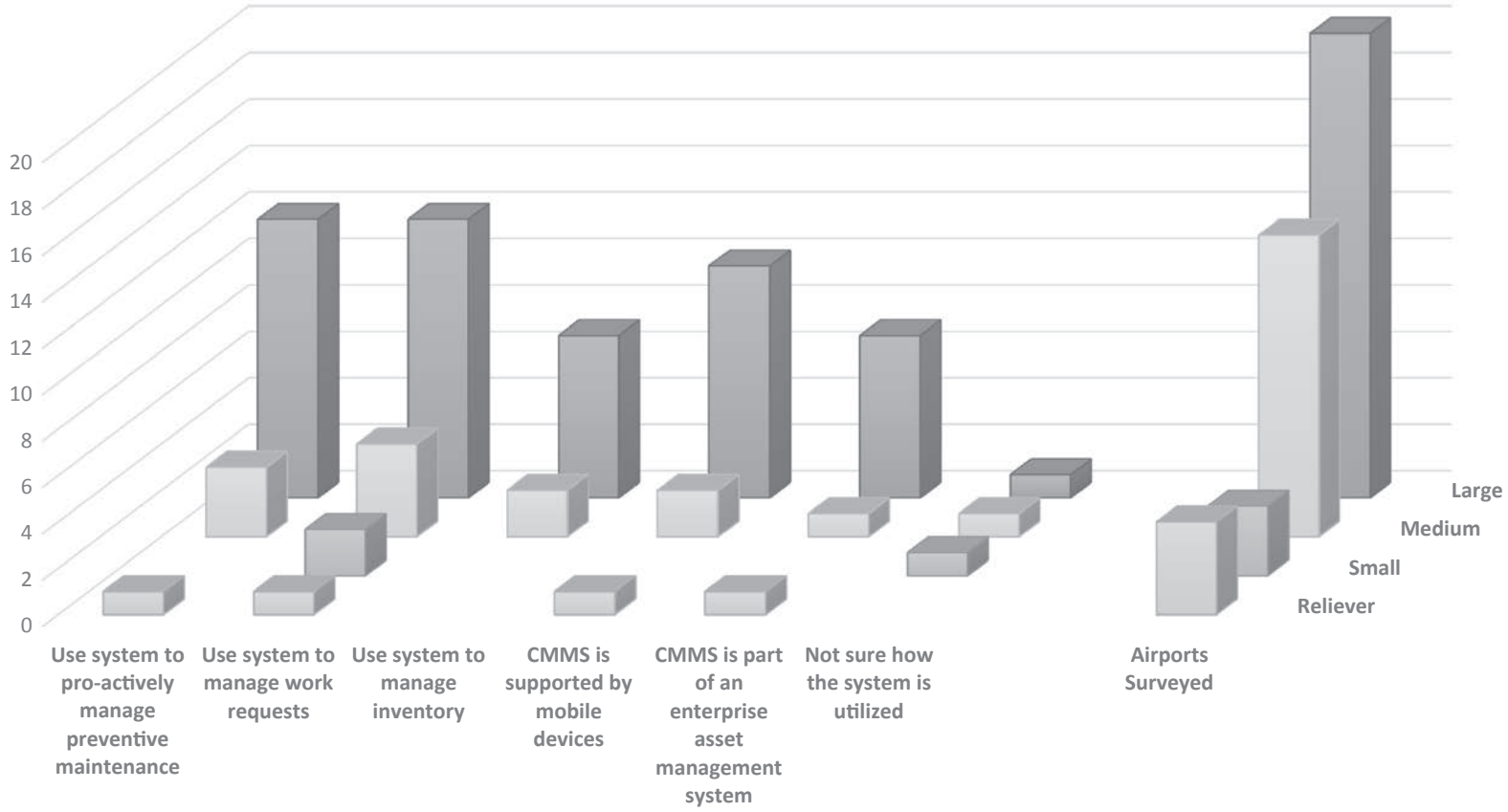


Figure 2-4. Statements that best describe CMMS utilization, by airport classification.

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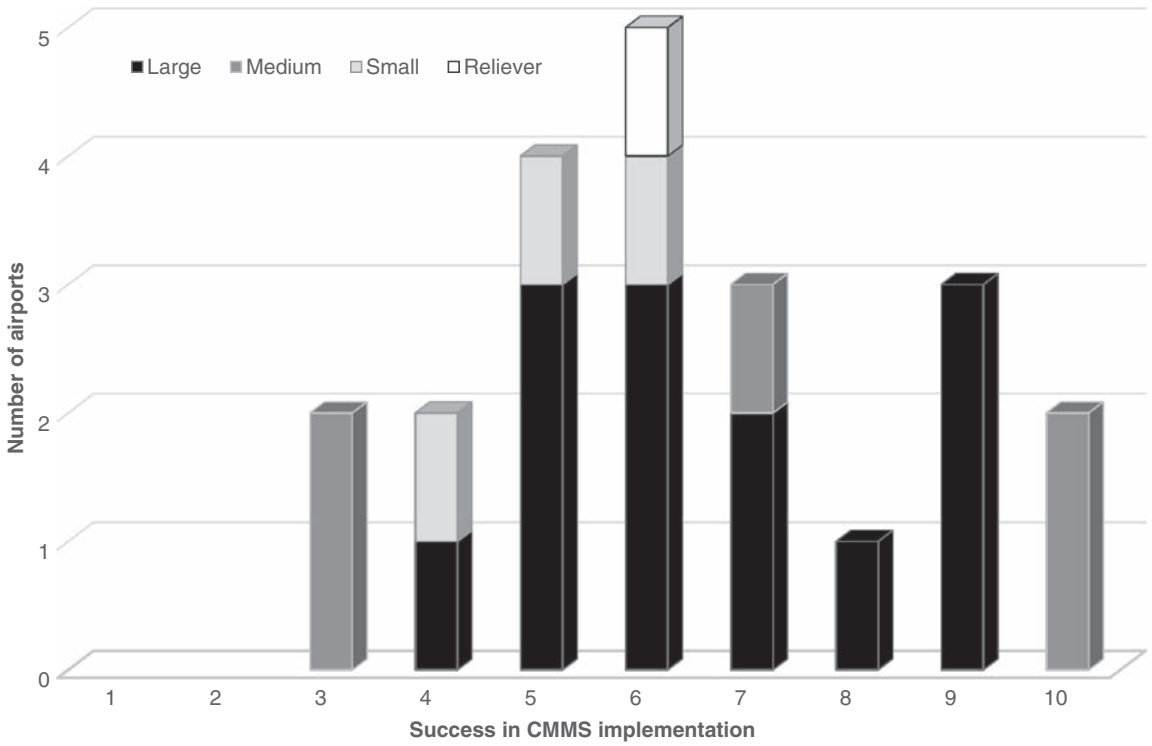


Figure 2-5. CMMS implementation success rating at airports surveyed.

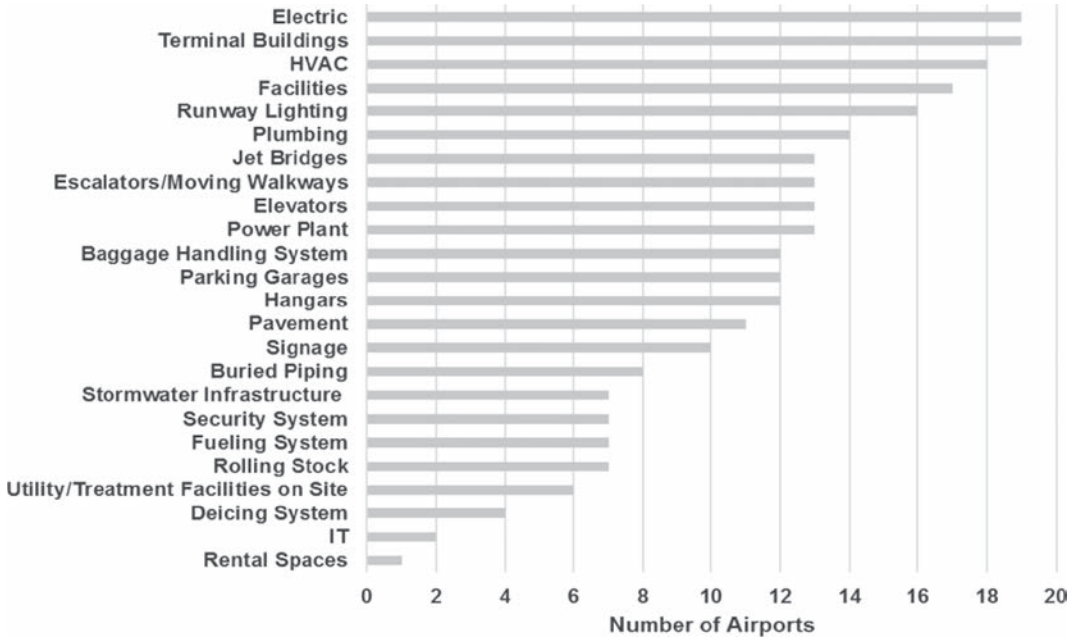


Figure 2-6. Ranked list of assets managed through CMMS (all respondents).

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Table 2-1. Ranking of top asset systems in CMMS (all respondents).

System	Rank
Electric	1
Terminal buildings	2
HVAC	3
Facilities	4
Runway lighting	5
Plumbing	6
Elevators	7
Escalators/moving walkways	8
Power plant	9
Jet bridges	10

and runway lighting, a higher percentage of smaller airports also focuses on assets such as security systems. Comments provided by airport respondents included the following:

- "We are just starting to implement the system; we have a small staff and will start with the basics: airfield and airport-owned facilities and hangars." (Reliever non-primary airport)
- "Minimalist implementation. Aside from airfield GIS platform, data generated is largely unused or unusable. For HVAC, there are great opportunities to improve quality of data throughout the 'legacy' system." (Commercial large hub airport)

The survey asked respondents to score the assets listed in Figure 2-6 in terms of the benefits received for managing them within a CMMS. The scores scale the benefits from 1 to 10 (where 1 is low benefit and 10 is high benefit). Figure 2-10 presents the average score received for each asset. None of the assets received an average score below 5. It can be assumed that the top 10 assets were the ones that directly affected passenger experience (e.g., jet bridges, HVAC,

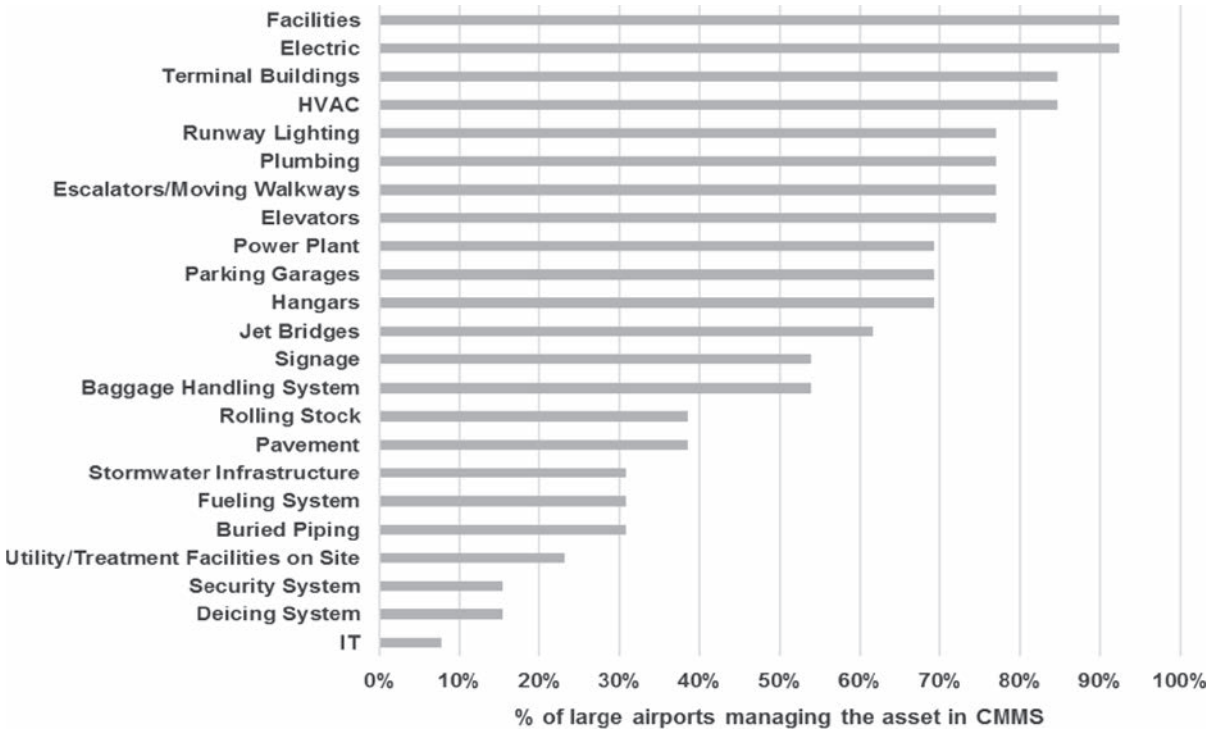


Figure 2-7. Ranked list of assets managed through CMMS (large airports).

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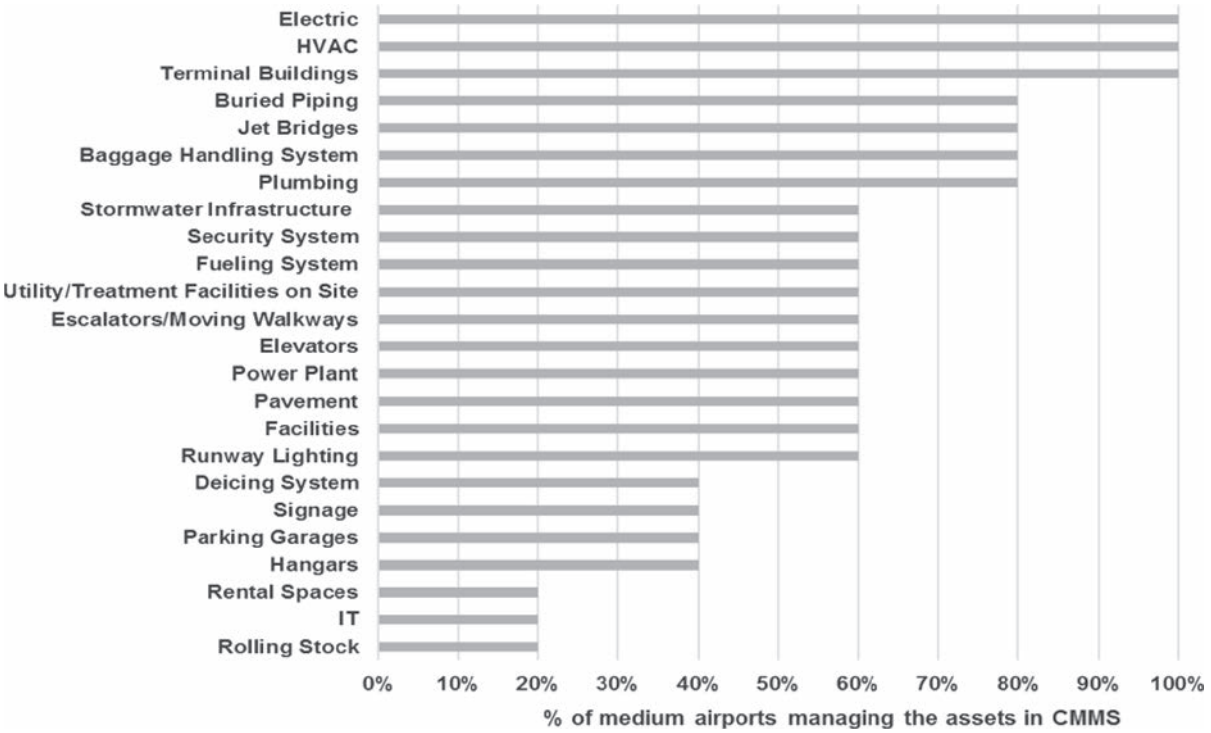


Figure 2-8. Ranked list of assets managed through CMMS (medium airports).

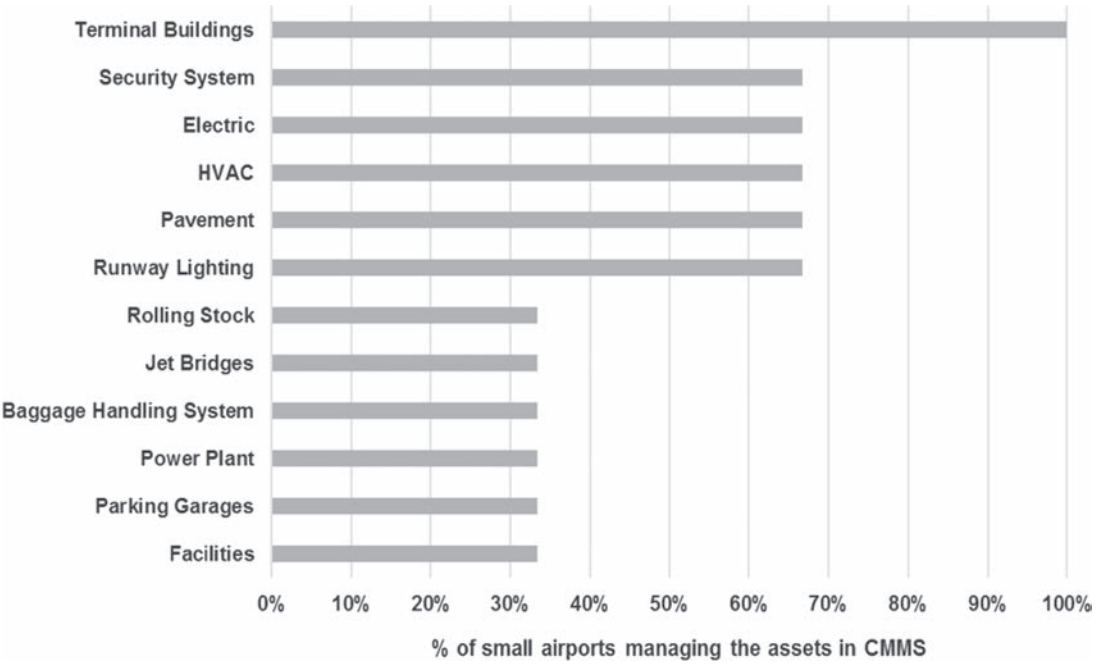


Figure 2-9. Ranked list of assets managed through CMMS (small airports).

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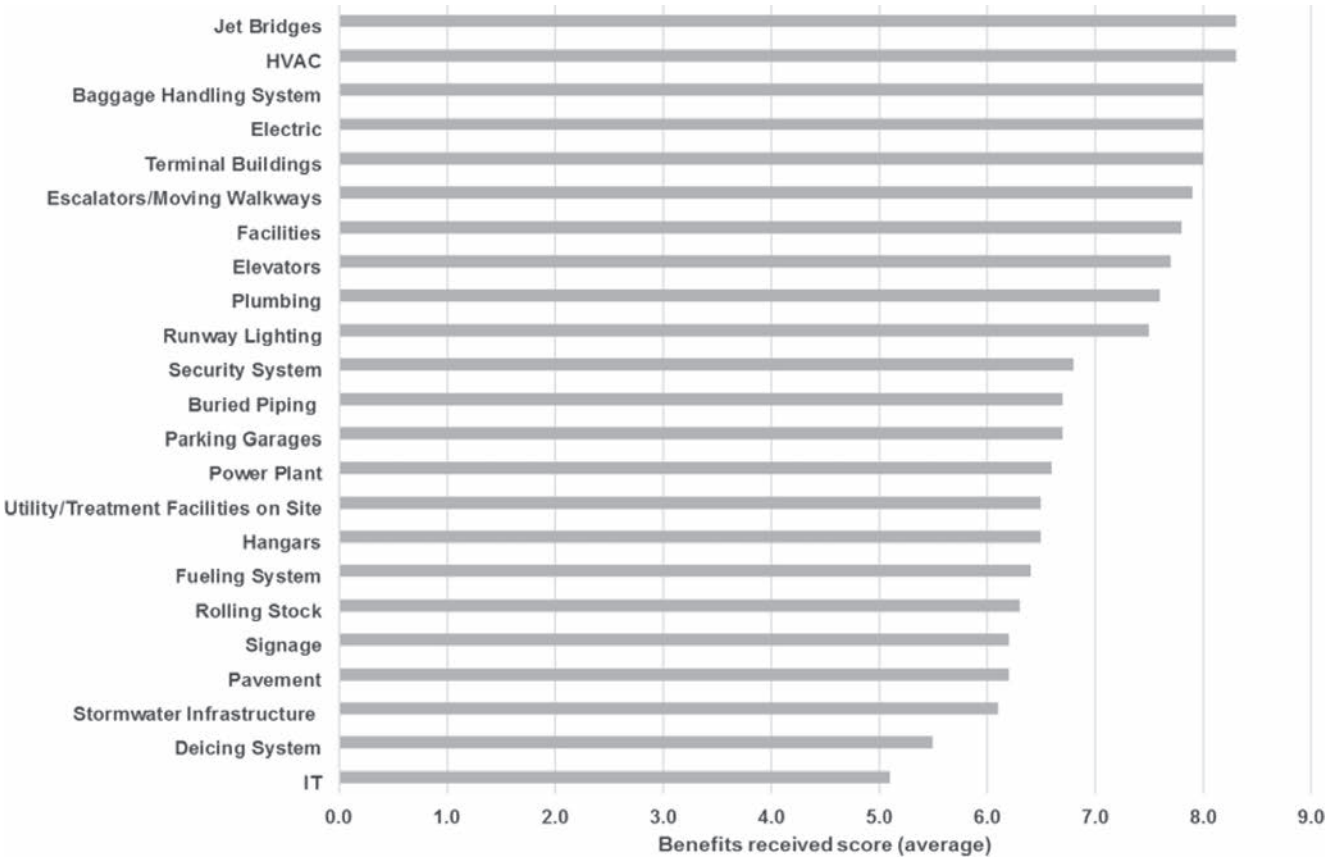


Figure 2-10. Average score of benefits received from managing asset in a CMMS.

baggage handling system). Another reason could be that many of these assets require more frequent maintenance than assets that were ranked lower.

Integration of CMMS with Other Systems

The survey asked participating airports for their insights on integrating a CMMS with other business systems used, such as a financial system, storeroom system, or GIS. About 55% of the airports separate the various systems, and 45% have partially integrated CMMS with a few systems. None of the airports reported having fully integrated a CMMS with other business systems.

The survey results indicate which systems are most likely to be integrated with a CMMS. Figure 2-11 ranks the integration of systems from greatest to least, as reported by the survey respondents. The top five systems that are integrated with a CMMS are: (1) financial software, (2) FAA Part 139 inspection application software, (3) GIS, (4) storeroom systems, and (5) airfield lighting systems. The rankings also suggest that few airports are likely to consider existing security systems and resource management systems as options for integration. This result may reflect airport priorities given resource constraints in conjunction with the relative difficulty of integrating such systems into the CMMS.

Airports tend to contract out many services to third parties, especially for tenants and maintenance activities. These activities generate data points and can be tracked in a system and connected to the main CMMS at the airport. The data can be entered into the CMMS in several ways. Tenants/contractors may be granted access to the system, airport staff may input data supplied by the tenants/contractors, or data may be input by a combination of airport staff and tenants/contractors.

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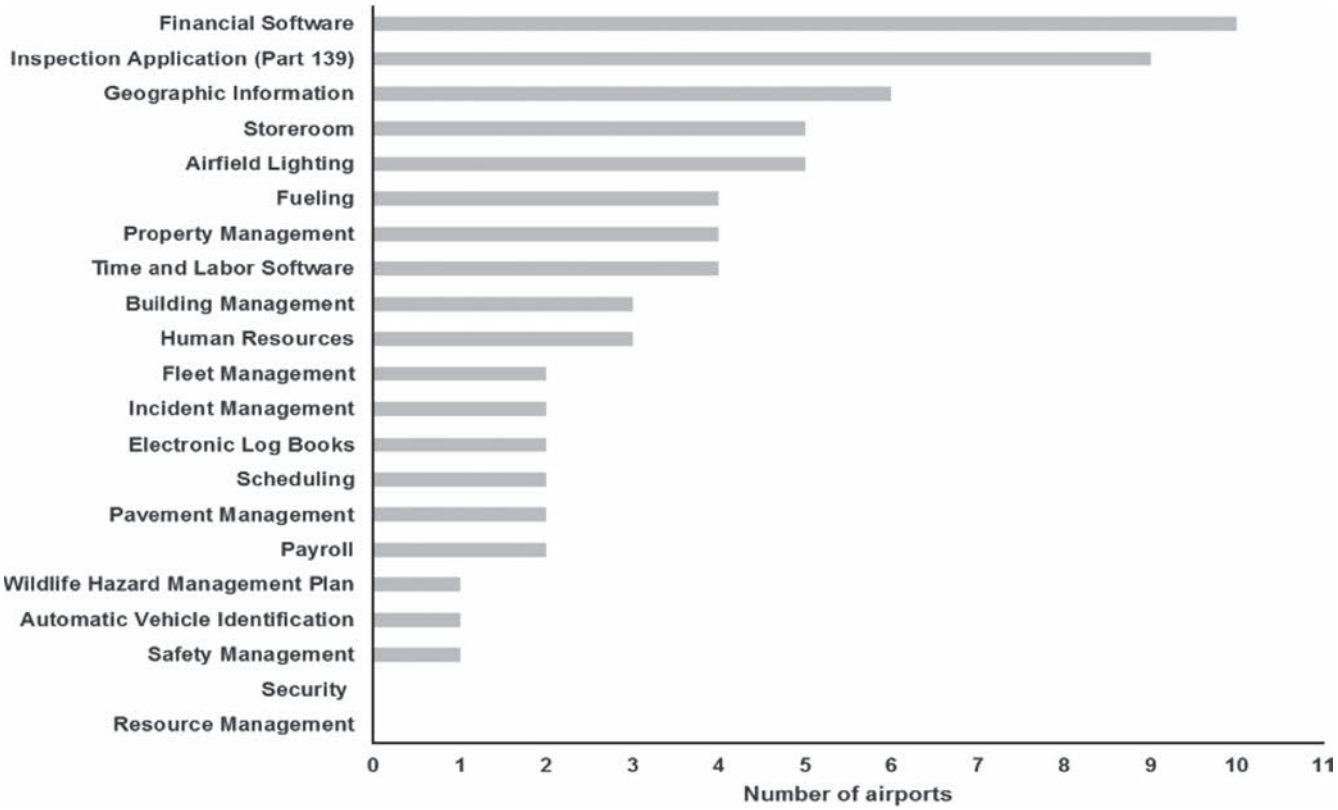


Figure 2-11. Ranked integration of existing systems with CMMS.

Data Management and Reporting

A major driver in improving O&M practices is collected data. The data life cycle has six main stages: (1) identification, (2) collection, (3) cleansing, (4) storage, (5) integration, and (6) utilization. A lapse at or between any of these stages can have a significant impact on the quality of the decisions that depend on this data.

Most of the airports responding to the survey (55%) capture, analyze, and use CMMS data for reporting and benchmarking. Other airports (18%) capture the data and analyze it but do not use it for reporting or benchmarking. Additionally, some airports (14%) reported capturing the data but not analyzing or using it. This group, together with the remaining airport respondents (totaling 23%) is still working on developing better data collection and utilization. Some airports (4%) did not respond to this question in the survey. Figure 2-12 illustrates the utilization of CMMS data as reported by airports in different airport classifications.

Many airports (45%) report using the data from a CMMS to modify and influence maintenance activities and strategies. Only 24% of airports reported using the data to modify and influence operational activities and strategies, and 26% of airports reported using it to influence capital projects.

Comments received from airports describing data captured in a CMMS included the following:

- "Data is captured and analyzed, but the analysis is not extensive or ingrained in the culture as we desire." (Commercial large hub airport)
- "No operational experience yet, however, the intent will be to capture and analyze (at minimum) to better manage resources."(Reliever non-primary airport)

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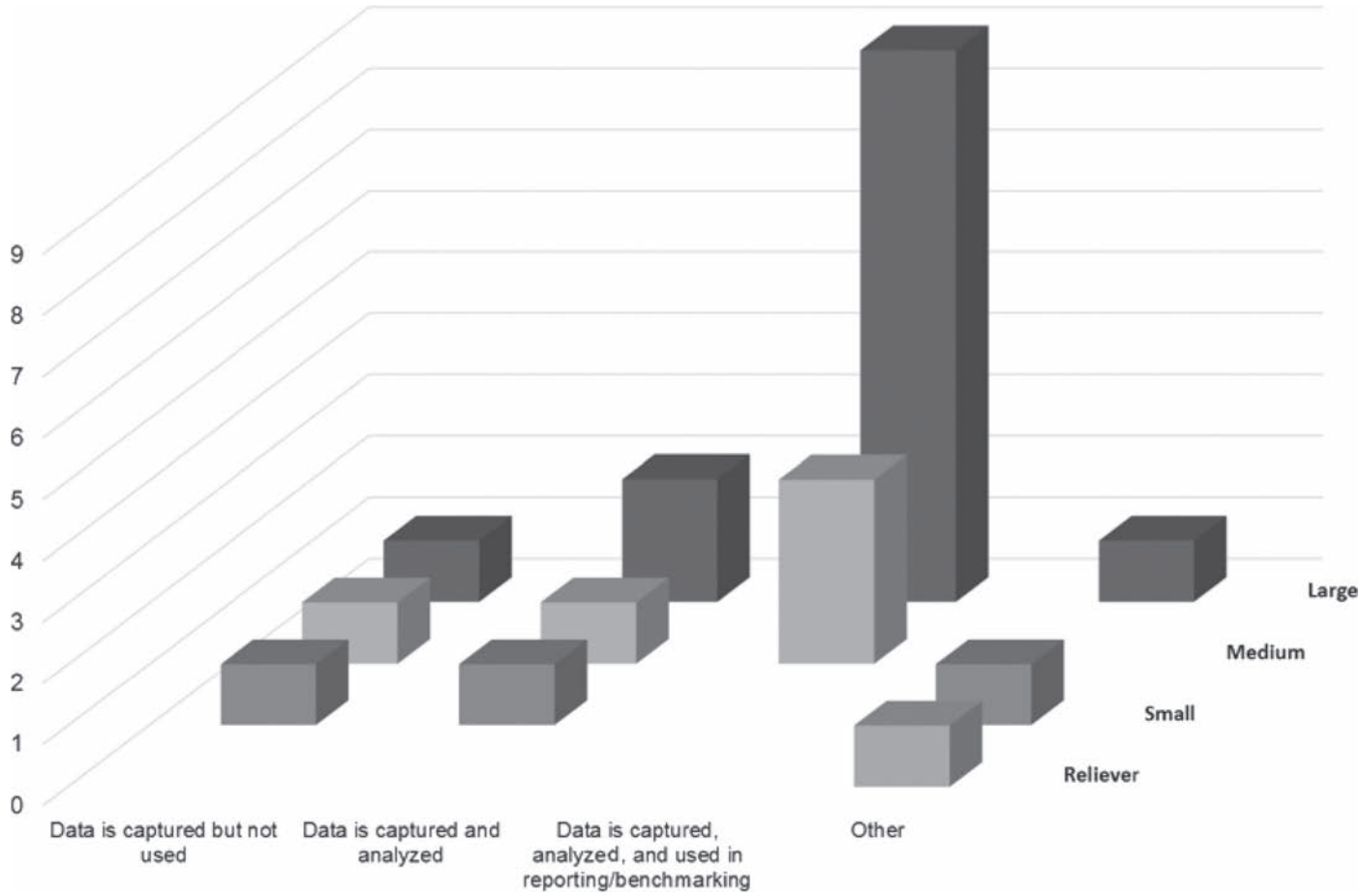


Figure 2-12. Statements that best describe data captured in CMMS, by airport classification.

Large, small, and reliever airports also contributed comments to the "Other" response option. These comments included:

- "Data is not captured."
- "The airport is in the process of evaluating how to better use the CMMS system."

Additionally, the survey results show that 30% of airports use airport staff to enter the data, and 48% of airports utilize a combination of staff and tenants to enter the data. Responses indicate that 22% of the airports do not track maintenance data in the system. The airports that do not track data fall across several airport classifications.

Metadata is essential for CMMS users to effectively access, search, and retrieve data. The metadata includes the asset hierarchy, which should include both geographical and system locations, numbering, naming, and other related asset information. Figure 2-13 presents the effectiveness rating of the CMMS metadata at different airports. An effectiveness rating of 6 or higher was selected by 65% of the airports; however, some airports were not pleased with the effectiveness of their metadata and thought it needed improvement. Among the comments received about this topic were:

- "It works well if you know the system but isn't intuitive." (Commercial large hub airport)
- "Was poor due to many variations; system is being rebuilt."(Commercial medium hub airport)

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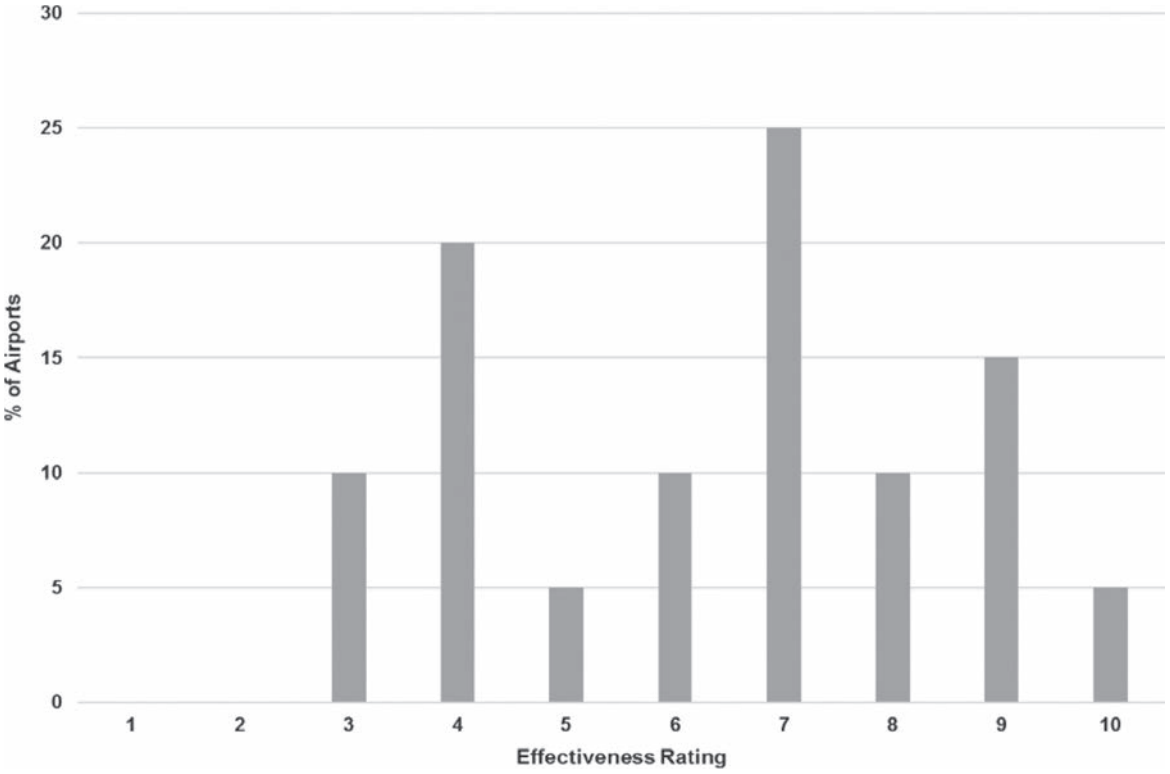


Figure 2-13. Effectiveness of CMMS metadata.

Data from a CMMS often is used for calculating KPIs, which can be shared and communicated with leadership and other business units and can otherwise be used as part of capital and operational decision making. The data can be presented and communicated in many ways, including:

- Periodic briefings
- Lists of KPIs
- Reports
- Dashboards
- Scorecards

Figure 2-14 shows the various ways respondent airports present and share CMMS data. Many airports use multiple ways to communicate the data. Most preferred are lists of KPIs and reports. Some airports reported that they do not use CMMS data for performance metrics; however, these airports mentioned that KPIs were under development. Some airports reported that they do not use CMMS data for performance metrics; however, these airports mentioned that KPIs were under development.

As a follow-up to the ways they choose to present data, the airports described their maintenance KPIs by selecting from among specific phrases, as shown in Figure 2-15. The majority of responding airports indicated that their KPIs are developed based on specific needs. Moreover, at many airports, management uses KPIs to effect change. On the other hand, a few airports indicated that they do not have KPIs. Among the comments received from these airports were the following:

- "Still [a] work in progress. We need more staff with understanding of the software to ensure full utilization of the capabilities of the software, but we aren't there yet." (Commercial large hub airport)

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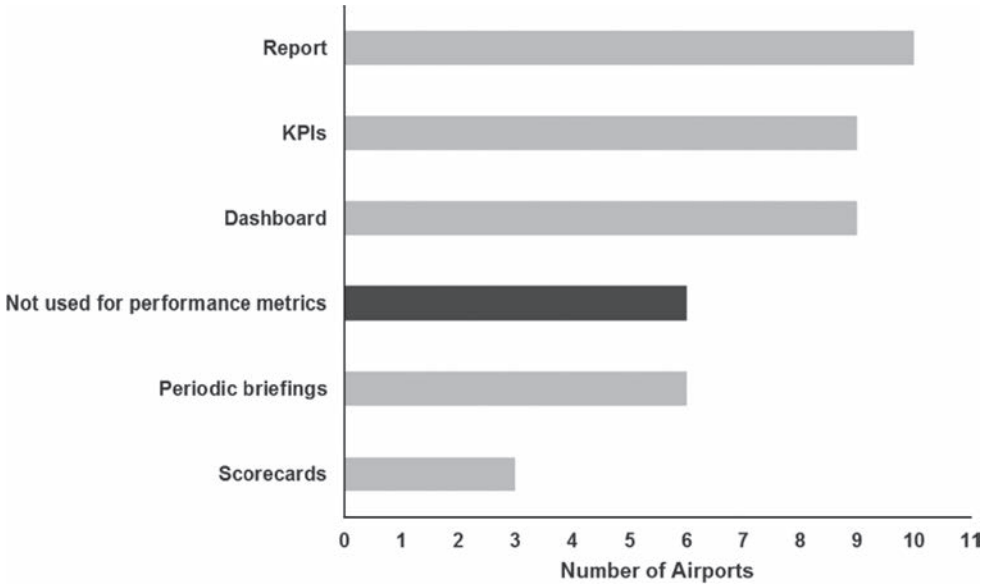


Figure 2-14. Methods of presenting CMMS data.

- "Many KPIs are contract driven to monitor contractor performance, mostly for ground support equipment (GSE) and [the] baggage handling system (BHS)."(Commercial large hub airport)

Discussion and Conclusions

The objective of the survey conducted for ACRP Project 09-14 was to benchmark and document the status of the airport industry in implementing a CMMS.

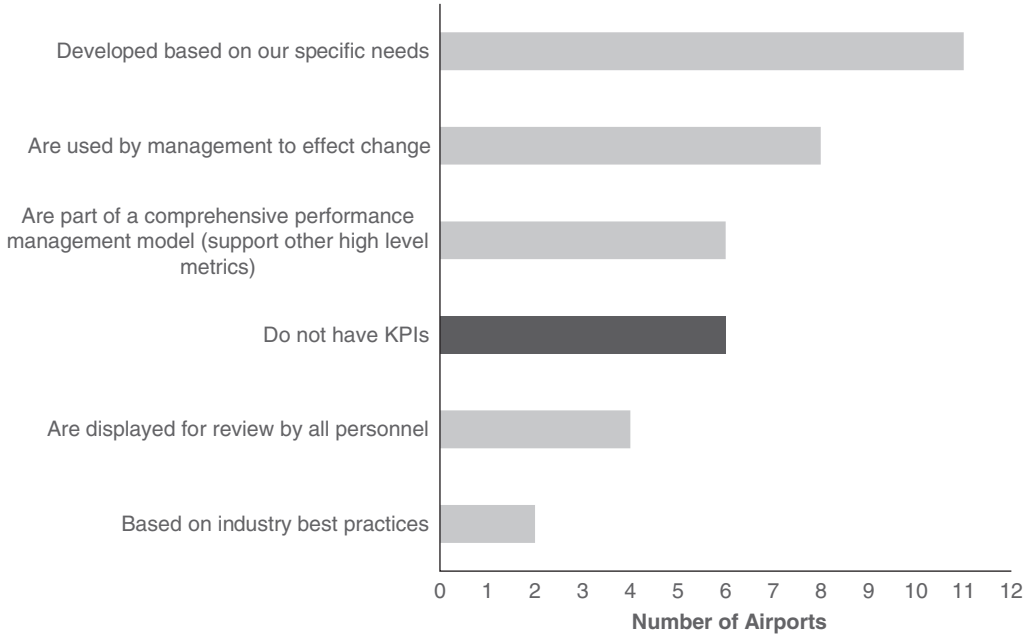


Figure 2-15. Description of airport's maintenance performance indicators.

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Twenty-five (25) airports participated in the survey. Most of the airports were in the United States, and two airports were in Canada. The responding airports spanned all FAA classifications and all regions of the United States and Transport Canada.

The research team analyzed the responses from the airports participating in the survey, including scores and comments. The responses indicated that airports are at different stages in the journey of implementing a CMMS and its integration with other systems. The results indicate that airports want to implement systems to improve their maintenance, operational, and capital strategies, but that airports, regardless of classification, perceive many challenges in that implementation. The perceived challenges include lack of resources, funding, understanding their systems, leadership buy-in, and more.

Based on the survey results and additional comments from the airports, it can be concluded that the larger the airport is, the higher the chances are that it will consider and successfully implement a CMMS. Airport size also impacts the number and types of assets that are available and considered to be managed using a CMMS; however, airports that have not implemented a CMMS indicated they have systems in place to track and manage their maintenance activities and that they are in the process of considering and/or implementing a CMMS.

Airports are becoming more proactive in managing their maintenance activities, yet many factors affect how they go about this, such as funding, qualified personnel, training, and available resources.

The majority of the airports that responded to the survey (55%) capture, analyze, and use the data from a CMMS in reporting and benchmarking. Other airports (18%) capture the data and analyze it, but do not use it in reporting or benchmarking activities. Many airports that have a CMMS manage it internally (i.e., using airport staff) whereas a small number of airports contract with a solutions provider to manage and maintain the system.

Communicating data and analysis to decision makers is critical to securing capital and operational funds. The two preferred methods to communicate data are through lists of KPIs and reports. KPIs are generally developed based on the airports' specific needs. KPIs are used by management to effect change. Across a variety of airport classifications, a few airports do not use CMMS data for tracking performance. These airports report that they do not use KPIs because of lack of resources and/or a lack of staff knowledge/understanding of the full CMMS capabilities.

Regardless of classification, the majority of airport respondents recognize the benefits of implementing a CMMS. The airports that have a CMMS in place indicated their CMMS is at a good to a satisfactory stage. These airports also indicated there is still room for improvement in areas such as resources, leadership buys-in, systems, and standards.

Implementing a CMMS has driven airports to develop and put strategies in place that include connecting and integrating the CMMS with other systems. The top six systems that are integrated with a CMMS are: (1) financial software, (2) FAA Part 139 inspection application software, (3) GIS, (4) storeroom systems, and (5) airfield lighting systems.

After completing the survey analysis, the research team, based on selection criteria, identified and approached representative airports in each airport classification to invite them to participate in the case studies. The selection criteria included:

- Experience in implementing a CMMS
- Using a CMMS for different types of assets
- Integrating a CMMS with other systems
- Reporting and communicating data to decision makers

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CHAPTER 3

Designing the CMMS with the End in Mind

This chapter provides an overview of the development and implementation strategies for a CMMS and the critical decisions that an airport should make when implementing a new CMMS or upgrading an existing system. The information is based on a variety of sources, including actual CMMS implementation experiences and airport feedback. The following topics are discussed:

- Key considerations in designing the use of a CMMS
- Choosing assets/systems to include in a CMMS
- Necessary steps to integrate the identified assets into a CMMS
- Steps to determine the critical data needed from a CMMS
- Development and utilization of KPIs, reporting, and business intelligence (BI) information (transforming data to actionable information)
- Considerations for integration of other critical airport systems into a CMMS

Airports have a basic understanding of some of the major strategies they should consider relative to the organization's goals and objectives when embarking on a CMMS implementation. As apparent from the case studies, these goals can include staffing justification, identifying the proper level of maintenance and replacement schedules, or PM to preclude asset failure. Many goals are not airport-specific; rather, they are global to many types of organizations. Experience with airports and other industries has enabled numerous best implementation practices to emerge, and these practices can help others embarking on the journey to adopt a CMMS.

Key Considerations in Designing a CMMS Program/System

The first step in the overall process of designing a CMMS is to put together a CMMS project team. This team should include personnel from all levels of the airport management and maintenance business units. The airports that have experienced the highest level of success with their CMMS programs are those that formulated the design and selection of their CMMS by integrating both top-down (strategic) and bottom-up (tactical) perspectives. The airports that use only one approach do not produce the overall organizational buy-in that is required to ensure a successful implementation. The case study findings support the contention that a cross-departmental steering committee is essential to implementation success.

The major goal of the CMMS project team is to define an organization-wide CMMS strategy and to align it with the airport's vision for short- and long-term asset management. The project team should ensure that the organization has an overall commitment to the implementation or upgrade of the CMMS. The team also should consider reviewing and updating the organization's existing processes and procedures as needed to meet the established vison and strategies. As part of this effort, it is suggested that the team consider a complete review of existing maintenance

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practices and procedures. The project team also should define the criteria for measuring short-term and long-term success.

It is important that the CMMS project team understand the main elements to be considered in the successful design and implementation of a CMMS. These elements, itemized below, include lessons learned by other airports during their journey through the design and implementation of a CMMS. Among the most important recognized challenges is organizational buy-in for change management. From the outset, the project team should determine if they will build and implement a change management process from within the airport or bring in outside assistance to help identify and implement change management practices and procedures.

Decision Making

- **Preparation and inclusiveness.** Too often, asset management decisions—especially those made reactively—are made in a vacuum. For example:
  - During the design of a facility, long-term O&M requirements often are not considered. Implementing the CMMS can allow the airport to collect the O&M cost data required for whole LCC estimation.
  - Organizations often fail to consider the consequences of failure of an asset.As a result, too much rigor can be placed on some assets and not enough on others. Determining both an asset's likelihood of failure and the consequences of that failure makes it easy for an airport to assess risk more efficiently, and to more accurately define which assets are critical and should get the most attention.

  A project team that integrates all levels and management units of the organization can reach out effectively to obtain the information needed to plan and ensure that the CMMS will support effective decision making.
- **Performance.** A list of assets should be established for inclusion in the system and a program built to optimize these assets' useful life.
- **Stakeholder education.** All organizational stakeholders should understand the overall vision and goals of the organization as they relate to the successful implementation or upgrade of a CMMS. It is important to ensure that all stakeholders understand the big picture.
- **Data management and transparency.** It is better to make decisions with no data than with bad data. Date should be treated as an asset in terms of data collection, validation, verification, storage, security, maintenance, retrieval, auditing, and disposal. The project team can establish and maintain an overall data management strategy and use internal data analysts to help in the effort.
  - Internal staff and external vendors should capture and record all asset data in accordance with an established data management strategy.
  - External vendors should be required to record all maintenance activity data into the owner's system.
  - Facility and other asset data should be treated as an asset in and of itself, and it should be maintained, audited, and safeguarded.

Organizational Change Management

- Enterprise vision and strategies should be communicated at all levels within the airport.
- Change can be required for many reasons, including:
  - Government regulation,
  - Market changes,
  - Growth of the airport,
  - New technologies within asset construction, installation, replacement, refurbishment, retrofitting, repurposing, decommissioning, and disposal,
  - Bonding requirements,

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  - Airlines, and
  - Tenants.
- All departments should be engaged in the process and included in the "change" discussions.
- At times, opportunities may exist to change accountabilities within the organization, aligning them with the functions of the new CMMS. The project team can:
  - Deploy planning and scheduling staff, tying accountabilities to the CMMS.
  - Look at sensible organizational structure changes (e.g., in two case studies, the airports re-organized by moving inventory accountabilities under the facilities group).
- Expectations for new and revised roles must be clearly set.
- Leadership has an important role by setting an example in collaboration and supporting change.

Enterprise-wide Systems Continuous Improvement Program

- **Plan.** The project team establishes and communicates policies and procedures.
- **Do.** The organization's staff follows the established policies and procedures.
- **Check.** Responsible leaders and managers ensure policy and procedure compliance.
- **Adjust.** The team (and organizational leadership) revises policies and procedures as required to maintain continuous improvement.

Competence Management

- Capture and document unwritten knowledge.
- Develop and implement a succession plan throughout the airport.
- Recruit talented and skilled professionals.
- Establish and implement a professional development program.
- Put plans in place to retain talent.

KPIs and Standards

- Establish success criteria for the overall CMMS implementation or upgrade.
- Determine and monitor KPIs for the CMMS implementation or upgrade.
- Determine and monitor O&M KPIs to track via the CMMS.
- Engage staff data analysts by having them extract and perform data quality analysis for cost trending and the KPI/performance reporting.

Like any major technology program implementation, it is critical to start off the program the right way; implementation of a CMMS is no different. Before implementing a system, an airport must address some major considerations. For example, the project team must:

- Define assets and distinguish important assets from "spare parts."
- Define which assets are critical to the airport and develop or enhance the plan for collecting information to populate the CMMS with those assets and the associated asset attributes.
- Define the KPIs/metrics the airport needs to make risk-based, data-driven business decisions relative to the assets and overall maintenance organization operations.
- Define the overall asset system hierarchy (asset system parent-child relationships).
- Define the goal-dependent hierarchies within the system.

Maintenance Business Processes and the CMMS Workflow

Another key factor affecting the CMMS will be the design and documentation of maintenance operational workflows. Most CMMSs include a workflow engine as part of the heart of the system. A workflow engine is a CMMS module that automates business processes. A workflow

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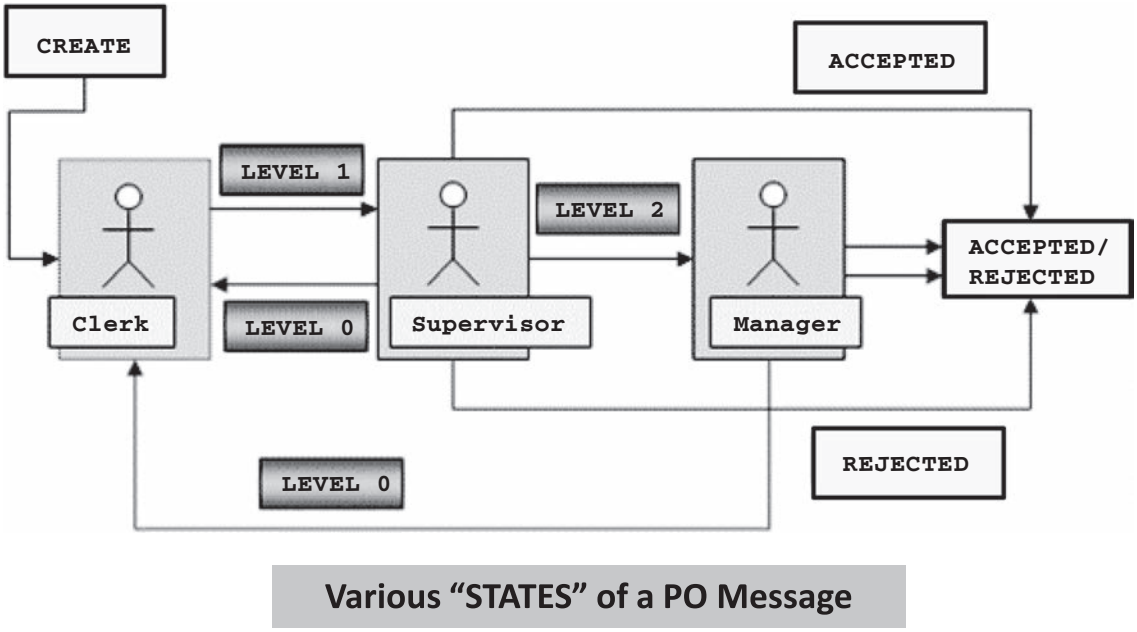


Figure 3-1. Purchase order workflow.

can be configured to manage and report on automated processes that are built and mapped to redundant day-to-day maintenance activities.

A typical workflow might move maintenance work requests from the requestor to the appropriate maintenance supervisor or craft group. Workflows also typically allow for building notifications and escalations. For example, if an action on a critical item is not taken in a timely manner, the responsible personnel can be notified via text, email, or another messaging system that the action is required. If no response is received, the item can then be escalated to another individual or manager. Workflows also can be built to automate the transition of an item through the overall business process, based on actions taken or not taken at a given time. For example, an approval process may be organized to send notifications/approval requests to individuals in parallel or sequentially. Figure 3-1 illustrates the workflow for a purchase order.

Choosing Assets/Systems to Include in the CMMS

Choosing which assets to include in a CMMS can be a stumbling point for many organizations. An organization may assume that every asset should be included in the CMMS with no real consideration of why; however, an airport has to keep in mind that both the asset and the associated data must be maintained. It is therefore advantageous to resist the tendency to over-burden the maintenance group with too much data management.

Before selecting the assets to include in a CMMS, an airport should complete a "consequence of failure" analysis of their assets. How does an asset failure impact the airport, customer interfacing, safety, cost, and the achievement of the airport's objectives? If the failure of an asset has the potential to significantly impact the achievement of the airport's objectives, this asset is considered a *critical asset*, as defined in ISO 55000 (issued in 2014). An overall maintenance program/philosophy will apply varying levels of rigor to the assets, reflecting the varying impacts of the assets' potential failure on the airport.

Many professionals shy away from implementing the Run-to-Failure (RTF) approach; however, if an asset's failure will have little impact on achieving the airport's asset management

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objectives, spending a significant amount of money on these assets is unjustified. If PM will not be performed on such assets, they may not need to be included in the CMMS; alternatively, the amount of attribute information stored in the CMMS may be limited to save time and costs. For example, non-critical assets can be combined together within the asset hierarchy. Using this approach, the CMMS can capture the cost of replacement when the RTF asset fails.

Airports also can consider how assets relate to each other. Single assets often are parts of an overall system. For example, the HVAC system in the airport generally will consist of numerous individual, interconnected, and interrelating assets. By considering how the assets relate to one another, the assets can be monitored and managed within a functional system hierarchy. The CMMS can then monitor the system rather than the individual assets. If the individual HVAC assets have been correctly set up as a system, when someone calls in a "hot" or "cold" report the maintenance department will know—based on a geographical location—which HVAC unit feeds that location and which electrical circuit it is tied to. Correctly implemented asset systems can give the airport the flexibility to incorporate more assets (and more levels of detail) in the CMMS.

Federal, state, and local regulatory reporting requirements relate to some airport assets. Obviously, those assets should be included in the CMMS to ensure regulatory compliance. Additionally, those assets that must be reported on relative to life safety and overall safety management/incident tracking must be included in the CMMS.

Necessary Steps to Integrate the Identified Assets/Systems into CMMS

Several schools of thought exist when it comes to the processes and procedures for importing or integrating asset data into a CMMS. First, the airport must identify where the necessary data about the asset is located. For some assets, data can be stored and maintained in different formats and even different locations. Useful data can be found in existing systems, in documents and spreadsheets, and in electronic, hard copy, or PDF formats. How best to capture and integrate the asset information into a new or upgraded CMMS can vary, as can the associated costs.

The importance of ensuring that the assets to be included have been reviewed for criticality bears repeating. It is equally important to be selective about which attribute and specification data to collect about the chosen assets. Making decisions with bad data can be worse than making decisions with no data, and choosing to collect all the data—as opposed to a strategically chosen subset of the data—comes at a cost.

The data integrated into a CMMS must be verified, validated, and properly maintained to ensure its accuracy and relevancy to the airport's overall CMMS vision and strategy. The cost-effectiveness of including specific asset data should be considered carefully. A good practice for asset data collection is to limit the data to only that which will be analyzed to inform decision making or to meet regulatory compliance. Another useful practice is to ensure that data is being collected cost-effectively and stored in a manner that ensures it will be available for analysis at the time it is required.

The actual methodologies used to integrate asset data into a CMMS vary greatly in approach and cost. Relative to new assets and systems, many organizations now incorporate building information modeling (BIM) into their asset data loading processes. If it has been properly specified in the airport's contract with the vendors, the asset data found in BIM can be directly imported into the airport's CMMS. That said, ensuring that data from BIM can be loaded into a

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CMMS takes a significant amount of coordination between the airport, its architects, and construction firms. It is the airport's responsibility to ensure that all parties involved in the design and build process understand which assets and which associated data must be included in the BIM, including their alignment with the airport's geographical, functional, and/or electrical location hierarchies.

If BIM is not used (and in many cases this is not possible, especially for existing assets), then the alternative is to conduct a manual asset walk down/data collection program. If the airport has an existing database with current information about critical assets and their related data, it may be able to export the data from the database to a new or upgraded CMMS.

No matter which method is used to populate the CMMS with asset and their related data fields, the first step should be to complete a quality control review of all existing data to ensure its accuracy and relevancy relative to the overall reporting/analytic needs of the airport. Most organizations, including airports, have a great deal of data but very little actionable information. This is one of the main reasons why, before defining the asset data needs, the airport's CMMS project team should conduct a complete review of the existing data requirements and look for any gaps in the data as it relates to the overall informational needs of the airport.

In most cases, if the data is not available in an existing database or if the airport project team finds gaps, then a complete manual asset walk down/data collection project may be required. Performing an asset walk down is time consuming and expensive. If done properly, however, it can provide the airport with accurate asset location and attribute information.

A recommended additional step is to ensure that assets are bar coded during the asset walk down. Bar coding or radio frequency identification coding of assets with the use of mobile devices can significantly reduce the time it takes to execute work orders against the assets and keep asset data current.

Steps to Determine the Critical Information Needed from CMMS

Determining the data needed in a CMMS is not an easy task. To ensure that the airport is making decisions based on useful/actionable information, a proper review of the overall, organization-wide information needs is best conducted prior to implementing or upgrading a CMMS. This review also helps ensure that inaccurate data will not be loaded into the CMMS. To understand the critical information the airport needs from the system, the CMMS project team should take the following steps:

- Identify and understand the questions stakeholders at all levels of the airport need to have answered. The needs and overall expectations of all airport stakeholders need to be understood. If the information needs of the airport are understood, the project team can then establish what types of asset data and attributes need to be associated to the assets and systems. It is impossible to produce quality reporting and high level analytics if the data in the system is wrong, not standardized, or out-of-date.
- Consider the types of maintenance operations required for each asset, and whether the maintenance operations are reactive or proactive. Certainly, most maintenance departments strive to be proactive and reduce the amount of emergency or break-in work. Having historical asset failure history available from the existing system is one of the keys that will help an airport move from reactive to proactive as it relates to maintenance activities. Including historical asset failure data in the CMMS significantly enhances the airport planner/scheduler's ability to realign the overall maintenance program.

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Steps to Implementing a CMMS

To implement a CMMS the airport will:

- Select the members of the implementation team wisely, including internal and external participants and ensuring cross representation of functions and levels in the organization.
- Establish a program vision statement that is inspiring to staff.
- Plan and account for the time key personnel must devote to the implementation.
- Consolidate, validate, and standardize all required data prior to system population.
- Develop and implement system maintenance procedures prior to roll out.
- Perform "just-in-time" training for personnel that will use and manage the system.
- Where possible, roll out the technology in modular form.
- Where possible, use off-the-shelf tools that require little or no modification to meet the airport's goals.
- Keep it simple.

The following phases outline a basic CMMS Project Implementation Plan:

- Initiate
- Define
- Develop
- Validate
- Deploy
- Operate

Figure 3-2 highlights the *Initiate* phase of the project. This phase covers tasks that address project logistics, project kick-off, team members' roles and responsibilities, project schedule development, and system immersion training. The purpose of system immersion training is to educate key stakeholders and decision makers on the basic fundamentals of the software solution. This education prepares the implementation team to make the necessary decisions in the second (*Define*) phase of the implementation project. If the airport will use internal IT resources for the technical development of CMMS, appropriate IT staff can be represented on the project team from the beginning of this phase. If the airport will use contracted IT professionals, initial steps to identify an appropriate contractor may begin toward the end of this phase.

Deliverables from this first phase of CMMS implementation involve both meetings and documentation, and include:

- Project kick-off meeting
- Team roles and responsibilities
- Baseline Project Plan (including an initial list of functional and technical requirements)
- Communication Plan
- System Implementation Plan
- System Immersion Training Plan and subsequent training

Figure 3-3 highlights the *Define* phase. This phase focuses on defining specific process and system objectives to cover the functional and technical requirements that have been identified in the Baseline Project Plan. During this phase, the implementation team conducts specific,



Figure 3-2. CMMS implementation phases: Initiate.

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Figure 3-3. CMMS implementation phases: Define.

topic-based workshops that will illuminate specific business processes and identify case scenarios within the existing systems. The workshop results are documented in a Functional Specification Document, which will serve as the basis for the technical development, testing, and training exercises to follow. Also during the *Define* phase, the implementation team will develop a Data Migration Plan, a User Training Plan, and other planning documents that support upcoming activities.

Deliverables from the second phase of CMMS implementation include:

- Functional Requirements Workshop outputs
- Functional Specification Document
- Data Migration Plan
- Configuration Management Plan
- Validation Plan
- User Acceptance Testing Plan
- User Training Plan
- Risk Matrix
- Risk Management Plan
- Updated Project Plan

Figure 3-4 highlights the *Develop* phase of CMMS implementation. This phase starts with the development of detailed technical specifications based on the Functional Specification Document. At this point, input from the IT members of the project team or from the contracted technical developers/analysts will be essential to ensure that the technical specifications align with the desired functional results. From there, the implementation team will divide the development process into scheduled "iterations" and configuration of the CMMS system will begin. Members of the implementation team should perform walkthroughs of the system development at regular intervals throughout this phase to provide iterative feedback to the technical development team.

Deliverables from the third phase of CMMS implementation include:

- Technical Specification Document
- Installed development environment
- Configuration of requirements within development environment
- Data migration into development environment
- Scenario-based test scripts/user acceptance test scripts
- Requirements Traceability Matrix
- Role-based training material
- System configuration documentation
- Operational Support Plan
- Deployment ("Go-Live") Execution Plan



Figure 3-4. CMMS implementation phases: Develop.

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Figure 3-5. CMMS implementation phases: Validate.

Figure 3-5 highlights the *Validate* phase of CMMS implementation. This phase has two stages. First is the performance of a Conference Room Pilot, in which the key stakeholders of the project participate in an end-to-end walkthrough of the various use case scenarios of the fully developed system. The purpose of this exercise is to validate the system. The pilot allows the system to be analyzed from a data, integration, security, and performance perspective. Second is the performance of user acceptance testing, during which users of the system execute test scripts in a preproduction environment. After the first round of user acceptance testing, any discrepancies are addressed, and a second round of testing occurs to specifically target the remedied discrepancies from the first round. Resolution of discrepancies is documented as part of the iterative development and validation process. In addition to the two stages, the Deployment Execution Plan is reviewed and updated as necessary to address any deviations that were captured during the user acceptance testing.

Deliverables from the fourth phase of CMMS implementation include:

- Configured system deployed to test environment
- User Acceptance Testing Summary Report
- Discrepancy resolutions documentation
- Updated Deployment Execution Plan

Figure 3-6 highlights the *Deploy* phase, which includes end user training and deployment of the new system configurations to a production environment. While training exercises are ongoing, the production environment is installed and configured according to the Deployment Execution Plan. Once ready, the existing computer system is cut over to the new CMMS system, which is released to the users for live usage. Also finalized during the *Deploy* phase are the Operational Support Plan and a Backup and Restore Plan to handle needs that may arise once the new system is operational.

Deliverables from the fifth phase of CMMS implementation include:

- Operational Support Plan
- Backup and Restore Plan
- End user training
- Executed deployment and cut over
- Live CMMS system

Figure 3-7 highlights the *Operate* phase of CMMS implementation, which includes operational support. This phase includes establishing a mobile "help desk" on site and the process by which the user community can communicate with the help desk. Any needed follow-up training is identified and scheduled, issues are electronically logged, and triage is performed as needed so that each issue is managed to resolution.



Figure 3-6. CMMS implementation phases: Deploy.

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Figure 3-7. CMMS implementation phases: Operate.

Deliverables from the final—and ongoing—phase of CMMS implementation include:

- Weekly status reports of outstanding issues
- Hands-on managed system support

Transforming Data to Actionable Information

Without a clear understanding of organization-wide information needs, the development of accurate and useful KPIs, reports, and BI tools is impossible. With today's CMMS software, it is very easy to build KPI dashboards, ad-hoc reports, custom reports, and use BI tools. However, if the necessary data fields are not in the system, then useful, actionable information cannot be extracted from the database into KPI dashboards or reports, or imported into BI tools. The key to the effective use of these management tools is having valid and verified data in the system.

Ideally, before developing any KPIs or reports, the CMMS implementation or project team will have met with all the airport stakeholders to obtain a true understanding of their reporting needs, including what types of information are needed daily to do the job. The data needed by maintenance supervisors will be somewhat different from the data needed by technicians or department managers. Having collected this information during the first or second phase of the CMMS implementation (or during comparable phases for a system upgrade), the team will have gained a better understanding of what data must be in the CMMS and can determine how to develop useful KPIs and reports and build effective BI tools. The data field configuration will be specific to the organization, and typically will take place during the *Develop* phase of CMMS implementation, with adjustments made as necessary during the *Validate* phase.

Useful KPIs that will require data field configuration include:

- Scheduled, completed, on hold (with reason), and past-due work orders
- Planned versus actual maintenance actions, labor time, parts utilization, and overall costs with delta (change)
- Mean time to asset failure
- Overall asset uptime versus downtime
- Mean time to asset repair
- Ratio of planned maintenance to reactive maintenance

Figure 3-8 shows a sample of what a help desk KPI/dashboard may look like.

Integration of CMMS with Other Critical Airport Systems

A CMMS can significantly improve the overall efficiency and overall productivity of the maintenance business unit. A CMMS can help the airport lower overall maintenance costs by extending asset life, improving wrench time, and improving the overall materials management processes. To fully achieve these efficiencies, however, integration between the CMMS and other critical airport systems is a must. Integrating a CMMS with other critical airport systems allows for the efficient, timely flow of data between systems and can significantly improve the overall maintenance process.

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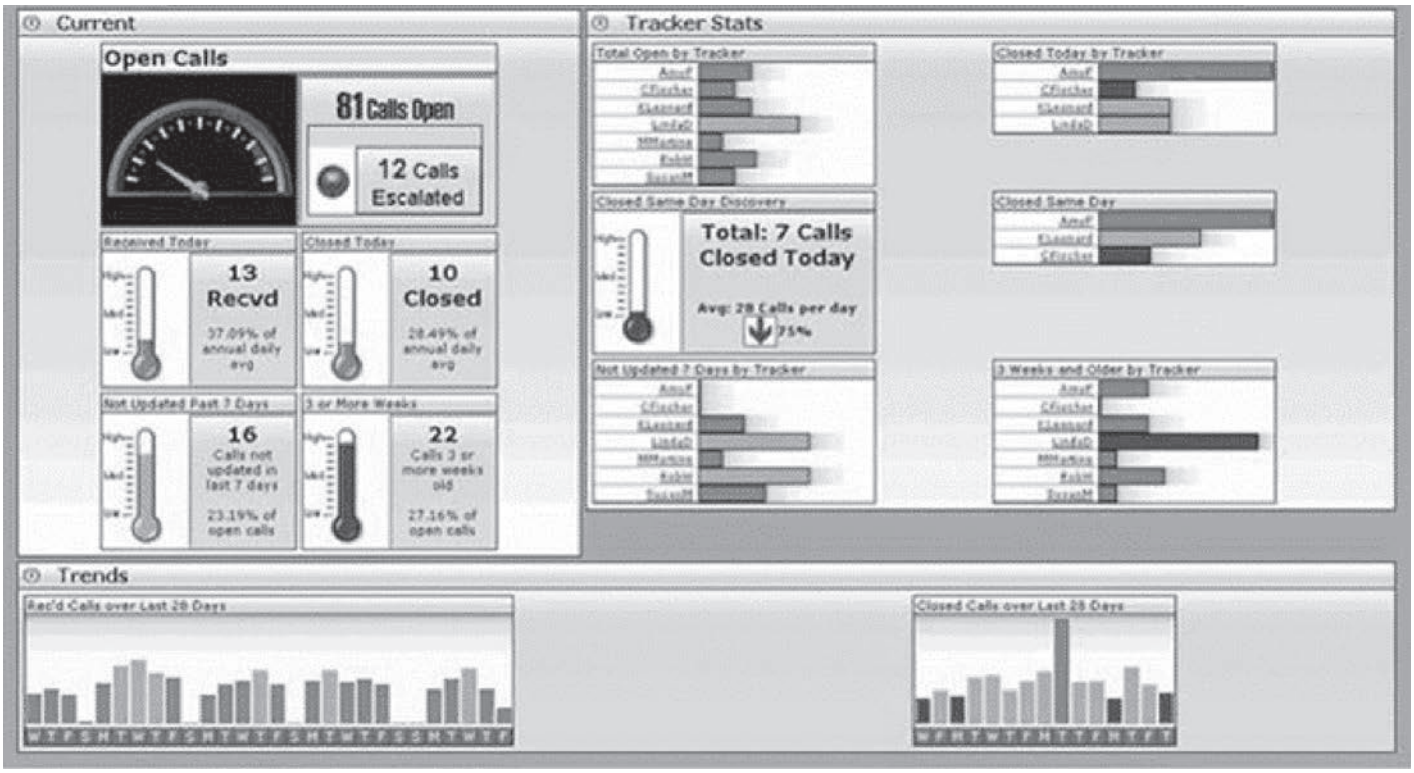


Figure 3-8. Example of a help desk KPI/dashboard.

Integrating a CMMS with another critical system involves establishing the *type* of integration needed based on the necessary data flows. Integrations can be one-way or bi-directional. What systems an airport integrates with the CMMS depends on how the airport is organized and how the various departments within the airport interact in terms of information needs and data sharing. Existing software or systems that are candidates for integration with a CMMS may address:

- Resource management
- Security
- Safety management
- Automatic vehicle identification
- Wildlife hazard management
- Payroll
- Pavement management
- Scheduling
- Electronic log books
- Incident management
- Fleet management
- Human resources
- Building modeling
- Time and labor
- Property management
- Fueling
- Airfield lighting
- Materials management/inventory
- GIS
- Airfield operations inspection
- Enterprise resource planning (ERP) financial software

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Today, the effort involved in integrating disparate systems is not as difficult as one may think. The bigger question is how to minimize the number of systems staff need to access to get the information they need to do their jobs.

One example of this is the ability to determine who is responsible for maintenance work and the associated costs via integration of the CMMS with the airport property management system. Many airport maintenance business units lose time and money by completing work for which they are not responsible. Maintenance business units hold high standards when it comes to good customer service. As a result, work often is performed at tenant locations. In some cases, work done by the airport either should have been done by the tenant or, if the airport was responsible for doing the work, should have involved a charge back to the tenant. Within-airport leases almost always define such maintenance responsibilities, and details about these arrangements may already reside in an airport's existing property management system. By completing a simple integration between the airport's property management system and the CMMS, when a work request comes in, the system can immediately identify if the work is a tenant responsibility, an airport responsibility, or an airport responsibility with a charge back. Additionally, if the interface is bi-directional, once the work is completed, the labor and parts costs can be sent back to the property management system so that the maintenance costs for the month can be added to the tenant or airline invoice. Once the integration has taken place, airport staff need only interact with the CMMS—the data coordination between the CMMS and the property management system essentially functions as a behind-the-scenes operation.

Key Considerations and Approaches for Small Airports

Effectively implementing and using a CMMS in the small airport environment requires overcoming a unique set of challenges, including the large, diverse set of assets and airport systems, and the complexity of features and capabilities of a CMMS. The challenges of data transfer and data sharing across systems also can be compounded by limited resources and a lack of supporting infrastructure. The ability to leverage a CMMS as a single "source-of-truth" data repository for maintenance and asset management strategy implementation and reporting is a ripe opportunity for many airports to enhance overall operations, maintenance, and business decision making. The right maintenance business processes, aligned with the right airport-specific CMMS, can streamline paperwork, integrate with other technologies, and help make operations more efficient.

The proper implementation, use, and maintenance of a CMMS program can be scaled for all size airports. Moreover, it affords small airports the ability to execute a world-class maintenance practice that meets the universally accepted work management business process flow illustrated in Figure 3-9.

Some common benefits of a well-implemented CMMS can be achieved by airports regardless of size. With the associated processes in place, a CMMS can facilitate:

- Use of a single source-of-truth data repository
- Improved management of preventive maintenance, work orders, and inventory
- Tracking events, categorize issues, and allow a variety of departments ready access to event or activity status
- Maintenance department performance reporting and justification of staffing levels
- Development of "most appropriate" asset replacement schedules
- Early identification of pending failures

To ensure robust reporting, the CMMS selection and implementation process should include carefully defined management reporting criteria.

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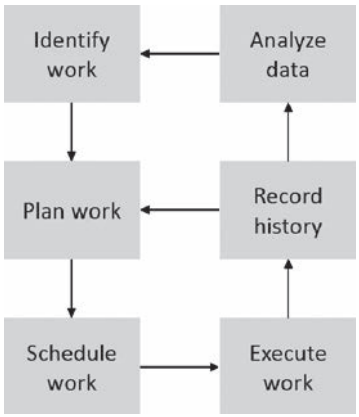


Figure 3-9. Work management process flow.

Selecting or Replacing CMMS Applications

Airport CMMS applications vary in cost and complexity. A "one-size" CMMS will not fit all airports. Small airports typically are the most challenged when trying to purchase and implement or replace a CMMS, primarily because of costs and resource constraints that may affect both IT and O&M staff.

The availability of CMMS Cloud-based subscriptions puts smaller airports in a better position to afford a CMMS, but such airports still face the challenge of dedicating resources to keep their asset data current and to administer and maintain the new or upgraded system. Beyond the initial investment cost, a typical challenge involve finding resources and staff time to convert from a manual, paper, or spreadsheet-based system to a new system that is consistent with FAA data standards. Apart from careful, targeted selection of the applications desired in the CMMS, small airports with resource challenges may consider front-loading their investment by requiring the CMMS vendor to perform a turn-key implementation. Such an implementation includes software implementation, hardware installation, data gathering, data loading, system training, and user training.

Airports of all sizes should consider the following basic applications (system capabilities) when determining system functionality. The CMMS should be able to:

- Track asset information (e.g., cost center, department, location)
- Allow for cost and repair information to be accessed at multiple asset levels
- Save work orders to the asset history as one line of information or in full detail
- Sort or build ad-hoc reports that allow all information in the asset history to be accessed
- Provide inventory capabilities for each asset
- Provide the ability to retrieve, select, sort, store, use, and report data relative to asset items/types

Efficient data management and accurate reporting require the establishment of a consistent asset taxonomy, hierarchy, and attributes. *Taxonomy* is an important term in CMMS. Defined as the science of classification, taxonomy is guided by ISO 14224 (issued in 2006, with an update issued in 2016). Classification is used to make indexing, grouping, saving, searching, and retrieving of digital data easier, faster, and more accurate. A benefit of establishing the asset hierarchy parent-child relationship is the ability to charge costs to the lowest possible asset level, thereby providing a means to allow field O&M staff to write work orders to identify and report where O&M dollars are actually being spent.

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Having a well-structured asset taxonomy helps an airport set up the CMMS accurately, reliably capture the data, and allocate resources (personnel, budget, tools, and schedules) to efficiently manage the system. In turn these efficiencies increase the reliability and maintainability of the airport's critical assets to meet operational goals.

Linking the CMMS to Decision Making and Performance Reporting

Performance measures are used in the daily operations of airports of all sizes. These measures (also called *metrics*) help airports track performance in various categories as they relate to either the organizational goals or an initiative's objectives. An organization's selection of metrics depends on these goals and objectives, as well as organizational structure, programs that are in place, and the availability of accurate data.

As the single source-of-truth repository, the CMMS is a key source of quantitative O&M data that can support developing and tracking performance measures. A data quality control plan must in place, which small airports can handle by limiting administrative access to only a few key individuals. The CMMS selection criteria also should include definitions of the required management reports as well as a review of the CMMS's flexibility in generating useful reports. Reporting criteria for a small airport to consider when selecting a CMMS may include the ability to:

- Produce interval reports (daily, weekly, monthly, yearly, or any user-defined interval)
- Generate summary reports, lists of information, or exception reports
- Create (write) customized reports in addition to the standard system reports
- Create reports designed to be used by CMMS managers, airport managers, or O&M staff
- Generate maintenance budget reports

Small airports can start with reports that cover a few basic performance measures. Once the reporting potential of the CMMS is seen, the list of reports usually grows significantly. Useful performance measures to consider include:

- Labor craft/crew costs
- Inventory costs
- Inventory usage
- Maintenance costs by asset type
- Total maintenance costs
- Percentage ratio between preventive and reactive maintenance
- Work schedule compliance (e.g., safety, preventive, corrective, emergency)
- Work backlog (e.g., safety, preventive, corrective, emergency)

Airport Asset Systems Managed Within the CMMS

The research has shown that many organizations stumble when choosing assets to include in a CMMS. It bears repeating that organizations may assume that every asset should be included in the CMMS without considering why it is being included. Airports, especially small airports, must keep in mind that both the asset and the associated data must be maintained. To help decide which asset systems to include in the CMMS, the airport project team can consider drivers such as federal, state, and local regulatory reporting requirements; FAA-specific inspection requirements; life safety; customer interfaces; and other systems deemed critical through an analysis process. It also is helpful to recognize which systems will have work orders generated against them and consider any asset that requires some form of preventive maintenance.

Table 3-1 presents an abbreviated list of airport asset systems developed from the results of the project survey, in which airport respondents indicated the top systems to be managed within a

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Table 3-1. Ranked list of asset systems in CMMS.

System	Rank
Electric	1
Terminal buildings	2
HVAC	3
Facilities	4
Runway lighting	5
Plumbing	6
Elevators	7
Escalators/moving walkways	8
Power plant	9
Jet bridges	10

Source: ACRP Project 09-14 survey.

CMMS. The complete list for airports can extend to 30 airport systems. (For more information and the complete list, see Chapter 4, Tables 4-1 through 4-5).

General CMMS Implementation Considerations for Small Airports

Small airports should decide on the implementation approach that best meets their needs. The number of assets, users, and other elements will differ from airport to airport; however, the actual, "How to" if implementation typically is the same for all.

A key consideration for a small airport implementing CMMS is to "keep it simple." Leveraging a CMMS implementation steering or governance team is a way to share the workload and build the system and associated processes together. For a small airport with limited technology resource capabilities, another consideration could be to require the CMMS vendor to provide a fully documented implementation plan and support team made up of software experts, maintenance experts, training experts, and airport operations subject matter experts.

If a small airport needs to pursue a phased implementation, the following modules and processes can be considered for an initial phase:

- Work orders
- Preventive maintenance
- Materials/inventory management
- Cost capture (labor and materials)
- Reporting

Some typical critical success factors have been compiled from the research. Factors that should be considered as a part of a more-detailed CMMS Project Implementation Plan include:

- Select the implementation team wisely and include cross-departmental involvement from operations, maintenance, and leadership.
- Estimate the staff time needed to collect and load data into the CMMS.
- Consolidate, validate, and standardize all required data prior to CMMS population.
- Develop and implement CMMS maintenance procedures prior to system roll out.
- Perform initial training for staff who will use and manage the CMMS.
- Perform modular-centric training for staff that will use specific modules.
- Where possible, roll out the technology in modular form.
- Where possible, use off-the-shelf tools that require little or no modification.

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CHAPTER 4

Case Studies

Case study subjects for this project represent four airports, an airport authority that manages six airports (three large hub), and a non-airport organization.

They are:

- Port Authority of New York and New Jersey (PANYNJ), United States
- Salt Lake City International Airport (SLC), United States
- Seattle-Tacoma International Airport (SEA), United States
- Toronto Pearson International Airport (YYZ), Canada
- General Mitchell International Airport (MKE), United States
- Gwinnett County Department of Water Resources (GCDWR), United States

Some of the CMMS questions in the case studies were adopted and modified from Terry Wireman's book, *Successfully Utilizing CMMS/EAM Systems* (Wireman 2016). The complete case studies are presented in Appendix A.

Discussion

Most of the case studies were conducted on site over a 2-day period, with the rest conducted as telephone interviews. CMMS and asset management implementation practices within each organization were examined, as well as the benefits to the organization from a CMMS.

The research team found that every organization's CMMS journey has been unique in terms of approach, implementation, software selection, communication, and data usage. Case study participants had common goals, which were to become more efficient and effective in managing their assets through a CMMS, increase their assets' uptime, and provide high quality services to their stakeholders based on the available resources.

Many lessons can be learned from these case studies. Throughout their respective journeys, the participants did not always get things right the first time; however, each organization is learning from its experiences and introducing adjustments to its processes and technologies. The participants also are constantly learning from others.

Strategies

Across all the organizations (airports and non-airport) that took part in the case studies, an important driver is to have visible business strategies, goals, and objectives that cascade to the different business units to make sure that everyone works under a unified strategy. A common theme was that every business unit—including maintenance and asset management—should have a plan or a roadmap that assists them in detailing how they are part of the overall strategy and how they will achieve it.

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Every airport's strategy is to provide an excellent service and experience to its stakeholders by providing safe, secure, and efficient facilities while enhancing its competitiveness and investing in world-class infrastructure.

Asset management and maintenance are key elements in achieving these strategies. At PANYNJ, the maintenance business unit strategy is to work toward meeting current maintenance challenges, maintaining high customer satisfaction, and operating a safe and reliable airport. At SEA, maintenance's main strategy is having a safe operating environment, increasing availability of reliable assets, and minimizing cost. At GCDWR, the facilities team's vision is to achieve "**M**aintenance, **O**perations, and **R**eliability **E**xcellence (MORE)."

Challenges

Every organization faces challenges in implementing and achieving its strategies. Across the case study organizations, some challenges were similar and included:

- Budgetary constraints for capital and operational projects
- Continuing to operate airport facilities at a normal operational level while constructing new facilities and upgrading existing ones
- Capacity constraints on facilities' ability to meet increased demand
- Retirement of highly skilled workers that have a rich organizational knowledge
- Development and harnessing of institutional knowledge
- Striving to continuously improve through personnel commitment, innovation, efficiency, and education
- Meeting stakeholders' growth and modernization demands
- Changes in leadership and reporting structures

Maintenance business units also face many challenges, such as:

- Budgetary constraints
- Scheduling of maintenance activities around airport operations
- Overutilization of assets
- Maintaining asset condition, performance, and availability to meet increasing demands
- Achieving an 80:20 ratio of proactive to reactive maintenance activities
- Lack of available, reliable asset data
- Lack of talented technicians to meet maintenance needs
- Retirement of highly skilled workers
- Alignment of workforce talent with the airport's strategies and goals
- Increasing participation in the planning, design, construction, and commissioning of assets

CMMS

The implementation and utilization of a CMMS is a continuous process, and it always has room for improvement. The collected data significantly impacts the organization's operations and the flow of asset-related information across the organization. A CMMS assists airports in increasing their assets' uptime, availability, and reliability.

The case studies indicate that some of the main drivers to consider in implementing a CMMS are:

- Effective management of budgets
- Effective management of resources
- Reliable work order management
- Reduced maintenance costs through PM

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- Improved wrench time
- Elimination of basic frequency-based maintenance
- Implementation of maintenance best practices, such as performing real-time asset health analytics and availability of failure modes effects analysis (FMEA) and root cause analysis
- Documentation of personnel's activities
- Improved access to critical information
- Improved quality of information
- Increased safety
- Regulatory compliance
- Aging workforce and retirement
- Streamlined business and operational processes
- Efficiencies gained in overall asset management

The case study participants were at different stages with their CMMS. Some were upgrading their existing systems, and others were changing the software altogether by replacing it with a more robust option.

Software-related concerns prompted maintenance departments to upgrade or change the CMMS software. Some concerns with existing systems include:

- Reports are limited (e.g., existing system cannot run reports for individual labor; reports are bulky and cannot be formatted; existing system is not user friendly for work orders).
- The system is not web-based and does not allow real-time updates.
- The system is unable to integrate with other systems.
- The system is not GIS-centric; individual assets are not mapped, and the system is not enterprise-wide.
- The system does not allow the airport to house work order information, warehouse inventory, data for certain financial details, or other information.

Tables 4-1 through 4-5, provided at the end of this chapter, compare the case study participants' CMMS selection, implementation, assets, and integration with other systems. As revealed by the tables, the participants' CMMS implementation methods, approaches, and types of assets considered have been similar. Additionally, participants have used the CMMS to report to decision makers about the assets' condition.

The participants chose asset classes to be part of a CMMS on the basis of the assets' criticality to the different operational activities, reliability, inventory owned and operated by the organization, frequency of maintenance, and on the organizations' ability to develop the real maintenance cost.

Once the CMMS was put in place and properly implemented, the maintenance business units used its data and reports. For example, at PANYNJ, aviation utilizes a CMMS 100% of the time to allow the department to charge activities back to the airlines. SLC utilizes a CMMS 95% of the time in planning and scheduling maintenance activities.

Types of reports produced by the CMMS produces for maintenance departments include:

- Work Order Costs Report
- Completed Work Order Performance Report
- Work Order Backlog Summary
- Work Order Waiting Report
- Asset Repair History
- Identify Asset Failure Modes Report
- Asset Maintenance Costs Repair Report
- PM Overdue Report

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Given that a CMMS is basically a maintenance database, it can be linked with other systems, such as resource management, scheduling, fueling, elevators and escalators, BHS, fleet services, and so forth. With a CMMS, these systems can be accessed through an integrated platform. The organizations in the case studies selected which existing software or systems to integrate with or develop as part of a new or upgraded CMMS. Overall, the the asset systems prioritized for CMMS by the case study organizations were:

- Terminal buildings
- Runway lighting
- Facilities
- HVAC
- Electric
- Plumbing

Integrating a CMMS with other systems results in positive outcomes, such as eliminating redundant work, ensuring accuracy of information, capturing actual costs, increasing data visibility, increasing productivity by reducing to one the number of systems personnel must access, transparent decision making, improving maintenance activities, better budget allocations and savings, and increasing system efficiency and data yield.

Additionally, a CMMS assists organizations in making robust and informed decisions based on accurately collected data and having it correlated with other parameters. This information advises other business units throughout the organization about parameters that might be of interest and value to their activities. This data can be shared through reports, KPIs, scorecards, etc.

Lessons Learned

The research team developed a list of the common lessons learned by the case study organizations in terms of implementing a CMMS. Highlights of the list include:

- Involve stakeholders (business units and ranks) as part of the CMMS selection and implementation discussions and obtain their support to ensure its success.
- Involve leadership within and across the different levels of implementation.
- Establish organizational partnerships early.
- Include champions and game changers as part of the CMMS project team.
- Invite leadership to sponsor the different projects (executive sponsorship).
- Demonstrate value by linking the CMMS implementation to the strategic plan.
- Establish and utilize KPIs to inform the process and the improvement of the assets' performance.
- Identify the various work order priorities.
- Conduct a criticality assessment of assets.
- Ensure that once new assets have been commissioned by a contractor, they are ready to be managed through the CMMS.
- Educate and empower staff (communicate the "whys").
- Solicit feedback (even negative feedback).
- Start small and continuously improve on the system.
- Ensure that contractors providing information adhere and comply with the organization's CMMS standards.
- Evaluate legacy system data accuracy and relevance before transferring the data to the new CMMS.
- Practice flexibility in implementation and learn from other organizations' challenges (peer exchange).
- Update based on business need, not because it is "Tuesday."
- Conduct scheduled audits on the CMMS and on the data.
- The most dangerous phrase in maintenance is, "We've always done it this way."

Additional lessons learned are provided with the detailed case study information in Appendix A.

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Table 4-1. Case studies: CMMS selection comparison (shaded cells indicate planned/future implementation).

Question Category		CMMS Selection	Airports					Non-Airport
			PANYNJ	SEA	SLC	MKE	YYZ	GCDWR
1	Which of the following statements were considered in selecting CMMS?	Tracks asset information such as cost center, department, location, etc.	●	●	●	●	●	●
		Allows for cost and repair info to be accessed at multiple asset levels	●	●	●	●	●	●
		Work orders can be saved to asset history as one line of info or full detail	●	●	●	●	●	●
		All information in asset history can be accessed by sorting or building ad hoc reports	●	●	●	●	●	●
		System provides a bill of materials for each asset	●	●		●	●	●
		System provides user-defined screens for storing info relative to asset items/types, which also can be selected and sorted by reports	●	●		●	●	●
2	Which of the following preventive maintenance statements were considered in selecting CMMS?	Schedule PM actions (PMs) by calendar, meter readings, operational parameters (real time)	●	●	●	●	●	●
		System allows for unlimited numbers of PMs per asset	●	●	●	●	●	●
		System allows for multiple crafts scheduled on a PM	●	●		●	●	●
		System has the ability to schedule PMs for any specific date and/or day of the week	●	●	●	●	●	●
		System has the ability to forecast labor resources, material requirements, or special tools for PMs due for any specified period	●	●			●	●
		System will schedule PMs for assets or facilities		●	●	●	●	●
		System will combine all PMs that are due for an asset automatically, manually, or not at all	●	●		●	●	●
		System generates the PM work orders daily, weekly, monthly, or at user-defined intervals	●	●	●	●	●	●
		System reports overdue PMs by number, incomplete PMs, or results of PM inspections	●	●	●	●	●	
3	Which of the following work orders statements were considered in selecting CMMS?	System tracks information at the work order task/step level for planned versus actuals, for labor, materials, tools, contractors and safety	●	●	●	●	●	●
		System produces a report of all work orders which can be sorted by their current status in the work order workflow	●	●	●	●	●	●
		System is capable of tracking work order backlog by craft, crew, department, planner, supervisor, or other defined categories	●	●	●	●	●	●
		System updates work order status either manually or automatically via workflow configurations	●	●	●	●	●	●
		When planning a work order, the user can access stores, personnel, tools, contractors, etc., w/o leaving the work order module	●	●	●	●	●	●
		System provides the ability to create notifications, escalations, and [can] send these items to personnel internally or externally	●	●		●	●	

(continued on next page)

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Table 4-1. (Continued).

Question Category		CMMS Selection	Airports					Non-Airport
			PANYNJ	SEA	SLC	MKE	YYZ	GCDWR
4	Which of the following MRO inventory and procurement statements were considered in selecting CMMS?	System generates a Spares Reorder Report when the quantity on hand drops below a minimum level	●	●		●	●	●
		System has the ability to manage multiple warehouses	●	●		●	●	●
		Inventory system is integrated (or can be) with the vendor's own purchasing system	●	●			●	●
		If a reorder point is reached, a purchase requisition is generated either manually or automatically	●	●			●	●
		When the part is received, the system notifies the planner, automatically or manually, which work orders can now be filled	●	●			●	
		System produces performance reports for purchasing section that include overdue POs, inactive parts, or inventory valuation		●		●	●	●
		System automatically tracks costs to the work order on an itemized basis	●	●		●	●	●
5	Which of the following management reporting statements were considered in selecting CMMS?	System produces reports daily, weekly, monthly, yearly, or at user-defined intervals	●	●	●	●	●	
		Reports produced by the system are summary reports, lists of information, or exception reports	●	●	●	●	●	
		System has a report writer in addition to standard system reports	●	●		●	●	
		Reports were designed to be used by system managers, managers, or maintenance personnel	●	●	●	●	●	
		System has a maintenance budget reporting module	●	●		●	●	
		System reports and forecasts asset downtime	●	●			●	
		System tracks asset downtime costs	●	●			●	

MRO = Maintenance, repair, and operating supplies.

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Question Category		CMMS Selection	Airports					Non-Airport
			PANYNJ	SEA	SLC	MKE	YYZ	GCDWR
6	Which of the following implementation statements were considered in selecting CMMS?	Vendor performs full turn-key implementation, software implementation, hardware installation, data gathering, data loading, system training, and user training	●	●	●	●	●	●
		Vendor has fully documented installation plans	●	●		●		
		Vendor has software experts, maintenance experts, training experts, and airport operations subject matter experts on staff	●	●		●	●	●
		System needs no customization, some customization, or extensive customization	●	●	●	●	●	●
7	Which of the following maintenance software analysis statements were considered in selecting CMMS?	System to be operated by maintenance personnel	●	●	●	●	●	●
		System has the necessary modules to meet the organization's needs (PMs, work orders, inventory, integration, etc.)	●	●	●	●	●	●
		System has single, semi-multi, or multi-user capabilities	●	●	●	●	●	●
		System modules are closely integrated	●	●	●	●	●	●
		System can archive and retrieve files for reports spanning long time periods	●	●	●	●	●	●
		Software vendors update the software on a regular basis	●	●	●	●	●	●
		System has ability to display analytical information a graphic format	●	●		●		
		System can be integrated with payroll, accounts payable, general ledger, ERP system, time keeping, etc.	●	●		●	●	

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Table 4-2. Case studies: CMMS implementation comparison (shaded cells indicate planned/future implementation).

Question Category	CMMS Implementation	Airports					Non-Airport
		PANYNJ	SEA	SLC	MKE	YYZ	GCDWR
1. Which of the following statements were considered during CMMS implementation flow processes?	Establish steering committee for CMMS system	●	●	●	●	●	●
	Establish the site implementation team	●	●	●	●	●	●
	Establish site project teams	●	●	●	●	●	●
	Promote kick-off meeting for site and project teams	●	●	●	●	●	●
	Install hardware and the CMMS software	●	●	●	●	●	●
	Initialize CMMS software (Create multiple system environments including, development, test, train and production)	●	●	●	●	●	●
	Provide initial training for the site and project teams	●	●	●	●	●	●
	Define organizational procedures for CMMS use	●	●		●	●	●
	Develop written CMMS procedures	●	●		●	●	●
	Restructure the organization/business processes (if needed)	●	●		●	●	
	Begin developing various data collection formats		●		●	●	●
	Use the formats to start data collection process (asset, stores, PMs, personnel, purchasing, accounting)	●	●		●	●	●
	Begin data entry process	●	●	●	●	●	●
	Begin user training on CMMS system	●	●	●	●	●	●
	Establish stores stock levels	●	●			●	●
	Begin using work order system		●	●	●	●	●
	Monitor all usage of CMMS system	●	●		●	●	●
	Audit CMMS system for 30 days	●	●		●	●	
	Continue 30-day audits for the first 6 months	●	●		●		●
2. Which of the following issues have you faced during CMMS implementation?	Assessing current and future needs			●	●	●	●
	Properly document the system user requirements or “get user input”	●		●	●	●	●
	Lack of management support						●
	Conduct a good search in the CMMS market place			●		●	●
	Opted to develop an in-house system						
	Assess vendor’s qualifications	●		●		●	
	Software testing	●		●		●	
	Put an implementation plan together	●		●	●	●	
	Obtaining sufficient training and/or documentation	●		●		●	●
	Correctly estimating time needed to collect and load data	●		●	●		●

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Table 4-3. Case studies: Assets in CMMS comparison (shaded cells indicate planned/future implementation).

Question Category	Assets in CMMS	Airports					Non-Airport
		PANYNJ	SEA	SLC	MKE	YYZ	GCDWR
Which assets are currently part of CMMS? (Present)	Terminal buildings	●	●	●	●	●	
	Runway lighting	●	●	●	●	●	
	Facilities	●	●	●	●	●	
	Hangars	●		●	●	●	
	Parking garages	●	●	●	●	●	
	Pavement (airfield)	●	●	●	●	●	
	HVAC	●	●	●	●	●	
	Electric	●	●	●	●	●	
	Power plant		●	●	●	●	
	Plumbing	●	●	●	●	●	
	Elevators	●	●	●	●	●	
	Escalators/moving walkways	●	●	●	●	●	
	Baggage handling system	●	●	●	●	●	
	Jet bridges	●	●	●	●	●	
	Rolling stock/fleet	●	●	●		●	
	Utility/treatment facilities on site	●	●	●	●	●	
	Fueling system			●	●	●	
	Security system		●	●	●	●	
	IT		●		●		
	Signage	●	●	●	●	●	
	Buried piping (water, sewer, gas, electric, etc.)	●			●		
	Stormwater infrastructure (culverts, swales, inlets, basins, etc.)		●	●	●	●	
	Deicing system	●		●	●	●	
	Fire protection	●	●	●	●	●	
	Airfield lights	●			●		
	Environmental management	●			●		
	Terminal connecting bridge		●		●		
	Tide gates	●			●		
	Glycol recovery			●	●		
	Chemical expensing			●	●		

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Table 4-4. Case studies: CMMS integration comparison (shaded cells indicate planned/future implementation).

Question Category	CMMS Integration	Airports					Non-Airport
		PANYNJ	SEA	SLC	MKE	YYZ	GCDWR
Is CMMS integrated with any of the following systems?	Resource management	●	●	●			
	Security				●		
	Safety management	●		●	●		
	Automatic vehicle identification						
	Wildlife hazard management plan	●		●	●		
	Payroll		●				
	Pavement management	●		●	●		
	Scheduling	●	●		●		
	Electronic log books				●		
	Incident management	●			●	●	
	Fleet management	●	●	●			
	Human resources	●					
	Building management					●	
	Time and labor software	●		●		●	
	Property management	●		●	●		
	Fueling	●	●	●		●	
	Airfield lighting	●		●	●		
	Storeroom	●		●	●		
	Geographic information	●		●	●		
	Inspection application (FAA Part 139)	●		●	●		
	Financial software	●		●		●	
	Elevators and escalators		●		●		
	Baggage handling system	●	●		●		
	Fire protection	●			●		
	Asset health analytics					●	

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Table 4-5. Case studies: CMMS decision-making comparison (shaded cells indicate planned/future implementation).

Question Category	Decision Making	Airports					Non-Airport
		PANYNJ	SEA	SLC	MKE	YYZ	GCDWR
Which of the following reports do you produce? What type of data is in the report?	Work order priority analysis	●			●		●
	Planner performance	●					●
	Scheduling activities	●	●				●
	Inventory	●	●		●	●	●
	Procurement	●	●				
	Supervisor work order performance	●			●		
	Supervisor/skill work order performance	●			●		
	Work order costs report	●	●	●	●		
	Completed work order performance	●	●	●	●	●	●
	Work order backlog summary	●	●			●	●
	Asset repair history	●		●	●		●
	Asset maintenance costs repair	●	●	●	●		●
	Asset maintenance cost exception	●					
	Safety work order backlog	●		●	●		●
	Stock item usage report	●	●			●	●
	Work order waiting report	●	●	●	●	●	●
	Preventive maintenance	●	●		●	●	●
	Preventive maintenance overdue report	●	●	●	●	●	

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CHAPTER 5

CMMS Integration into Business Decision Making

This chapter provides guidance to airport CMMS developers, integrators, and users that will improve their ability to implement and manage the CMMS as it relates to and supports a performance management program, and thereby improves data-based decision making. The chapter includes information on basic performance management principles, but it is not intended as instruction for developing a performance management program. Readers can refer to *ACRP Report 19: Developing an Airport Performance-Measurement System* for information related to developing a performance management program (Infrastructure Management Group, Inc., et al. 2010). This material also is intended to supplement and support *ACRP Report 19A*: *Resource Guide to Airport Performance Indicators* (Hazel et al. 2011). *ACRP Report 19A* presents the results of an extensive research project conducted to identify a set of KPIs found to be most effective and used across the airport industry.

A CMMS usually includes some level of KPI tracking and reporting, which can be valuable to understanding the maintenance organization's performance. This chapter presents an overview of *ACRP Report 19A* sufficient to provide basic understanding of how and why KPIs are developed, the types of KPIs available, and the information sources that support KPI development and integration into business decision making. The chapter also provides guidance on how to use the *ACRP Report 19A* information with a CMMS to provide accurate data for performance tracking to inform capital and operational business decisions.

It is important to highlight that, although much information is available from a CMMS, it typically is not the only data source supporting the performance management system.

The chapter also outlines the steps necessary for development of a performance management program, including developing a performance management strategy, developing a KPI model, setting up the CMMS to support performance management, leveraging the CMMS data to inform decision making, and communicating KPI analysis and results to decision makers.

Overview of *ACRP Report 19A: Resource Guide to Airport Performance Indicators*

Under ACRP Report 19A, Hazel et al. (2011) reviewed literature related to airport performance measures; reached out to trade associations, government agencies, and other experts in the field; and conducted two workshops to solicit industry views on airport performance indicators (APIs). Through this process, the researchers developed an extensive list of APIs that can be used in an airport performance measurement system. To assist airport practitioners in selecting the APIs that are most appropriate for their particular airport functional interest, each API listing presents and defines each API, suggests methods for

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collecting relevant data, and, generally, supports the implementation of an airport's performance measurement system.

The report provides a list of 840 APIs grouped into categories labeled "Core," "Key," and "Other." Figure 5-1 shows the number of APIs in each category. Their details are as follows:

- **Core APIs (29).** These strategic APIs are important for airport overall operation or otherwise important at the airport executive level (chief executive officer and aviation director) and/or the airport's governing board.
- **Key APIs (132).** These strategic and tactical APIs are important for the operations of key airport departments or functions (e.g., finance and maintenance).
- **Other APIs (679).** Not considered as useful for overall airport operation, to the executive level, or to key airport departments/functions, these tactical APIs can be useful as secondary departmental unit APIs at or below the manager level.

Few, if any, airports have the need, the resources, or the data to track and report on 840 individual APIs. Even among airports with similar characteristics, managers will have different views on which APIs are most important, and over time these views are likely to change as new issues and challenges arise.

Based on the results reported in *ACRP Report 19A*, airports should use between 6 and 20 Core APIs. Accordingly, an organization needs to fully understand its mission, goals, and business objectives as it evaluates its performance measuring options and builds a realistic, sustainable performance model.

A subset of the APIs identified in *ACRP Report 19A* will require data that are typically stored within a CMMS. Identifying which APIs can be supported by the CMMS is covered in this chapter in the section titled "Setting up CMMS to Support Performance Management."

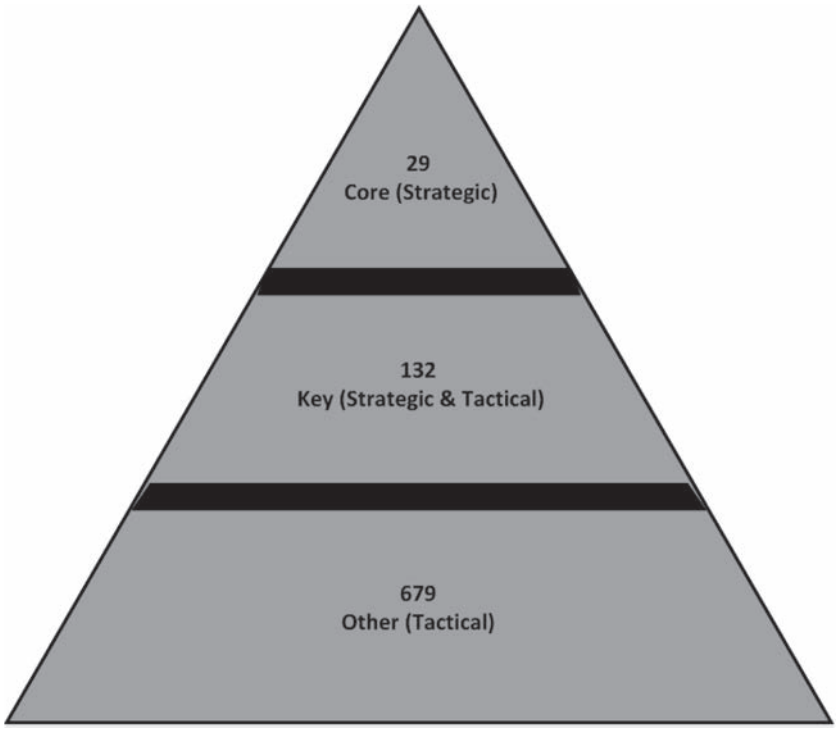


Figure 5-1. Breakdown of airport performance indicator categories.

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Leveraging CMMS Data to Inform Decision Making

The potential volume of information and data that can be contained within a CMMS can be both powerful and intimidating. Airports implementing or upgrading a CMMS may be tempted to include most or all available data, reasoning that if the data is not included in the CMMS, it cannot be used for analysis, and therefore cannot be used to inform decision making for risk-based, data-driven decisions. As was discussed in Chapter 3 of this guidebook, however, the CMMS should be designed with the end in mind, including making decisions about which assets, data, and information to include.

To leverage the CMMS information for improved decision making, the data requirements must be mapped from the decision-making process back to the CMMS. Given that most business decisions are (or should be) actively informed from a performance management system, it makes sense that if the CMMS has been configured to support the data requirements of the performance management system, it will support the decision-making process. In addition to supporting the performance management system, the CMMS data provides value to other analysis processes associated with conducting FMEA, RCM, etc.

A CMMS user or integrator addressing the data requirements of the CMMS should begin the process with establishment of a performance management model to include the performance measure calculation and data sources shown in Figures 5-1 and 5-2. Figure 5-1 presents a list of initial maintenance and reliability program performance measures selected for tracking and reporting by a maintenance department, and Figure 5-2 presents a list of future potential performance measures the maintenance department might be interested in tracking. It is important to consider likely future performance measures so that the data being collected today will support the needs of the future performance management model. It also should be remembered that the data that feeds the performance management system will usually come from multiple sources, including the CMMS.

Some short-term performance measures are:

- Assets missing PM plans and/or job plans
- Total assets with spare parts indicated
- Total assets missing spare parts
- Mean time from failure report to correction
- Planned work orders to unplanned (PM to CM ratio)
- Work Order backlog (critical versus non-critical assets; assumes organization has classified assets by consequence of failure)

Some mid-term performance measures are:

- Asset re-work work orders
- Asset failures (bad actors)
- Mean time between failures
- Warehouse turns by part/asset classification
- Warehouse material valuation
- Work order placed on hold due to waiting materials

Some long-term performance measures are:

- Craft utilization (time associated with work order versus non-wrench time)
- Craft training time
- Labor hours planned versus actuals
- Part utilization, planned versus actuals
- Maintenance costs, planned versus actuals
- Internal maintenance costs versus vendor maintenance costs

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The information presented in Figures 5-4 and 5-5 includes the calculation and data sources for each performance measure. As indicated by the shaded panels, a CMMS is a major source of information for this maintenance and reliability performance management model.

Prioritizing the population and management of CMMS data is critical for improving business decision making. By understanding the needs of the performance management system, the CMMS administrator can identify which CMMS data will be used for decision making, which information needs to be of high quality, and what information might be required in the future.

Developing a Performance Management Strategy

Use of performance measures is growing in one form or another in most airports in the United States and internationally. Often called *benchmarking*, or simply as financial and operational data, performance measures are being put in place and used in the daily operations of airports of all sizes. From GA to large hub airports, the need for and relevance of performance monitoring has been manifested. Most commonly, financial and operational data are tracked due to FAA requirements (Infrastructure Management Group et al. 2010).

At a minimum, effective performance measurement programs do the following:

- Measure only areas that fall inside the airport's mission area
- Measure activities, products, services, and outcomes that move the airport toward its strategic goals
- Measure areas that have been identified as environmental, business, structural, or other barriers to success
- Involve an inclusive set of short- and long-term, leading and lagging, and operationally diverse measures
- Inform management decisions by linking strategic planning to budgeting, resource planning, and other areas of managerial importance

The purpose of a performance management system is to provide an organization with the tools it needs to measure performance in various categories at a strategic level and a tactical level. In general, the strategic level performance measures gauge performance against the stated organizational goals, and the tactical level performance measures measure performance of initiative objectives.

An organization's selection of performance measures depends on the organization's goals and objectives, as well as the organizational structure, the programs that are in place, and the availability of accurate data/information.

In accordance with *ACRP Report 19A* the main characteristics of a well-defined performance measurement system are the following:

- Organizational performance measures are gathered and displayed in a system or systems
- Performance measures are designed to ensure that past practices are analyzed, and that lessons are learned from prior decisions
- Performance measures are used to gauge whether current actions are moving the organization toward a desired future state
- The organization may use performance measures from a largely reactive and tactical perspective, but the long-term goal is for more proactive, strategic use
- The organization consciously uses data to challenge management decisions, pre-conceived notions, and past practices
- The organization combines quantitative information and qualitative information to provide accurate and rich interpretations of organizational activities, outputs, and outcomes

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A performance management strategy begins with establishing a clear airport mission, establishing goals that will work to accomplish the mission, defining objectives that support reaching the goals, and developing an action plan that will ensure the strategy is transformed into performance improvement.

Mission Statement

A performance management system should be built on a clear airport mission. Airports generally develop mission statements based on their enabling laws and rules. According to the National State Auditors Association (2004), the airport should ensure that those mission statements:

- Identify who the entity is and what it does
- Identify why the things the entity does are important
- Make sense to the average stakeholder
- Enable the stakeholder to understand why public dollars are being spent on these efforts

Goals

Goals describe the results desired to be realized as the airport successfully implements various improvement initiatives. Goals can be thought of as quantitative descriptors for each leg of an airport's journey to its destination, whereas the mission describes more broadly the "why and how" of the journey. In addition to supporting the mission, each goal should:

- Represent a desired, measurable result
- Be realistic and achievable
- Make sense to others outside the airport

Objectives

Objectives represent management's commitment to achieving specified results and spell out the quantity and quality of performance to be achieved in a specified period of time. Objectives are individual targets for the performance improvement initiatives that will move the airport closer to a goal and to its ultimate destination.

Action Plan

The action plan is a documented series of tactical activities that, when executed, implement the performance management strategy and, when successfully implemented, ensure that the organization eventually meets its mission. The performance measures and standards development should be part of the action plan.

A CMMS is a key source of quantitative data that can support the development and tracking of performance measures. Therefore, the action plan will work to identify various CMMS data required to meet the intent of the performance management strategy. If the team implementing or upgrading the CMMS is not aware of the needs of the performance management system, the quantity and accuracy of the resulting CMMS data may be inadequate to support the desired level of performance measurement. As with any data intensive analysis, a performance management system requires multiple years of accurate data in order to establish baselines and trends against those baseline measurements. Because performance improvement actions are based on the analysis of the data, bad data only produces bad decisions.

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To ensure that the right data is collected and the quality of the data is sound, four basic concepts need to be applied:

1. When establishing the data requirements for the CMMS, consider the needs of the performance management system. If a performance management system is not in place, use guidance provided in this report to ensure that the appropriate data is being documented within the CMMS to support future efforts.
2. Consider the CMMS data needs of the entire airport, including financial, safety, environmental, production, maintenance, human resources, payroll, asset performance, and asset management performance.
3. Create asset documentation and formal work management processes that will encourage the consistent and accurate documentation of asset and work execution information.
4. Train airport staff and management on the importance of the CMMS data for effective decision making throughout the organization.

Development of a Performance Measurement Model

The development of a performance management model includes creating the mission statement, defining the goals that support that mission, and identifying the lower level objectives that will drive the organization and departments to meet its goals.

As illustrated in Figure 5-2, tactical measures can provide input to multiple strategic measures. A key opportunity when building a performance management model is to limit the number of tactical measures to the fewest that will provide an accurate measurement of performance. Doing this limits the amount of work for collecting and entering the data, analyzing the data, and reporting the outcomes.

Composition of a Performance Measurement Model

A performance measurement model is constructed as a cascading structure in which overall performance is presented at the top of the model with the performance of supporting activities nested below. Several model structures can be considered, with many consistent in using a cascading architecture. A common presentation of performance measurement is the pyramid, as shown in Figure 5-3.

The *Guide to Airport Performance Measures* provides the following guidance for establishing performance areas in an airport (ACI 2012):

- **Core.** These are the core measures used to characterize and categorize airports, such as the number of passengers and operations. Although airports may have little control over these core indicators, especially in the short term, they are important indicators of overall airport activity, and important drivers and components of other indicators.
- **Safety and security.** These are the most important airport responsibilities, and therefore they are categorized separately.
- **Service quality.** This increasingly important area reflects the evolution of airport management from having a primary focus on facilities and operations to having a strong customer service focus in an increasingly competitive environment. Service quality includes the way in which asset performance delivers comfort for airport customers through provision of water, cooling, heating, sanitation, and other infrastructure services.
- **Productivity/efficiency.** These measures are closely related/overlapping measures of an airport's performance. They are sometimes separated into productivity measures, which track output on a non-cost basis (e.g., passengers per airport employee or departures per gate)

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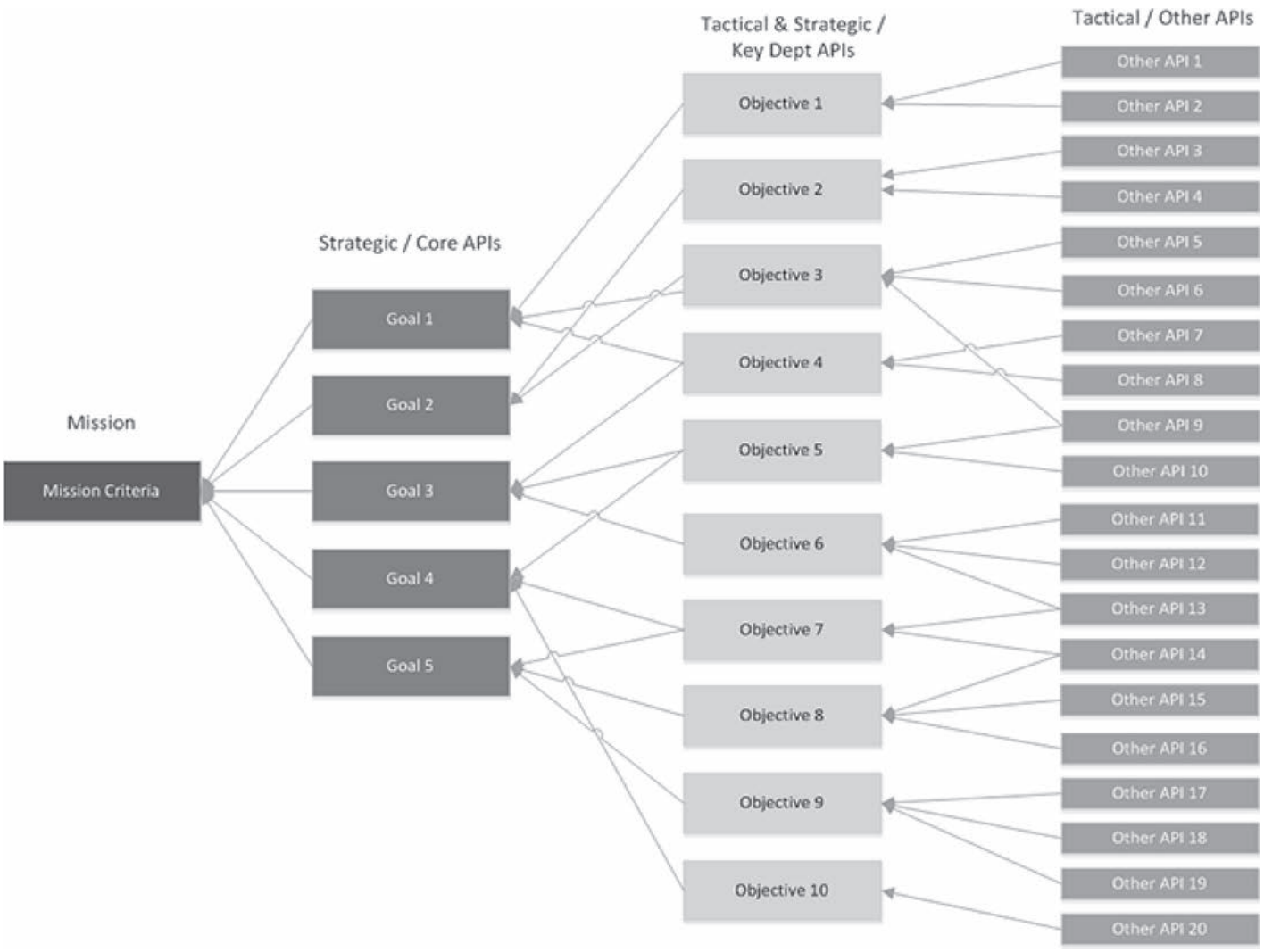


Figure 5-2. Airport performance measurement model.



Figure 5-3. Performance measurement pyramid.

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and efficiency measures, which track output on a cost basis (e.g., total or operating cost per passenger). Productivity/efficiency includes asset management performance, as defined in Clause 9.1 of ISO 55001.
- **Financial/commercial.** This includes measures relating to airport charges, airport financial strength and sustainability, and the performance of individual commercial functions.
- **Environmental.** This evolving area has become a strong focus for airport managements striving to minimize environmental impacts.

Characteristics of a Good Performance Measure

As mentioned previously, an effective performance management model will contain the minimal number of measures needed to provide the necessary level of information to manage performance. In turn, each measure should meet the following SMART criteria to be considered an effective measure (McNeeney 2005):

- **Specific.** The measure is clear and focused to avoid misinterpretation.
- **Measurable.** It can be quantified and compared to other data.
- **Attainable.** It is achievable, reasonable, and credible under conditions expected.
- **Realistic.** The measure fits into the organization's constraints and is cost effective.
- **Timely.** It can be achieved within the given time-frame.

Performance Measurement Categories

Categories of performance are as follows (Governmental Accounting Standards Board 1990):

- **Input.** Input shows the amount of resources, either financial or otherwise, used for a specific service or program. Input measures include labor, materials, equipment, and supplies. Demand for governmental services may also be considered an input indicator.
- **Output.** Output shows units produced or services provided by a service or program. Output measures include the amount of products or services provided, the number of customers served, and the level of activity to provide services.
- **Outcomes.** Outcomes show the results of the services provided. Outcome measures assess program impact and effectiveness and show whether expected results are achieved.
- **Efficiency.** Efficiency reflects the cost per unit of output or outcome.
- **Quality.** Quality shows the effectiveness in meeting the expectations of customers, stakeholders, and other groups. Quality indicators show the quality of the services delivered.

Setting Up CMMS to Support Performance Management

A CMMS provides functionality to track physical assets, spare parts, and the transactional activities associated with both O&M of those assets. Within the execution of these functions, the system acts as both a work management tool and a data analysis tool. Most CMMS users focus on using the work management functionality, with little appreciation for whether data is being documented within the system, or being documented at all. As a result, the full value of a CMMS is not realized.

To leverage the CMMS data for improving performance, the necessary data should be accessible and accurate. Given that the options for data fields in a CMMS are endless, the CMMS user must clearly understand which fields are important and which are not. If a current performance management structure is in place, the performance measure requirements should dictate which CMMS fields should be populated and maintained accurately to support the program. If a performance management system is not in place, the CMMS user can use some of the metrics shown in Figure 5-4 to get started.

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Performance Metric	Performance Target	Calculation	Data1/Source	Data2/Source	Data3/Source	Data4/Source
Total Annual Maintenance Costs / RAV	3%	$Labor + $Material + $Contractor / RAV	Maintenance Labor Cost CMMS	Material Cost CMMS	Contractor Cost CMMS	Replacement Asset Value (RAV)
% Inventory Value / RAV	1%	Inventory Holding Value / Replacement Asset Value x 100	Value of stock Held in Inventory CMMS	Replacement Asset Value Plant		
Critical Equipment Availability	97%	Total Targeted Production Time - (unscheduled unavailability + scheduled unavailability hours)/ Total Targeted Production Time x 100	Targeted Production Time (99.5%?)	Unscheduled unavailability	Scheduled unavailability	
Normalized Mean Time Between Failure (nMTBF)	NA - Increasing Trend	Total Operating Time (for population) / Number of failure events (for population)	Total Operating Time for population	Number of Failures for population CMMS		
% Proactive labor hours vs total labor hours (Proactive + Reactive)	80%	((PM hrs + PdM hrs + PMR hrs + PdMR hrs) / Total Hours) x 100	(PM hrs + PdM hrs + PMR hrs + PdMR hrs) CMMS	Total Proactive and Reactive Hours CMMS (proactive + reactive)		
% Schedule Success	90%	Count of scheduled Labor Hrs Completed / Total scheduled Hrs x 100	Total Scheduled Hrs Complete CMMS	Total Scheduled Hrs CMMS		
% PdM Inspection labor hours vs total labor hours (PdM + PM + PdMR + PMR+ Reactive)	15%	PdM hrs / Total Hours	PdM hrs CMMS	Total Proactive and Reactive Hours CMMS (proactive + reactive)		
Total Maintenance Costs / MGD	NA - Trend	$Labor + $Material + $Contractor / MGD	Maintenance Labor Cost CMMS	Material Cost CMMS	Contractor Cost CMMS	Production in MGD
Maintenance Contractor Costs / MGD (6 Month delay in reporting)	NA - Trend	$Maintenance Contractor Costs / MGD	Contractor Cost CMMS			
% Total Compliance Annual - Maintenance	100%	Total Compliance Commitments Completed/ Total Compliance Commitments for calendar year	Total Compliance Commitments Completed CMMS	Total Compliance Commitments CMMS		
% of Emergency & Urgent Labor Hours vs Total Labor Hrs (All Priority)	Downward trend (If % of total -Less than 7.5%)	Total Emergency Labor Hours (Order Priority "X") / Total Labor Hours x 100	Total Actual Hours on WOs with Priority X / CMMS	Total Actual Hours on WOs / CMMS		
Avg Mean Time To Repair MTTR	Decreasing Trend	Sum of Time to Repair per Work Order / Total Work Orders	Time to Repair per WO CMMS	Total Work Orders CMMS		
% PM Hours by Operations	NA-Trend	PM Hours OPS / PM Hours Total x 100	PM Hours OPS CMMS	PM Hours Total CMMS		
Avg time from Purchase Request to Material Receipt	NA-Trend	Material Receipt Date - Purchase Request Date	Material Receipt Date	Purchase Request Date		
Number of Stock Outs for critical equipment maintenance	<2%	Number of times a critical spare is needed but not in stock / number of times critical spare is needed x 100	Number of times a critical spare is needed but not in stock CMMS	Number of times critical spare is needed CMMS		
% Hrs for Skills Training	2 - 3% (40 hrs per employee per year)	Count of Hours of Maintenance Skills Training / Total Available Working Hours x 100	Hours of Maintenance Skills Training CMMS	Total Available Labor Hours		
% of Compliance Related Calibrations Completed YTD	100%	YTD Compliance PM Calibrations Completed / Total Count of Compliance PM Calibrations For Year	YTD Compliance PM Calibrations Completed CMMS	Total Count of Compliance PM Calibrations Scheduled For Year CMMS		
% of Compliance Related Inspections Completed YTD	100%	YTD Compliance PM Inspections Completed / Total Count of Compliance PM Inspections For Year	YTD Compliance PM Inspections Completed CMMS	Total Count of Compliance PM Inspections Scheduled For Year CMMS		
% YTD Maintenance Spend / Maintenance Budget	95% - 105%	YTD Maintenance Spend / Annual Maintenance Budget x 100	YTD Maintenance Spend CMMS	Maint.Budget		
Average Response time to Operations request by WO Priority	Need to establish - Trend	AVG (WO Start Date & Time - Customer Request Date & Time)	WO Start Date & Time CMMS	Customer Request Date & Time CMMS (If provided with work request or pushed to CMMS)		
% Planned Work orders vs Total Wos (Planned + Unplanned)	95%	Planned & Sched WOs / Total WOs x 100	Planned & Sched WOs CMMS	Total WOs CMMS		
Work Backlog (crew weeks)	4 - 6 weeks	Total Backlog Labor Hours / Total Available Crew Hours per Week	Total Backlog Hours CMMS	Total Available Crew Hours per Week CALCULATED		
On Time PM Comp. Rate	95%	Count of PM WOs completed on or before due date / Total PM WOs Scheduled x 100	Total PM WOs with status "complete" and complete date equal to or less than due date CMMS	Total PM WOs scheduled CMMS		

Figure 5-4. Initial maintenance and reliability metrics model spreadsheet.

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Performance Metric	Performance Target	Calculation	Data1/Source	Data2/Source	Data3/Source	Data4/Source
% of Work Orders Returned to Planning	0%	Count of WOs returned to planning / total WOs Planned x 100	Count of WOs returned to planning CMMS	Total WOs Planned CMMS		
Maintenance Labor Costs	NA - Trend	$Maintenance Labor Costs / MGD	Maintenance Labor Cost CMMS			
Material Costs	NA - Trend	$Material Costs / MGD	Material Cost CMMS			
% of Work Orders by Priority	NA - Trend	Count Work Orders by Priority / Total Work Orders x 100	Work Order Count by Priority CMMS	Total Work Orders CMMS		
% Est Hrs vs Actual Hrs For Work Order	Less than 10% deviation	Total Estimated Hours / Total Actual Hours x 100	Total Estimated Hours on WOs CMMS	Total Actual Hours CMMS		
% Scheduled Hours vs Available Hrs	100%	Scheduled Labor Hours / Total Available Labor Hours x 100	Scheduled Labor Hours on WOs CMMS	Total Available Labor Hours CMMS		
% of Break-In Labor Hours vs Total Scheduled Labor Hrs	Downward trend (If % of total - Less than 5%)	Total Schedule Break-In Labor Hours / Total Scheduled Labor Hours x 100	Total Schedule Break-In Hours SCHEDULER	Total scheduled Hours SCHEDULER		
After Hour Call In Hours (premium pay)	NA - Decreasing Trend	Total Number of After Hour Call In Hours	Total Number of After Hour Call In Hours CMMS			
% of Labor Hours by work category (proactive vs reactive vs support)	80% proactive	Total Emergency Labor Hours (Order Type "X") / Total Labor Hours x 100	Total Actual Hours on WOs with Type X / CMMS	Total Actual Hours on WOs / CMMS		
Failure Code Trends	Decreasing Trend	Sum of failures by failure code	Count of Failures by Failure Code / CMMS			
% of Closed Work Orders with labor hours recorded	100%	Count of WOs with Labor Hours / total WOs x 100	Count of Wos with Labor Hours CMMS	Count of Total WOs CMMS		
% of Closed Non-PM Work Orders with failure codes recorded	100%	Count of Non-PM WOs with Failure Codes / total WOs x 100	Count of Non-PM WOs with Failure Codes CMMS	Count of Total WOs CMMS		
Avg time WO in WMATL status	NA-Trend	AVG (WO WMATL Date / Time - WO WMATL Date / Time)	WO WMATL Date / Time CMMS	WO WMATL Date / Time CMMS		
% of Total WO Backlog Awaiting Materials (WMATL)	NA-Trend	WOs in Backlog in WMATL status / Total WOs in Backlog x 100	Total WOs in Backlog in WMATL status CMMS	Total WOs in Backlog CMMS		
% Inventory Accuracy	98%	Number of pieces in stock / Expected Pieces in Stock x 100	Number of pieces in stock CMMS	Expected Pieces in Stock CMMS		
Turns Ratio of Spare Parts / Year	> 2 to 3	Value of stock issued in a year / value of stock held at the time of measurement	Value of stock issued in year CMMS	Value of stock Held in Inventory CMMS		

Figure 5-5. Future maintenance and reliability metrics model spreadsheet.

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The amount of asset information that can be documented within a CMMS asset record is great, and the amount of time and resources necessary to find and record "additional" information can be overwhelming. Therefore, a key to increasing the value of the CMMS quickly is to limit information gathering to fields that are critical to the performance management system. It also is important to consider the organization's ability to maintain the data's accuracy over time, as a loss of trust in the data renders it valueless.

A list of generic CMMS data fields, arranged in three broad information categories, can be considered for data entry or data collection to support a current or future performance management system. The following information categories and CMMS data field descriptions provide a sample arrangement of of basic information that can be used for effective analytics.

Information Category: Asset Information

- **Functional location.** This data field addresses the location of each asset/piece of equipment within the organization in functional, geographical, and ownership senses. Many CMMSs employ a *functional location table* in which this information resides. The asset table is arranged in a hierarchy and includes links to other information on each asset. Data collected or entered as functional data may include:
  - Organization identification (ID)
  - Organization description
  - Site ID
  - Site description
  - Process ID
  - Process description
  - Functional location ID
  - Functional location description
  - Physical location ID/room number
  - Physical location/room description
  - General ledger account/other financial data
  - Status
  - GIS reference data
  - Criticality
  - Asset design life/useful life
- **Physical attributes.** This data field addresses physical characteristics of the asset that will remain consistent regardless of where the asset is installed or located. This information resides in the asset/equipment table and asset hierarchy. Data collected or entered as physical attributes may include:
  - Asset classification
  - Asset sub-classification
  - Manufacturer
  - Model
  - Serial number
  - Size descriptors (dependent on asset type), such as horsepower, diameter, and flow
  - Weight
- **Performance.** This data field addresses characteristics of the asset that describe its functional performance based on design, construction, and testing within its current operating context. The specific performance attributes depend on the asset's classification, but data collected or entered as performance attributes may include:
  - Capacity in units produced or processed
  - Quality of units produced or processed
  - Electrical input or output

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  - Speed or load
  - Safety ratings
  - Environmental ratings
- **Financial.** This data field addresses financial data associated with functions executed and tracked within the CMMS, including:
  - Financial asset ID (unlike the physical asset ID, one financial asset ID may account for multiple physical asset IDs)
  - Asset value
  - Asset purchase date
  - Asset install date
  - Capital project references
  - Labor rates
  - Estimated asset overhaul cost
  - Estimated asset replacement cost

Information Category: Inventory Information

- **Physical attributes.** This data field addresses physical characteristics of the inventory item. Data included in this data field will reside in the inventory items table and may include:
  - Item commodity type
  - Item description
  - Supplier/vendor
  - Supplier/vendor number
  - Manufacturer
  - Model number
  - Serial number
  - Special ratings
- **Movement/usage.** This data field addresses information related to inventory entry or exit from the inventory system in terms of both physical and financial tracking. Data collected or entered under movement/usage may include:
  - Usage location
  - Minimum quantity stored
  - Maximum quantity stored
  - Inventory purchase date
- **Financial.** This data field addresses financial data associated with the inventory item. Data collected or entered in this field may include:
  - Inventory value
  - Inventory purchase date
  - Capital project references

Information Category: Work Order Information

- **Work assignment.** This data field addresses documentation of work against a specific asset and/or functional location. Accurate assignment of work is critical for performing effective asset management. Data collected or entered in this field may include:
  - Functional location ID
  - Asset ID
- **Work activity.** This data field addresses information related to "as found" condition, work performed, and "as left" condition. Data collected or entered in this field may include:
  - Work type
  - Work sub-type

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  - Work priority
  - Work request description
  - Work details description
  - Failure codes (problem, cause, remedy)
  - Work close out comments
  - Shutdown start time
  - Shutdown end time
- **Labor.** This data field addresses information related to the description and amount of labor resources necessary to complete the work. Data collected or entered in this field may include:
  - Crew ID
  - Estimated labor hours to complete
  - Actual labor hours to complete
  - Craft type
  - Craft count
- **Parts and materials.** This data field is a listing of planned or added parts and materials to complete the work. Data collected or entered in this field may include:
  - Inventory item ID
  - Quantity required
  - Parts and materials cost
- **Schedule.** This data field collects dates associated with the identification/request for work through completion of the work. Data collected or entered in this field may include:
  - Work request date
  - Work schedule date
  - Work start date
  - Work complete date

Effective Communication

Communicating the performance measure analysis and results is a key element of the data life cycle (collect, review, preserve, analyze, document, and communicate). Communication can take many forms, such as:

- Dashboard (see Figure 5-6)
- Briefing
- Metrics heat map (see Figure 5-7)
- Presentation
- Report (see Figure 5-8)
- Scorecard

The performance measure type (institutional, strategic, or operational) influences the communication format—the way the information is received or viewed by each decision maker (data recipient). For example, day-to-day metrics may be arranged for viewing on a desktop dashboard (supervisory control and data acquisition, or SCADA), whereas more detailed monthly and quarterly staff updates may be provided in a report format.

As important as the communication format, the content of performance measure communications also needs to be considered in terms of the recipients. Everyone in the organization does not need to be advised of all performance measures. The reporting should be tailored to the needs of the recipients. With that tailoring in mind, all performance measure reporting should include:

- **Performance measure relationships.** In a cascading performance measure model, the performance shown in any higher level strategic measure is always influenced by tactical performance

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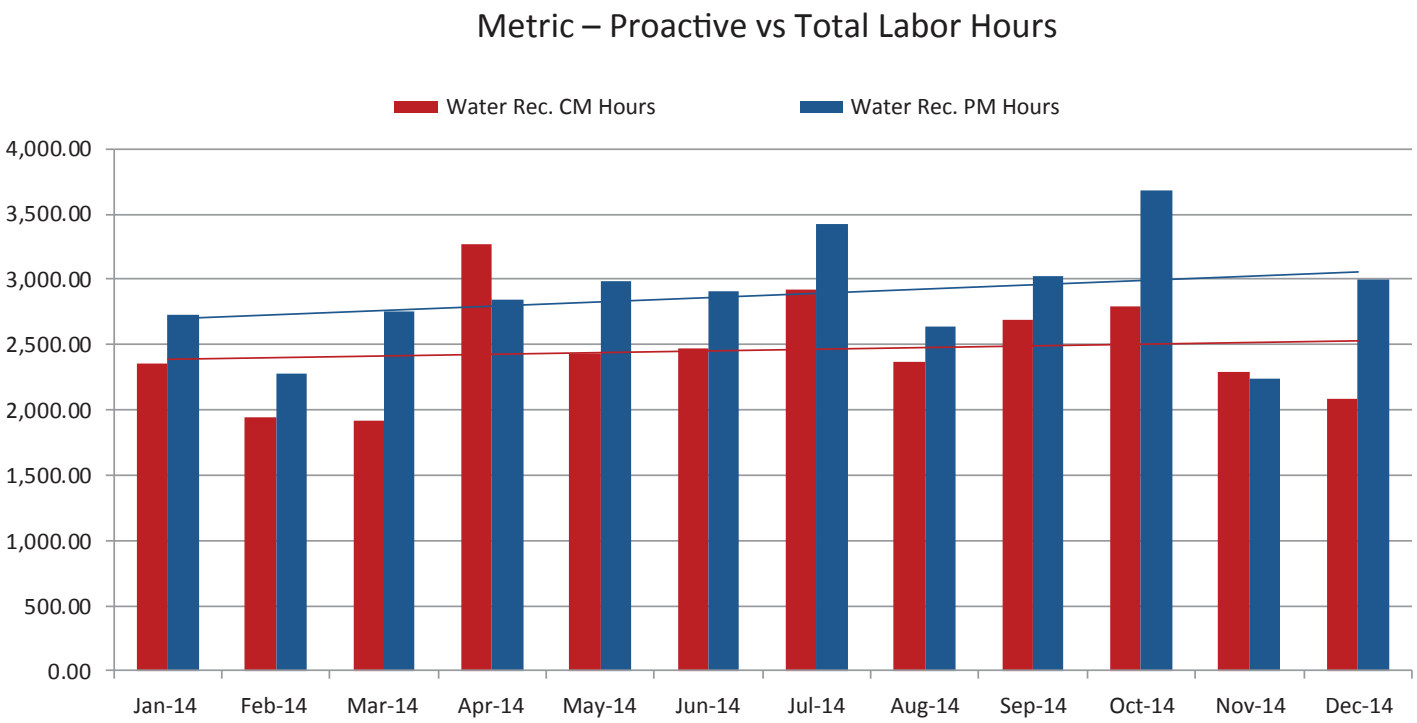
Ref	Performance Metric	Performance Target	Utility Overall	Water Ops	Lanier	Shoal Creek	WW Ops	Yellow River	Crooked Creek	FW Hill
A	Total Annual Maintenance Costs / RAV	<3%	5%↑	3%↑	2.50%	3.50%	6%↑	3%	7%	8%
B	% Inventory Value / RAV	<1%								
C	Critical Equipment Availability	97%>								
D	Normalized Mean Time Between Failure (nMTBF)	NA - Increasing Trend								
E	% Proactive labor hours vs total labor hours (Proactive + Reactive)	80%>								
F	% Schedule Compliance	90%>								
G	% PdM Inspection labor hours vs total labor hours (PdM + PM + PdMR + PMR+ Reactive)	15%>								
H	Total Maintenance Costs / MGD	NA - Trend								
I	Maintenance Contractor Costs / MGD (6 Month delay in reporting)	NA - Trend								
J	% Total Compliance Annual - Maintenance	100%								
K	% of Emergency & Urgent Labor Hours vs Total Labor Hrs (All Priority)	< 7.5%								
L	Avg Mean Time To Repair MTTR	Decreasing Trend								
M	% PM Hours by Operations	NA-Trend								
N	Avg time from Purchase Request to Material Receipt	NA-Trend								
O	Number of Stock Outs for critical equipment maintenance	<2%								

Figure 5-6. CMMS metrics dashboard.



Figure 5-7. CMMS metrics heat map.

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Calculation - ((PM hrs + PdM hrs + PMR hrs + PdMR hrs) / Total Hours) x 100

Type – Leading and Lagging

Purpose – to measure the proactive nature of the executed maintenance strategy.

Basis for Tracking – Proactive maintenance has been determined to be less costly than reactive maintenance and more effective for ensuring higher reliability.

Options for improving 1. Increase level of proactive maintenance labor hours 2. Decrease level of reactive maintenance hours	Leading Metrics 1. Avg Mean Time To Repair (L) 2. % Planned Hours vs Total Actual Hours (R)
Recommended Action(s) 1. Continue the expansion of predictive maintenance inspections 2. Improve maintenance efficiency thru improved planning and scheduling to reduce MTTR	

Figure 5-8. Metrics report example.

as measured at lower levels. The performance measure report must clearly link all strategic performance measures to tactical activities so that the recipient can easily drill down in the report to understand what is driving the performance indication, either positively or negatively.

- **Goals/targets/expectations.** Each performance measure in a report must include an associated expectation of performance (goal or target). Some performance targets are easily expressed with a quantitative numerical value, are typically defined by industry best practices, and are easily measurable. Other performance goals are less obvious, and some can be expressed in terms of expected trends (increasing or decreasing). In all cases, the communication of expectations must be clear so as not to overstate or understate the relationship between the performance measure and the organization's objectives.
- **Corrective action options.** For each performance measure being tracked and reported, it is beneficial to identify the corrective action options that are available when performance does

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not meet expectations. This provides value in terms of the organization being able to react effectively when performance suffers, but it also will assist with selecting performance measures that provide the greatest value. If a corrective action cannot be easily identified for a performance measure, the performance measure may be of low value.

Finally, although performance measure communication is essential for influencing performance within the organization, communication to individuals who are not educated in the purpose and meaning of the performance measures *can negatively influence performance.* Therefore, the most important aspect of communicating performance is to ensure that, whether communicating or receiving performance measure information, participants at all levels are fully trained on the entire performance measurement process and how each individual's performance impacts the overall mission.

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CHAPTER 6

Best Practices for Establishing Consistent Asset Taxonomy, Hierarchy, and Attributes

One of the most important features of a CMMS is data accessibility. The user should be able to easily access, accurately retrieve, quickly index, and efficiently develop reports for the assets. These functionalities can only be achieved if the system (database) has been designed and configured accurately. Asset taxonomy, hierarchy, and attributes are critical to establishing a robust database for a CMMS and allowing this accessibility. Additionally, they make linking and integrating different databases easier and more accurate.

Taxonomy and Hierarchy

Defined as the science of classification, taxonomy also can be described as a *controlled vocabulary with a hierarchical structure.* An important term as applied to asset management and in a CMMS, taxonomy refers to a classification system that makes indexing, grouping, saving, searching, and retrieving of digital data easier, faster, and more accurate. Figure 6-1 visually illustrates a taxonomy.

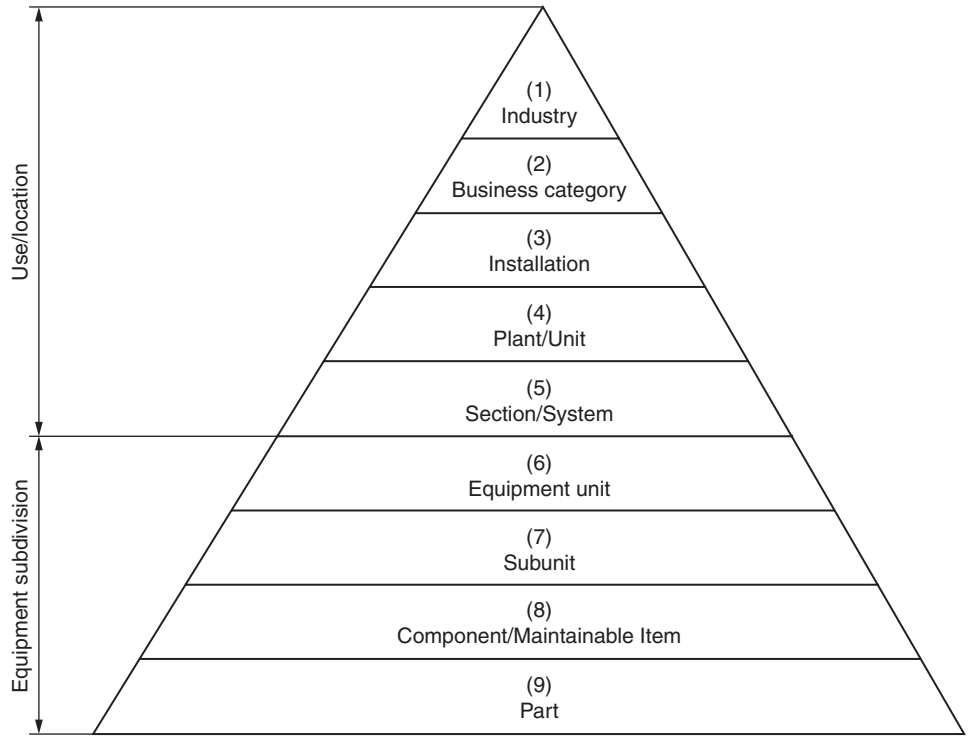
In asset management, the hierarchy embedded in a well-structured asset taxonomy assists airports in accurately configuring the system, reliably capturing the data, and allocating resources (personnel, budget, tools, schedule) to efficiently manage the system. The taxonomy increases the reliability and maintainability of the assets to meet the operational goals and performance measures and provides the ability to roll up the data on expenses from parts and components to sections/systems and business categories. Further, through taxonomy, CMMS users benefit from the ability to manage the assets under a unified, categorized content system with a controlled vocabulary. Table 6-1 shows an aviation taxonomy.

Hierarchy is the system organization of an asset group that shows the relationships between the highest level of the asset down to the various components of the asset. The relationships between these levels are noted as *parent-child* relationships.

Figure 6-2 presents a passenger boarding bridge hierarchy at Lester B. Pearson International Airport in Toronto, Canada.

A benefit of establishing the asset hierarchy parent-child relationship is the ability to charge costs to the lowest possible asset level, thereby providing a means to identify where maintenance dollars are actually being spent. Additionally, once the asset hierarchy parent-child relationship has been established, field technicians and engineers can write work orders to the correct asset level and not to a general area. This helps develop historical data that can be used to identify the actual "bad actors" that are creating the maintenance burden (Langan 2013). The taxonomic hierarchy also helps with scheduling: For example, if a planner schedules maintenance on a pump station, then all assets associated with that pump station (pumps, SCADA instrumentation, electrical systems,

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Source: ISO 14224

Figure 6-1. Illustration of a taxonomy.

piping, mechanical, and pumping system) also can be considered for maintenance to be scheduled during the time the pump station is offline.

In maintenance, the taxonomic hierarchy should be related to the maintenance data. For example, every equipment unit has a certain failure mode, while the failure mechanism is related to the lowest level within the hierarchy. Table 6-2 illustrates a guidance regarding the relation between maintenance levels and taxonomy.

Some best practices in taxonomy include:

- Follow taxonomy standards (e.g., ISO 14224)
- Follow a logical hierarchy
- Label categories and levels in a way that various users across the organization will find easy to understand and use
- Do not use acronyms or abbreviations
- Ensure the taxonomy can be used across different systems

Attributes

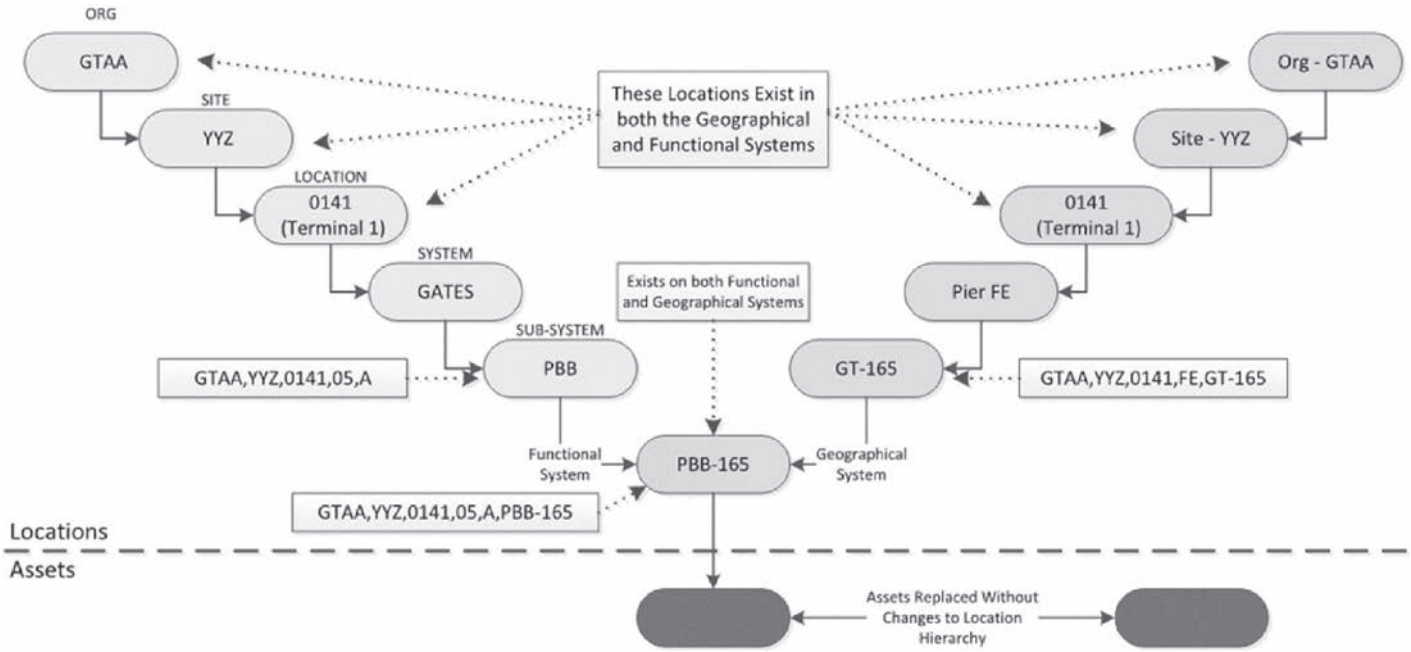
Attribute data provides the information that describes the properties of a field within the database. Each attribute provides a piece of information about the asset (e.g., identification, location, parts, inspection date, failure mode). Many benefits accrue to incorporating this information as data in attribute format rather than as free text. These benefits include:

- Efficiency and consistency in entering, storing, and retrieving data
- The ability to validate and verify data at the time of data entry
- Confidence in staff queries and data analysis based on data consistency, validation, and verification

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Table 6-1. Example of an aviation taxonomy.

Main Category	Taxonomic Level	Taxonomic Hierarchy	Definition	Example
Use/Location	1	Industry	Type of main industry	Aviation
	2	Business Category	Types of business or process	Airport
	3	Installation Category	Type of facility	Terminal
	4	Plant/Unit Category	Type of unit	Gates
	5	Section/System	Main section	Passenger Boarding Bridges (PBB)
Equipment Subdivision	6	Equipment Class/Unit	Class of similar equipment units; each equipment class contains comparable equipment units (e.g., pump, valve, chair, sprinkler)	Rotunda, Tunnel, Lift Column, Wheel Bogie, Cab
	7	Subunit	A subsystem necessary for the equipment unit to function	Visual Docking Guidance System (VDGS), Potable Water Cabinet (PWC), Pre-Conditioned Air (PCA), Ground Power Unit (GPU),Tunnel System
	8	Component/ Maintainable Item	The group of parts of the equipment unit that are commonly maintained (repaired/restored) as a whole	Lift Columns, Control Panels, Tunnel System Accessories, Heating and Cooling Subunits
	9	Part	A single piece of equipment	Hydraulic Oil, Lubricant, Gasket, Tubes, Tire, Switch, Fuse, Lightbulbs, Carpet



Source: Lester B. Pearson International Airport, Toronto, Canada

Figure 6-2. An example of an asset hierarchy (passenger boarding bridge).

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Table 6-2. Maintenance parameters in relation to taxonomic level/hierarchy.

Recorded Maintenance Data	Taxonomic Level/Hierarchy				
	4 Plant/Unit	5 Section/ System	6 Equipment Unit	7 Subunit	8 Component/ Maintainable Item
Impact of failure on safety	X				
Impact of maintenance on safety	X				
Impact of failure on operations	X	X			
Impact of maintenance with regards to operations	X	X			
Failure impact on equipment			X	X	X
Failure mode		X	X	X	X
Failure mechanism			X	X	X
Failure cause				X	X
Detection method		X	X	X	X
Subunit failed				X	
Component/maintainable item failed	X	X	X		X
Downtime	X				
Active maintenance time			X	X	X

Source: ISO 14224

Asset attributes may be entered using predefined codes in the database. These generic codes provide accurate descriptions regarding the data input and are more inclusive; however, sometimes more precise information and details are needed for an attribute. In those cases, a comment area can provide this extra information.

The relationships and links between the various attributes are detailed and designed to fit the database *architecture*. The four common models for database architecture are:

- Hierarchical
- Network
- Relational
- Object

Within a *hierarchical* model, data field relationships typically resemble a family tree in which each level represents a particular attribute of data. Hierarchical models can be organized around functional, electrical, or geographical systems, as seen in Figure 6-3. Figure 6-4 diagrams the asset hierarchy boundary for the HVAC system. Notice that in Figure 6-3, notations indicate locations that coexist in the "Functional, Geographic, and Electrical Systems," with one of the Functional Systems being HVAC. The coexisting locations act much like traffic intersections in the database. Although it also reflects the hierarchical model, the boundary diagram in Figure 6-4 details attributes that have been defined exclusively within the HVAC system. Figure 6-5 diagrams the HVAC asset system relationship, and Figure 6-6 presents a technical drawing for an air handler system (AHS). All of these figures show ways of organizing attribute information housed in a hierarchical database.

Like all organizational structures, the specific architectural model chosen to construct a database ideally reflects the organization's priorities in focusing on the information. For example,

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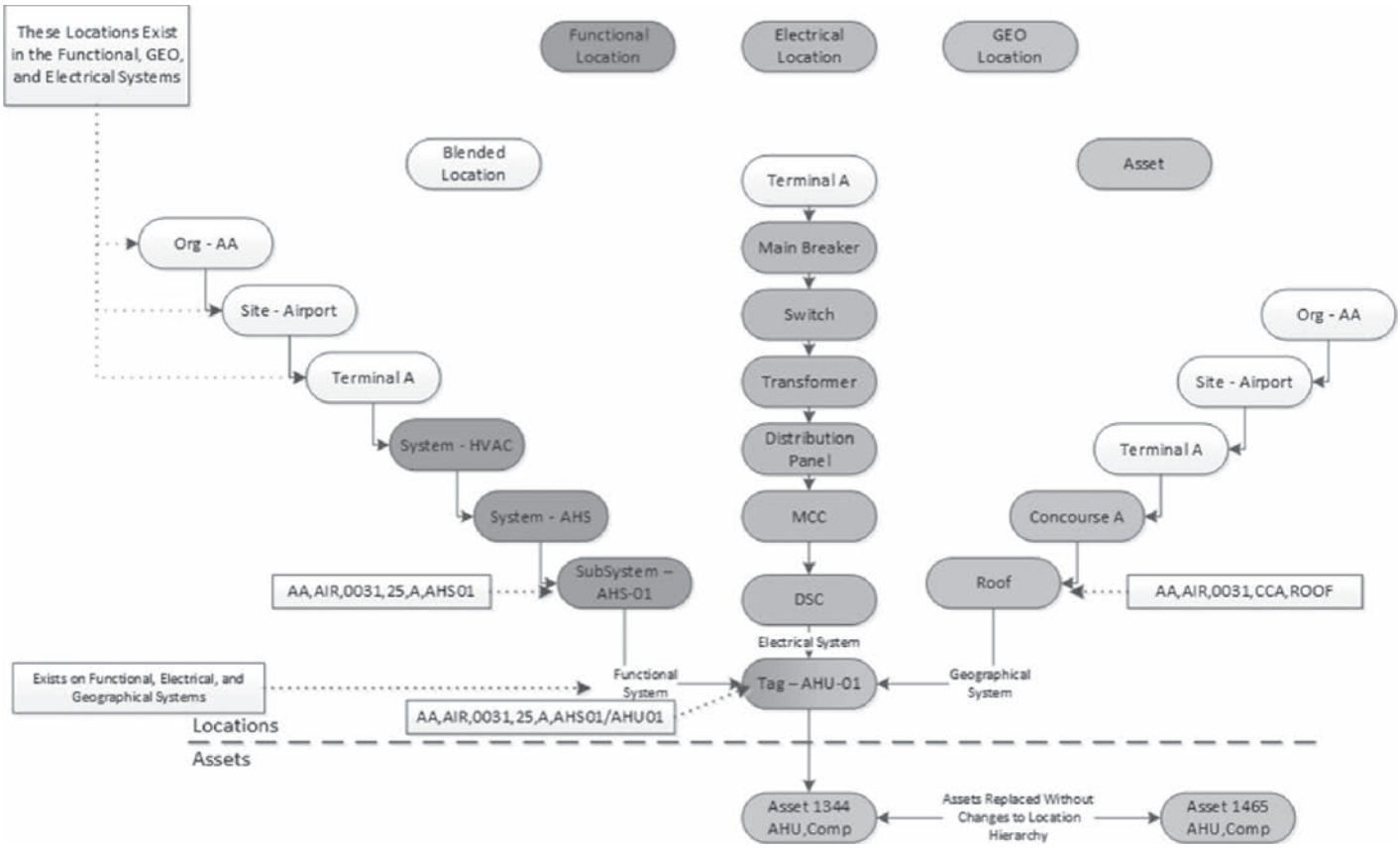


Figure 6-3. Example of overall HVAC hierarchy showing functional, electric, and geographical systems.

the *network* model allows attributes to be assigned to more than one parent. The *relational* model constructs the database from tables of data elements (attributes). The access path is not predefined, and all types of data manipulation are in tabular format. The *object* model consists of multiple objects, each of which has a structure and an interface. The interface itself provides the link between the different objects. This model allows for the database design to be very flexible, extendable, reusable, and easy to maintain. The object model is the most popular database model because of its interface.

In many organizations, separate legacy systems may be based on hierarchical, network, or relational, and/or object models. When integrating such legacy systems with a new or upgraded CMMS, developers create system tools and instructions that allow users to channel the information from the originating, legacy databases, into and across the new or upgraded CMMS structure.

Both figures show information drawn from a hierarchical database. The diagrams in Figures 6-3 through 6-6 are provided for reference only. Based on its overall CMMS goals and the availability and practicality of incorporating any legacy systems, each airport should decide if one or more models (and one or more databases) will be needed as part of the CMMS implementation.

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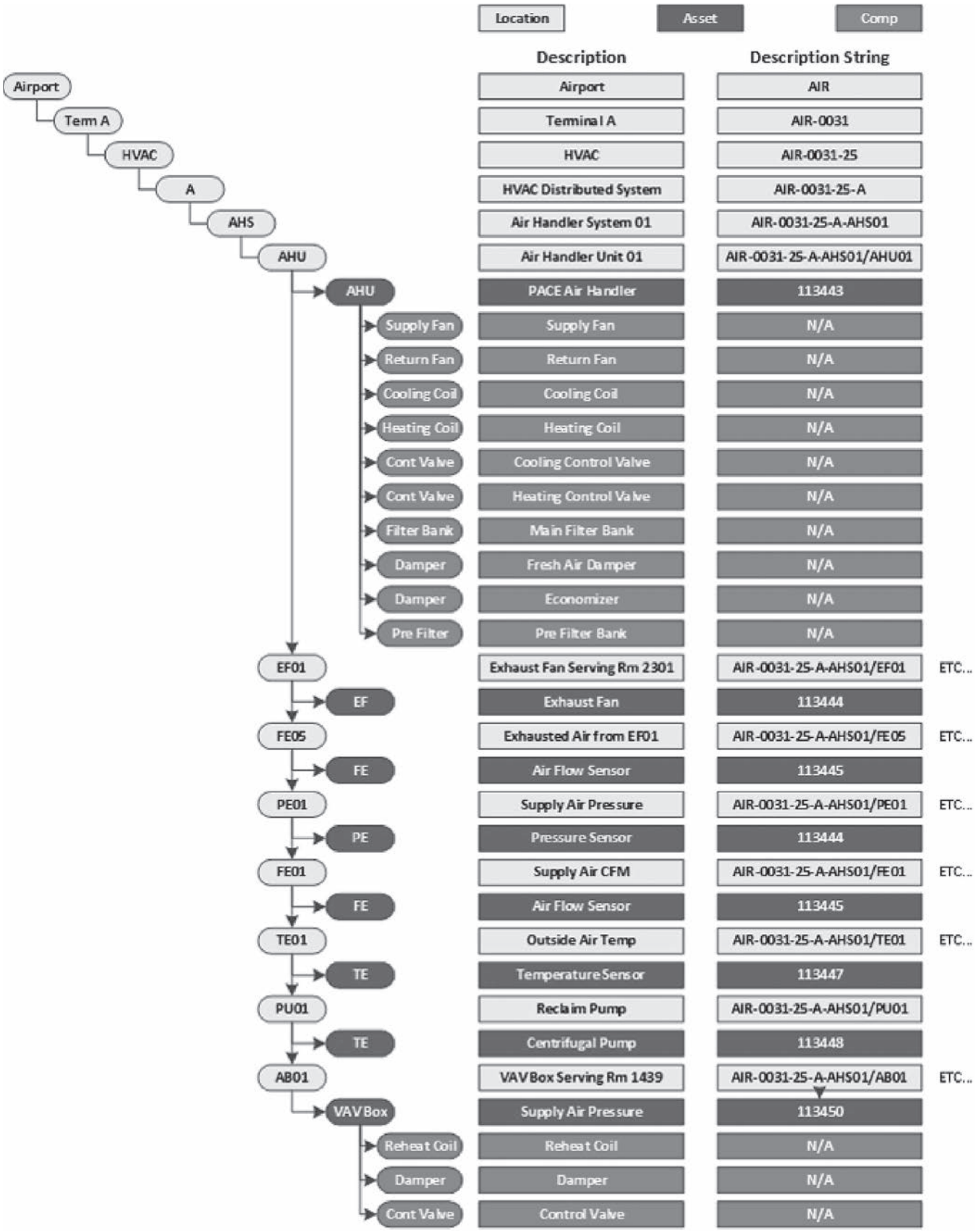


Figure 6-4. Sample asset hierarchy boundary diagram for HVAC system.

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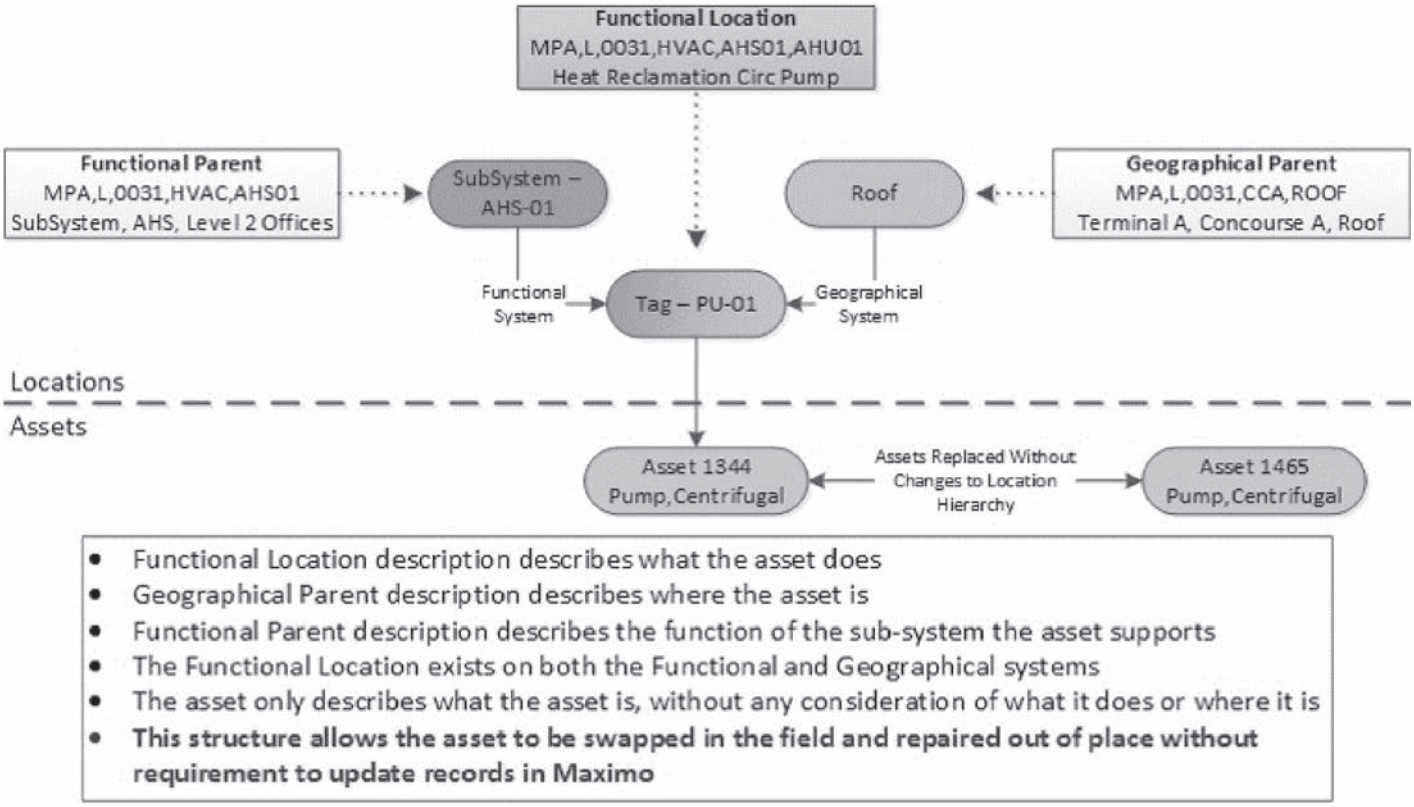


Figure 6-5. Relationship diagram for an HVAC asset system.

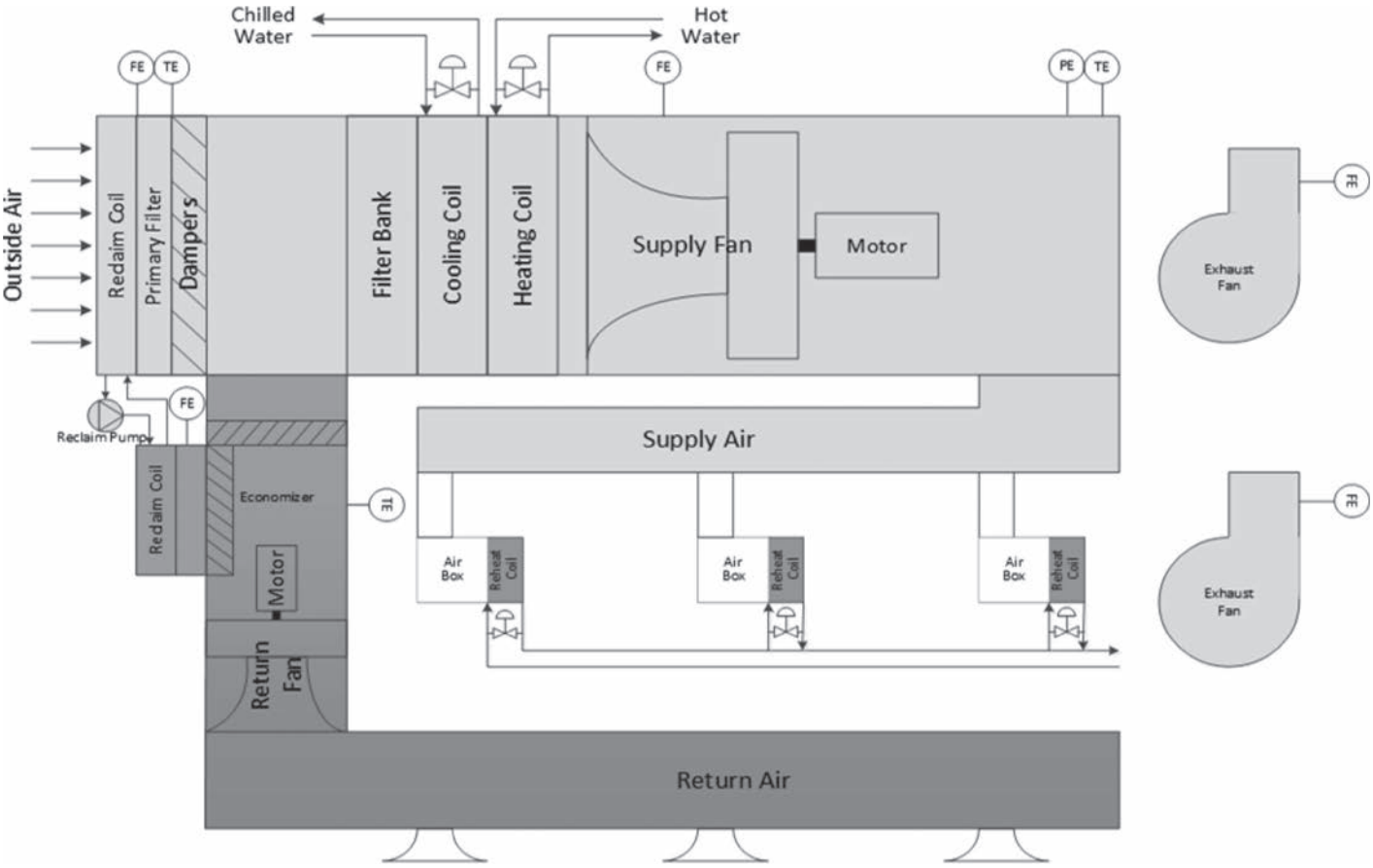


Figure 6-6. Example of an AHS technical drawing.

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CHAPTER 7

CMMS Implementation and Compliance with ISO 55001

The ISO 5500X Series

ISO 55001 is the international standard for asset management. This international standard specifies the requirements for establishing, implementing, maintaining, and improving a set of practices that together make up an AMS. The international standard for AMS further specifies that the requirements it puts forth must be implemented within the context of the organization. The reference to context has significant ramifications; although the requirements of an AMS are the same for every industry sector, the detailed operational procedures for an AMS will be very different for an airport than they are for a manufacturing plant. The AMS will also vary for airports of varying sizes. These factors mean that it is best for people working within an organization to develop and customize its AMS, using the requirements of ISO 55001 as a guide.

ISO 55000:2014 provides an overview of asset management, its principles and terminology, and the expected benefits from adopting asset management. It also provides context for ISO 55001 and ISO 55002.

ISO 55002:2014 provides guidance for the application of an AMS in accordance with the requirements of ISO 55001.

CMMS and Asset Management Fundamentals (ISO 55000, Clause 2.4.2)

The international standard of asset management (overview, principles, and terminology) describes asset management as being based on four fundamentals: alignment, assurance, leadership, and value.

Implementation of a CMMS will assist an airport with a more efficient realization of three out of the four of these fundamentals; specifically, alignment, assurance, and value.

Alignment

The purpose of asset management is to translate organization and business objectives into technical, financial, and operational decisions, plans, and activities. This effort includes the implementation of planning and decision-making processes through asset management plans (AMPs) and the integration of asset management processes with functional management processes. A CMMS facilitates this implementation and integration with respect to the airport's maintenance activities. PM activities are entered into a CMMS, establishing the majority of the maintenance AMPs; the CMMS functionality allows for

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integration of asset management processes with the following functional management processes:

- Finance, through cost capture and general ledger allocation
- Human resources, through timesheet confirmation and skills apportionment
- Logistics, through inventory management and materials allocation

The functionality of a CMMS assists airport maintenance teams with achieving alignment.

Assurance

ISO 55000 states that asset management gives assurance that assets will fulfill their required purpose. This includes the assurance that organizational processes are managed across all life cycle stages of the asset, resources are allocated for asset management activities, and monitoring is conducted for assets, asset management, and the AMS. A CMMS provides airport asset management staff with the ability to perform all of these functions. The maintenance team can provide auditable records of their asset management activities, allocate human and physical resources for PM tasks, and store the data used for monitoring assets, asset management, and the AMS.

Value

ISO 55000 stipulates that assets exist to provide value to the organization and its stakeholders, and it includes the use of a life cycle management approach to realize value from assets. This is where a CMMS is a useful software tool to implement: the life cycle management approach that the airport has determined, considered, and decided upon, can be efficiently planned and implemented within the CMMS. Further to this, the CMMS can collect and store the data required to monitor the life cycle management approach, making it easier to determine whether the chosen approach is realizing value for airport assets.

CMMS and ISO 55001

The functionality of a typical CMMS, to plan and implement maintenance activities, allows airport staff to record data for the tasks they are completing. The recording of this data is configured within a CMMS to ensure that the data sets are linked in a manner that provides additional value to this data. This additional value includes the ability to inform management decision making and provides the evidence required to show compliance, including compliance with ISO 55001.

For example, when staff implements an asset inspection, they record identified nonconformities with asset performance. Inspection results and CM expenses are recorded in the assets' history. Furthermore, CM work progress is tracked, and quality assurance/quality control (QA/QC) documents are recorded within the same database. Linking multiple data sets, recorded by different staff at the airport, allows management to analyze the effectiveness of their staff to conduct management of airport assets; determine more effective interventions and intervals for maintenance; identify high-risk, critical airport assets; and monitor the effects of these changes to ensure continual improvement. Because the information stored in the CMMS can be analyzed, evidence-based decision making can be conducted, providing management confidence that their decisions are based on sound empirical data.

Information collected and analyzed using a CMMS can demonstrate to stakeholders that airport staff are planning and implementing maintenance activities appropriately, and that airport assets are meeting the performance requirements to deliver the required LOS. A CMMS can be an effective tool for maintenance staff to prove their AMS is effective for managing airport assets, and that the AMS meets international standards through compliance with ISO 55001.

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The next section of this chapter focuses on the specific clauses of ISO 55001 and how the implementation of a CMMS assists with providing evidence that the AMS is compliant with ISO 55001. The discussion has been written with reference to the guidance document ISO 55002 (issued in 2014).

Planning to Achieve Objectives (ISO 55001, Clause 6.2.2)

ISO 55001, Clause 6.2.2, stipulates the requirement for an organization to determine how to integrate its planning to achieve its asset management objectives and details the requirements to be recorded for planning. Most of the information required to be recorded for maintenance planning can be done within a CMMS. This information includes: what activities will be done, the resources required, who will be responsible, when the activities will be completed, the financial and non-financial implications of the plan(s), and actions that will be taken to address risks and opportunities associated with managing assets. A CMMS facilitates documentation of the maintenance AMPs in the following ways:

- Maintenance planning information stored in a CMMS is available for retrieving and reporting.
- Information duplication is prevented. For example, if the strategy for PM is updated from time-based maintenance to condition-based maintenance, the change is required only once—in the CMMS—and the distribution of the change is completed at the issuance of the CMMS maintenance work orders.
- A CMMS stores detailed information, and links this information in a manner that allows detailed AMPs to be determined and documented. For example, a work plan can be linked to reoccurring PM work orders, which are further linked to an asset, the responsible staff member, required skill-sets, completion dates, and a cost center. The asset is further linked to failure codes, health and safety requirements, cost estimations, budgets, and details about asset attributes. The ability to link this information provides maintenance staff with the functionality required to meet the international standards outlined within ISO 55001.
- O&M risk and asset risk can be managed through a CMMS, and the associated inspections and CM information retained within the CMMS database as evidence.
- Storing the maintenance AMPs within a CMMS allows the AMP to be functional. Rather than being in a static report format, the CMMS AMP is in a format that staff can directly use for the management of their maintenance activities.

The points listed summarize most of the requirements of Clause 6.2.2; consequently, documenting the AMP in a CMMS provides an airport the ability to meet the majority of requirements of ISO 55001, Clause 6.2.2. The remaining Clause 6.2.2 requirements, including documenting the methodology and criteria for decision making and prioritizing of maintenance activities and resources, are best handled as operating procedures recorded within the document management system of the AMS. It is the outcomes of decision making that are best stored within the CMMS database and reported from the CMMS when the ISO 55001 auditor requires evidence of the maintenance AMPs.

Internal and External Communication (ISO 55001, Clause 7.4)

ISO 55001 requires that an organization determine and implement communication protocols both internally and externally. A CMMS enables an organization to implement these protocols, and the configuration of the CMMS workflow enables airport staff to provide evidence of compliance with this clause. For example, a CMMS can provide the evidence required

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for compliance with this clause through a CMMS configuration for escalating maintenance activities, work execution management notifications for internal staff, performance reporting to senior management, and automatic electronic maintenance requests for outsourced contractors.

Documented Information (ISO 55001, Clause 7.6)

ISO 55001, Clause 7.6 states that an organization's AMS shall include documented information for legal and regulatory purposes, determined by the organization, and required by ISO 55001. The clause states that the information shall have an appropriate identification, description, format, and review/approval process, and that the information shall be controlled to ensure its availability, suitability, and protection.

When configured and controlled correctly, a CMMS provides the ability for maintenance departments to manage the AMS to meet this clause within the international standard. Upon determining the information that staff are required to collect, the CMMS can be configured to ensure control of the information, including ensuring that the information staff report has the required metadata associated with it and that it is managed through the airport's review and approval process. When staff members report information into a CMMS, it is secure yet accessible, and this information can be retrieved when required, including during an ISO 55001 certification audit.

Operational Planning and Control (ISO 55001, Clause 8.1)

To meet international standards, an AMS is required to have in place operational planning, controls, monitoring, and documentation to manage the implementation of the AMPs (from ISO 55001, Clause 6.2.2).

A CMMS is specifically designed to do this for maintenance activities and provides airport staff with the functionality required to implement the control of processes, in accordance with specific criteria; record and retain evidence that maintenance activities within the AMPs have been implemented; and record and retain evidence that risks have been treated and monitored by airport staff.

Management of Outsourced Activities (ISO 55001, Clause 8.3)

Outsourced activities can have a significant impact on achieving an airport's asset management objectives. To ensure that this impact is positive, and for ISO 55001 compliance, airport staff are required to control the outsourced activities, and management of the outsourced activities is required to be integrated into the AMS. After deciding which maintenance activities are to be outsourced and the allocation of responsibilities for contractor management, the CMMS can be used as contractor management software for the sharing of knowledge and information between the airport and contracted service providers.

Asset Performance Monitoring (ISO 55001, Clause 9.1)

Monitoring, measurement, analysis, and evaluation are some of the most important requirements of a modern AMS. To meet the requirements of the international standard, airports must determine which assets are to be monitored; the frequency of analysis; and, most importantly, the methodologies for monitoring, measurement, analysis, and evaluation, which will produce

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valid results. After this step has been completed to meet the international standard, the airport will be required to retain records as evidence of this information and report on the effectiveness of the process for risk management.

A modern CMMS can be configured to collect and store data for any monitoring point. The collection of measurement data can be conducted either manually by staff and contractors or through data integration with SCADA and similar instrumentation-based performance-monitoring systems.

A CMMS provides the database for storing asset measurements. The stored results can be used to identify and evaluate risks and opportunities arising from the management of physical assets, as required by ISO 55001.

Within the CMMS, data can be stored for individual asset condition/health, asset criticality, and the resulting risk of asset failure. Upon analysis of the data stored in the CMMS, an asset's deterioration can be mapped. Every time condition-monitoring results are entered into the CMMS, the accuracy of the deterioration curve can be refined. Figure 7-1 shows an example of the various condition-monitoring techniques used to determine an asset's deterioration and how condition monitoring refines the accuracy of the deterioration curve, which in turn, will improve the calculation of when asset overhaul or replacement is required.

A CMMS can provide a partial solution to asset performance monitoring under Clause 9.1, while the implementation of BI software for data analysis and evaluation addresses other sections of this clause. Reporting the stored data from a CMMS provides staff with the inputs for asset management analysis and the evidence required for asset monitoring.

AMS Monitoring

ISO 55001, Clause 9.1, also refers to monitoring and measuring the effectiveness of the AMS. KPIs for the maintenance processes within the AMS can be collected and stored in the CMMS, ready for analysis through BI software. Examples of AMS efficiency performance measures include staff customer-response times, AMS process completion times, and whether decision-making outcomes have met the requirement to align with asset management objectives. A CMMS enables

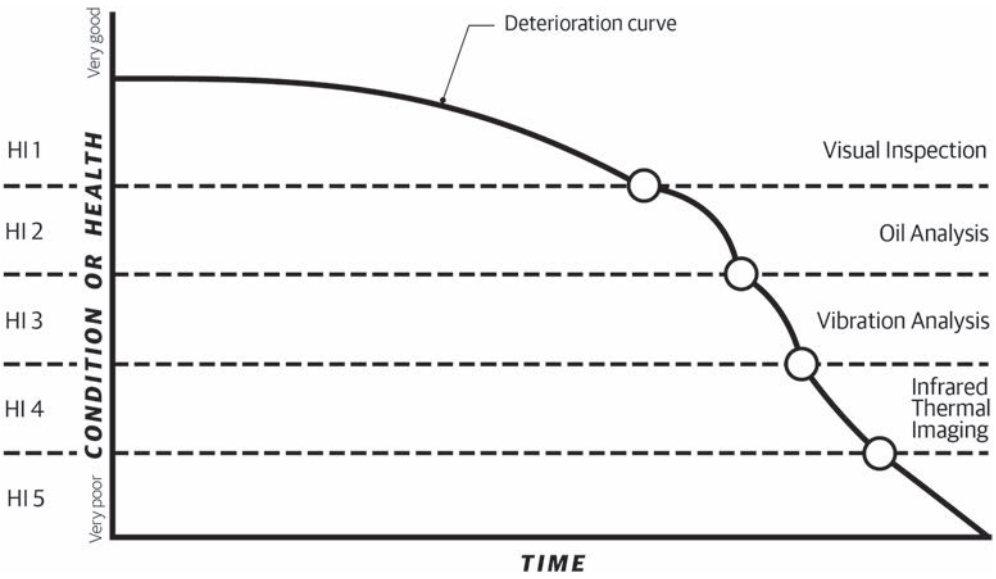


Figure 7-1. Sample asset deterioration curve with an ACM asset group example.

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agency staff to capture a selection of this information. Most notably, maintenance work execution system efficiency is recorded within the CMMS as a part of day-to-day management.

Asset Management Monitoring

ISO 55001, Clause 9.1, also deals with monitoring and measuring asset management performance. Referencing ISO 55000, asset management performance relates to realizing asset value; aligning decisions, plans, and activities with objectives; and having assurance that assets are fulfilling their required purposes. Meeting the standard entails monitoring, measuring, analyzing, and evaluating how well an organization is reducing expenditures, reducing risk, managing asset performance, and meeting its asset management objectives.

A CMMS provides the database for storing results from maintenance departments' asset management performance monitoring efforts. Examples of asset management performance measures will include financial and non-financial performance and may include measures of return on (asset) investment, labor utilization, service quality, and contractor performance. All maintenance work execution management is recorded within the CMMS, and many other asset management performance measures also can be managed through the CMMS.

Asset Nonconformity and Corrective Actions (ISO 55001, Clause 10.1)

ISO 55001, Clause 10.1, addresses the requirements for an organization to implement corrective actions when identifying an incident (nonconformity). An AMS that meets the international standard is required to have processes to manage nonconformities with its assets, asset management, and AMS, and to retain documented information. CMMS functionality assists with managing workflow for nonconformities and corrective actions for assets, but only for assets. Nonconformity response and management of corrective actions for asset management and AMS are best managed outside a CMMS.

When managing the response to asset nonconformities, including implementation of corrective actions, a part of the required work execution management is recording and retaining information. Staff can record CM information in the CMMS for the asset nonconformities discovered during inspections, asset monitoring, and from external parties such as tenants and visitors to the airport. For compliance with the international standard, the information collected must include details about the incident/nonconformity, the action taken to control and correct the consequences of the nonconformity, and details about further work to eliminate the cause of the nonconformity. If the information is collected during work execution management for managing both staff and assets, it can be reported from the CMMS for ISO 55001 certification.

Preventive Actions (ISO 55001, Clause 10.2)

ISO 55001, Clause 10.2, states that there is a requirement for organizations to have processes within their AMS to proactively identify potential failures in asset performance. Clause 10.2 links Clause 9.1 with Clause 10.1; that is, when a nonconformity is discovered during asset monitoring, a response will be implemented to deal with the consequences, a corrective action will be taken to eliminate the cause of the nonconformity, and the documented information will be retained.

The information collected and stored in the CMMS during work execution management can provide the required evidence for ISO 55001 compliance for organizations that administer processes for preventive actions through the CMMS.

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Summary

Implementing ISO 55000 and ISO 55001 into an airport's AMS through the functionality of a CMMS helps an airport with more efficient realization of asset management practices. ISO 55001 implementation focuses attention on the requirements of international standards, as stipulated by leaders who are physical asset management practitioners. It provides airport management and staff with the requirements for an integrated management system for asset management, which has an iterative feedback loop for continual improvement.

ISO 55001 implementation helps airport staff meet international asset management requirements for alignment, communication, policy, planning, control, monitoring, analysis, evaluation, auditing, competence, asset risk management, meeting stakeholder expectations, leadership, governance, organizational support, documentation, management of change, outsourcing, preventive action, corrective action, continual improvement, and the requirement to realize value from physical assets for the airport and airport stakeholders.

Implementation of the international standard helps airport staff realize good practices in their asset management programs.

Another way to describe how the content and clauses of ISO 55001 focus attention on the important elements of asset management is through describing the ISO 55001 sections that contain each clause. The balance of this chapter summarizes these sections of ISO 55001.

ISO 55001, Section 4: Context of the Organization

Section 4 ensures that external factors that impact the airport's AMS and programs are understood and acted upon. External factors define the reason for an organization to exist and the unique context in which the organization is to be managed. These external factors include legal, regulatory, political, social, economic, financial, environmental, and technological factors. Section 4 informs airport staff that the balance of external and internal stakeholder requirements, unique to each airport, will determine each airport's individual AMS and decision-making criteria.

ISO 55001, Section 5: Leadership

Section 5 establishes the asset management policy consistent with the organizational strategic direction. This section also requires leadership to commit to achieving the asset management policy by allocating sufficient resources, promoting collaboration between airport staff, and assigning responsibility to establish asset management objectives and a Strategic Asset Management Plan.

ISO 55001, Section 6: Planning

Section 6 requires airport management and staff to plan for success by defining success and documenting the plans to achieve it. With a focus on managing risks and opportunities, Section 6 requires staff to determine their team asset management objectives and document their processes, methods, and decision-making criteria for the asset management activities for which they are responsible. This section is applicable to every team responsible within the asset life cycle of an airport's asset portfolio: asset planning, procurement, project delivery, the asset commissioning team, operations, maintenance, and the airport staff responsible for asset monitoring, performance, replacement, disposal, and decommissioning.

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ISO 55001, Section 7: Support

Section 7 focuses on the requirements to support the planning and operation of asset management within the AMS. These are:

- Allocation of resources for the establishment, implementation, maintenance, and continual improvement of the AMS
- Managing the competence of airport staff who conduct asset management activities
- Raising awareness of the AMS, the importance of individual staff to contribute to the effectiveness of the AMS, and the requirement to conform
- Communication (specifically addressing what to, when to, how to, and with whom to communicate)
- Information requirements and the requirements for documented information

ISO 55001, Section 8: Operation

Section 8 also is applicable to every team responsible within the asset life cycle of an airport's asset portfolio (i.e., asset planning, procurement, project delivery, the asset commissioning team, operations, maintenance, and the airport staff responsible for asset monitoring, performance, replacement, disposal, and decommissioning). Section 8 provides the clarity airport staff requires to ensure that the asset management activities and plans determined in Section 6 and the actions identified in Section 10 are planned, implemented, and controlled for every airport team responsible for managing assets. This section includes the requirements for managing change, risk, and outsourced activities.

ISO 55001, Section 9: Performance Evaluation

Section 9 is separated into three clauses and describes the importance of monitoring, measurement, analysis, evaluation, auditing, and review. The first clause addresses the international standards for evaluating and reporting on asset performance, asset management performance, and the effectives of the AMS. The final two clauses discuss the requirements for auditing and reviewing the AMS itself.

ISO 55001, Section 10: Improvement

Section 10 provides the requirements for the processes and documentation for nonconformities, incidents, corrective actions, preventive action, and continual improvement. These requirements include improvements for assets, asset management, and the AMS.

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References

Airports Council International (ACI). 2012. Guide to Airport Performance Measures. ACI World, Montreal, Quebec, Canada.

Fortin, J., L. Alfaqih, and P. Bell. 2017. *ACRP Research Report 172: Guidebook for Considering Life-Cycle Costs in Airport Asset Procurement.* Transportation Research Board, Washington, D.C.

Governmental Accounting Standards Board. 1990. *Service Efforts and Accomplishments Reporting: Its Time Has Come.* Research Report. Governmental Accounting Standards Board of the Financial Accounting Foundation, Norwalk, CT.

Hazel, R. A., J. D. Blais, T. J. Browne, and D. M. Benzon. 2011. *ACRP Report 19A: Resource Guide to Airport Performance Indicators.* Transportation Research Board of the National Academies, Washington, D.C.

Infrastructure Management Group, Inc. 2010. *ACRP Report 19: Developing An Airport Performance-Measurement System*. Transportation Research Board of the National Academies, Washington, D.C.

Langan, P. 2013. *Finding the Hidden Value in Asset Hierarchy Validation*. Plant Engineering. http://www.plantengineering.com/single-article/finding-the-hidden-value-in-asset-hierarchy-validation/58e6bf336a75253c6c16374bedc97a18.html?OCVALIDATE=.

Mahaz, J., and L. Alfaqih. Subject matter discussion, March 10, 2017.

McNeeney, A. 2005. "Selecting the right key performance indicators." *Maintenance Technology,* Vol. 18 (April 2005). Available at: https://www.efficientplantmag.com/2005/04/selecting-the-right-key-performance-indicators/.

National State Auditors Association. 2004. *Best Practices in Performance Measurement: Developing Performance Measures.* A National State Auditors Association Best Practices Document. NSAA, Lexington, KY.

Neubauer, K., D. Fleet, and M. Ayres, Jr. 2015. *ACRP Report 131: A Guidebook for Safety Risk Management at Airports.* Transportation Research Board of the National Academies, Washington, D.C.

Price, M. 2014. *ACRP Report 116: Guidebook for Successfully Assessing and Managing Risks for Airport Capital and Maintenance Projects.* Transportation Research Board of the National Academies, Washington, D.C.

Turner, T. 2015. Why a CMMS Fails—Part 1. *ASSETPOINT* (August 25). Available at: http://www.assetpoint.com/why-a-cmms-fails-part-1/ (accessed July 29, 2016).

Wireman, T. 2016. *Successfully Utilizing CMMS/EAM Systems, Vol. 4.* Ft. Myers, FL. Available at: reliabilityweb.com.

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Abbreviations

ACY	Atlantic City International Airport
AMMS	Airport Maintenance Management System
AMP	Asset Management Plan
AMS	Asset Management System
API	Airport Performance Indicator
APTS	Airport Planning and Technical Services
BI	Business Intelligence
BIM	Building Information Modeling
CM	Corrective Maintenance
CMMS	Computerized Maintenance Management System
COO	Chief Operations Officer
EAM	Enterprise Asset Management
EWR	Newark Liberty International Airport
FMEA	Failure Modes Effects Analysis
GA	General Aviation
GCDWR	Gwinnett County Department of Water Resources
GIS	Geographic Information System
GTAA	Greater Toronto Airports Authority
HVAC	Heating, Ventilation, and Air Conditioning
ID	Identification
ISO	International Organization for Standardization
IT	Information Technology
JFK	John F. Kennedy International Airport
KPI	Key Performance Indicator
LCC	Life Cycle Costing
LEED	Leadership in Energy and Environmental Design
LGA	LaGuardia Airport
LOS	Level(s) of Service
MGD	Million Gallons per Day
MKE	General Mitchell International Airport
MMIS	Maintenance Management Information System
MORE	Maintenance, Operations, and Reliability Excellence
MRO	Maintenance, Repair, and Operating Supplies
O&M	Operations and Maintenance
PANYNJ	Port Authority of New York and New Jersey
PBB	Passenger Boarding Bridge
PM	Preventive and/or Predictive Maintenance
QA/QC	Quality Assurance/Quality Control

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RCM	Reliability-Centered Maintenance
ROI	Return on Investment
RTF	Run-to-Failure
SCADA	Supervisory Control and Data Acquisition
SEA	Seattle-Tacoma International Airport
SLC	Salt Lake City International Airport
SMART	Specific, Measurable, Attainable, Realistic, and Timely
SWF	Stewart International Airport
TEB	Teterboro Airport
YYZ	Toronto Pearson International Airport

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Glossary

A

Airport property. All property owned by the airport, whether purchased, leased, confiscated, donated, received by eminent domain, constructed, or annexed. Airport property may include supplies, real property, police property, capital assets, and controlled items.

Area. A subdivision of the airport campus; the combination of campus designation and area designation may define the general location of a facility.

Assessment. A systematic process of collecting and analyzing data to determine the current, historical, or projected status of an organization or system.

Asset. (1) An item of plant, equipment, or infrastructure that contributes to the function of a system so that the system can provide a service to a community or group of customers. An asset is continually maintained, refurbished, replaced, or upgraded so that the system provides an ongoing, minimum level of service (LOS). An asset has a capital replacement value and is depreciated over its useful life. (2) A physical component of a facility which has value, enables services to be provided, and has an economic life greater than 12 months. Dynamic assets have some moving parts, whereas passive assets have none. (3) A capital investment maintained by the airport's accounting system. The maintenance department typically refers to an asset as any item of physical plant or equipment. The term is used to describe items such as buildings, facilities, and systems and components that are controlled by the airport and from which a benefit is derived. For industry purposes, these items are considered fixed assets deployed or intended to be deployed in an operational environment.

Asset criticality. A ranking of assets according to potential operational impact. Criteria may include inherent safety and environmental risks, replacement cost, schedule, and redundancy.

Asset group. A group of like assets (e.g., valves), or a group of assets that make up a system (e.g., pump system).

Asset health. The overall condition of an asset measured in terms of reliability or technical assessments from predictive testing, operational monitoring, or maintenance history analysis.

Asset hierarchy. A framework that segments an asset base into appropriate classifications. The asset hierarchy can be based on asset function, asset type, or a combination of asset function and type.

Asset life cycle. The asset life cycle is the time span from when the asset is placed in service through its eventual replacement or disuse. How the asset's life cycle is managed will depend on specified asset management strategies and goals, which normally include training, maximizing utility, preventive maintenance, evaluation, and when use of the asset will stop.

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Asset management (AM). A systematic approach to the procurement, maintenance, operation, rehabilitation, and disposal of one or more assets. AM integrates the utilization of assets and their performance with the business requirements of asset owners or users. AM focuses on the continuous alignment of asset performance to meet service delivery outputs to deliver the desired outcomes. AM is both a management paradigm and a body of management practices that are applied to the entire portfolio of infrastructure assets at all levels of the organization, seeking to minimize the total cost of acquiring, operating, maintaining, and renewing the organization's assets within an environment of limited resources while continuously delivering the service levels customers desire and regulators require at an acceptable level of business risk to the organization. As defined by the ISO in Publicly Available Specification (PAS) 55, AM involves systematic and coordinated activities and practices through which an organization optimally and sustainably manages its assets and asset systems, their associated performance, risks, and expenditures over their life cycles for the purposes of achieving its organizational strategic plan. (Note: PAS 55 focuses primarily on physical assets and is being replaced as a formal specification by the more broadly focused ISO 55000; however, the information contained in PAS 55 remains useful as a reference.)

Asset management policy. An organizational policy that sets the framework for the management of airport infrastructure and assets. Most asset management policies include sections addressing (1) the organizational context and importance of asset management, (2) the overall vision and goals of the organization and supporting asset management vision and goals, (3) executive and key position roles and responsibilities, and (4) audit and review procedures.

Asset management strategy. A strategy for asset management covering the development and implementation of plans and programs for asset creation, operation, maintenance, rehabilitation/replacement, disposal, and performance monitoring to ensure that the desired LOS and other operational objectives are achieved at optimum cost.

Asset management steering committee. A committee that brings together people within the organization to design, build, and guide the asset management program. The steering committee establishes the program and enforce the ideas and practices within the organization.

Asset performance. The individual and collective contributions that an asset makes toward service delivery and/or business outputs. Asset performance is assessed by measuring how closely a set of predetermined outputs that arise from the existence and operation of the asset(s) meet assigned performance targets.

Asset registry. A record of asset information considered worthy of separate identification that includes inventory, historical, financial, condition, construction, technical, and financial information about each asset.

Asset replacement cost. The full replacement costs associated with a given asset, expressed in current dollars.

Assignment. A labor requirement on a work order record that has been matched to an appropriate laborer.

Attributes. Data items related to an asset.

Availability. The probability that an item or a system is operating satisfactorily, at point in time, when used under stated conditions.

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B

Backlog. All work that has not been completed by its specified "required by" date. The period for which a work order is "overdue" is the difference between the current date and the required by date. Work for which no required by date has been specified generally is included on the backlog. Backlog is generally measured in "crew-weeks," which are calculated by dividing the total number of labor hours represented by the work by the number of labor hours available in an average week to the crew responsible for completing the work.

Benchmark. A standard measurement or reference that forms the basis for comparison. When used to measure performance, the benchmark is recognized as the standard of excellence for a specific process.

Benchmark measures. A set of measurements (or metrics) used to establish improvement goals for processes, functions, products, etc. Benchmark measures often are derived from those used by other organizations to display "best in class" achievement.

Best in class. A descriptor used to highlight the outstanding performance of an organization within an industry.

Best practice. A superior or innovative practice that contributes to the improved performance of a process or an organization and is usually recognized as "best" by other peer organizations.

Best value. Achievement of the nearest possible match to the functional requirements for the best price. The best value is not about taking the lowest-priced option; rather, it is the optimum combination of whole-life cost and quality (or fitness for purpose) to meet the user's requirement.

Business process. The end-to-end sequence of activities that defines one or more business functions required to deliver goods or services to a customer. Customer-related activities are considered *external* business functions whereas support activities are considered *internal* business functions.

C

Capital asset. A physical asset that is held by an organization for its production potential.

Capital project. A project that involves new construction, major repairs, or improvements to an asset and for which the cost is capitalized rather than expensed. The asset is then depreciated.

Change management. The process of bringing planned change to an organization. Change management usually means leading an organization through a series of steps to meet a defined goal.

Computerized maintenance management system (CMMS). A software package used to assist in asset and work management functions. The CMMS tracks work orders, equipment histories, and preventive/predictive maintenance (PM) schedules. Usually it is integrated with support systems such as inventory control, purchasing, accounting, and manufacturing, and controls maintenance and warehouse activities.

Consequences of failure. The expected results of an asset failure, expressed in triple bottom line (social/community, economic, and environmental/regulatory) terms.

Construction. All constructed facilities, buildings, and infrastructure. A construction is defined as a physical setting used to serve a specific purpose. A construction may be within a building, or be a whole building, or be a building together with its site and surrounding

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environment. Alternatively, a construction may be a *constructed facility* that is not a building (e.g., a bridge, road, or railway).The term encompasses both the physical object and its use.

Corrective maintenance (CM). Maintenance tasks initiated as a result of the observed or measured condition of an asset or system, before or after functional failure, to correct the problem. CM can be planned or unplanned.

Critical assets. Assets for which the financial, business, or service-level consequences of failure are sufficiently severe to justify proactive inspection and rehabilitation. Critical assets have a lower threshold for action than do non-critical assets.

D

Data. A set of collected facts. The two types of data are *measured or variable data* and *counted or attribute data.*

Database. An organized set of searchable and retrievable records (e.g., a computer file composed of records, each containing fields together with a set of operations for searching, sorting, recombining, and other functions).

Data collection. The gathering and recording of facts, changes, and forecasts for analysis.

Data entry. The process that a technician uses to enter data directly to the system.

Data file. A collection of related data records or application data values that have been organized in a specific manner and stored after and separated from the user program area.

Data management. The process by which the reliability, timeliness, and accessibility to an organization's database is assured.

Decommission. To take an asset out of service, usually by following a set of prescribed activities.

Design life. The service life intended by the designer.

Document. In context, the creation of a physical or electronic "paper trail" (recorded steps and procedures) to support the CMMS functions. Documentation may include blueprints, receipts, and contracts, as well as .pdf files, pictures, and other computer-based files and images.

E

Economic life. The time period from the acquisition of the asset to the time when the asset, although physically able to provide a service, ceases to be the lowest-cost alternative to satisfy a particular need. At maximum the economic life equals the physical life of the asset, but obsolescence often ensures that the economic life is shorter than the physical life.

F

Facilities. Non-linear installations of assets that facilitate a process or function. Within a subsystem, assets can be classified as either linear or non-linear. Non-linear assets may be mobile (e.g., vehicles in a fleet) or fixed (e.g., a building), occupy a bound (non-continuous) space, and can be modeled as part of a parent-child hierarchy. Linear assets (e.g., underground pipes) are continuous with linear properties.

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Failure analysis. The process of analyzing a failure to determine its cause and to put measures in place to prevent future problems.

Function. What the owner or user of a physical asset or system wants it to do.

Functional failure. A state in which an asset or system is unable to perform a specific function to a level of performance that is acceptable to its user.

G

Gap analysis. Assessment of the current level of practices in asset management. The gap is the difference between the current practices and the best practices for that organization. Gap analysis is sometimes called as a *maturity assessment*.

General aviation (GA) airports. One of the two categories of civil aviation airports. GA airports cater to all flights other than military and scheduled airline and regular cargo flights, both private and commercial.

Geographic information system (GIS). A computer package that displays a map connected to a database. The package typically combines features (e.g., roads and sewers) from different maps, and can overlay them on the same screen. To qualify as a GIS, the system also should be able to reference a computer database for textual information (e.g., notes and dimensions) regarding the features displayed on the map.

I

Infrastructure assets. Stationary systems forming a network and serving whole communities, and for which the network as a whole is intended to be maintained indefinitely at a particular LOS potential by the continuing replacement and refurbishment of its components. The network may include normally recognized assets as components. Examples of infrastructure assets include roads, water supply, wastewater, stormwater, power supply, flood management, and recreational assets.

Infrastructure management. The discipline of managing infrastructure assets that underpin an economy.

Inventory. In accounting, a record of current assetsthat includes property and equipment owned (e.g., counting parts in stock, value of work in progress, and work completed but not sold). With regard to maintenance, the term *inventory* frequently is used to describe the list of equipment and spare parts held in stock.

Inventory analysis. Involves data collection regarding the products or services being investigated and quantification of relevant material and energy inputs and outputs.

Inventory management. The process by which inventory is controlled. Typically, this includes tracking usage of stock items, optimization of stock levels and control of costs.

ISO (International Organization for Standardization). An organization known for producing and publishing international standards for all subject areas in the world. This organization forms a network between the public and private sectors as it is not a government organization.

Item. A specific asset or part of an asset (e.g., a piece of equipment).

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J

Job. A work assignment, task, or series of related tasks, position classification, or quantity of work.

Job Plan. Instructional and supporting information provided to the person or entity that performs the work. The Job Plan addresses job-specific requirements, such as task descriptions sequenced in steps, permits (including any safety permits), procedures, drawings, and materials.

K

Key performance indicator (KPI). A quantitative measure or qualitative indicator of the performance of the service, efficiency, productivity, or cost-effectiveness of an agency, program, or activity. The KPI enables a performance comparison to be made against a standard target or norm for management purposes.

Knowledge management. The deliberate and systematic coordination of an organization's people, technology, processes, and organizational structure to add value through reuse and innovation. This value is achieved through the promotion of creating, sharing, and applying knowledge, and through the communication of valuable lessons learned and best practices.

L

Labor. Physical effort that a person expends to repair, inspect, or deal with a problem.

Large hub airport. A commercial service airport that has at least 10,000 passenger boardings each year and 1% or more of total passenger boardings within the United States during the most current calendar year ending before the start of the current fiscal year.

Level of service (LOS). (1) A measure of the level of asset service, determined by the quality and the quantity of services provided by an asset under consideration. (2) The service quality defined for a particular activity or service area against which service performance may be measured. Service levels usually relate to quality, quantity, reliability, responsiveness, environmental acceptability, and cost.

Life cycle. The life of an asset (or a project, product, system) from its conception through to end-of-life, decommissioning, or disposal.

Life cycle cost (LCC) analysis. Any technique that allows assessment of a given solution, or choice from among alternative solutions, on the basis of all relevant economic consequences over the service life of an asset.

Linear asset. An asset that is maintained in segments (e.g., a road, pipeline, or railroad track). Measurements are made along the linear asset to specify work, monitoring, metering, or placement of signs.

M

Maintainability. The ease and speed with which a maintenance activity can be carried out on an asset. It is a function of equipment design and maintenance task design.

Maintenance. All actions necessary for retaining an asset in, or restoring it to, a specified condition.

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Maintenance engineering. A staff function whose prime responsibility is to ensure that maintenance techniques are effective, that equipment is designed and modified to improve maintainability, that ongoing maintenance problems are investigated, and that appropriate corrective and improvement actions are taken.

Maintenance history. A chronological list of all maintenance activities performed on an asset.

Maintenance management. A method of allocating resources to accomplish predetermined LOS through planning, budgeting, scheduling, executing, and reporting and reviewing maintenance strategies and tactics.

Maintenance task. The maintenance effort necessary for retaining an asset in, or restoring it to, a specified condition.

Maintenance work order. A formal document, usually created in the CMMS, used to identify, control, and record maintenance-related work as it goes through the organization's work flow process.

Maintenance work order system. A means of requesting maintenance service; planning, scheduling, and controlling work; and focusing field data to create information.

Maintenance, repair, and operating supplies (MRO). Items used in support of general O&M (e.g., supplies, spare parts, and consumables used in supporting operations).

Management information system. A computer-based information system that produces routine reports and on-line access to current and historical information needed by managers mainly at the middle and first-line levels.

Management strategies. The operations, maintenance, and capital investment strategies determined for the planning period. These may include changes to maintenance strategies to implement more proactive maintenance, or they could include changes to emergency response plans for specific events.

Management system. A system to establish policy and objectives and to achieve those objectives. The management system of an organization can include a range of subsets that collectively constitute the whole. These subsets usually include a quality management system, a human resource management system, a fiscal management system, an asset management system, an information management system, and an environmental management system.

Manager, asset management services. A position with responsibility for management of the Asset Management Services Team. The team's role is the implementation of the organization's Asset Management Improvement Plan.

Maturity assessment. Evaluation of the degree to which an organization uses recognized best practices and the availability of information required to plan for, and implement, improvements. The ultimate value of maturity assessment can be realized only in the context of an overall process improvement and organizational change program.

Mean time between failures (MTBF). The average length of time between one failure and another for an asset or component.

Mean time to failure. The average length of operating time to failure of a non-repairable asset or component.

Mean time to repair. The average time needed to restore an asset to its full operational capabilities after a failure.

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Measure. A criterion, metric, or means by which a comparison with output can be made.

Metadata. Information about data, including details such as document length, author, dates, and summary.

N

Non-hub airport. Commercial service airports that have more than 10,000 passenger boardings each year but less than 0.05% of total passenger boardings within the United States in the most current calendar year ending before the start of the current fiscal year.

O

Operations and maintenance (O&M). Activities related to the performance of routine, preventive, predictive, scheduled, and unscheduled actions aimed at preventing asset failure or decline with the goal of increasing efficiency, reliability, and safety.

Optimized decision making. A process for considering and prioritizing all options to rectify existing or potential performance failures of assets.

Organizational change. A process of change in an organization as a result of change in the organization's business processes, organizational structure, or culture.

Organizational strategic plan. An overall long-term plan for the organization that is derived from and embodies its vision, mission, values, business policies, stakeholder requirements, and objectives, and the management of its risks.

P

Performance indicator. A qualitative or quantitative measure of service or activity used to compare actual outcome against a standard or other target. Performance measures commonly relate aspects such as safety, responsiveness, cost, comfort, asset performance, reliability, and sustainability.

Performance measurement. A process of assessing progress toward achieving predetermined goals, including information on the efficiency with which resources are transformed into services and outputs, the quality of those outputs, and how well they are delivered.

Performance monitoring. Continuous or periodic quantitative and qualitative assessments of actual performance compared with specific objectives, targets, or standards.

Preventive and/or predictive maintenance (PM). A schedule of planned maintenance actions aimed at the prevention of breakdowns and failures.

Proactive maintenance. Maintenance work that is completed to avoid failures or to identify defects that could lead to failures.

Process improvement. Application of the Plan-Do-Check-Act philosophy to processes to produce positive improvement and meet stakeholders' needs and expectations.

Q

Quality assurance. Implementation of all planned activities within a quality system that can be demonstrated to provide confidence that a product or service will fulfill requirements for quality.

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Quality control. Operational techniques and activities that sustain a quality of product or service that will satisfy specific needs.

Quality management system. A formalized system that documents the structure, responsibilities, and procedures required to achieve effective quality management.

R

Rehabilitation. A project to rebuild or replace parts or components of an asset to restore it to a required functional condition and extend its life, which may incorporate some modification. Rehabilitation henerally involves repairing the asset using available techniques and standards to restore it to its original LOS without resorting to significant upgrading or renewal.

Reliability. The probability that a system performs a specified function or mission under given conditions for a prescribed time.

Renewal. A project to upgrade, refurbish, or replace an existing asset with an asset of equivalent capacity or performance capability.

Repair. An action to restore an item to its previous functional condition after failure or damage.

Replacement. Complete replacement of an asset that has reached the end of its asset life, so as to provide a similar (or agreed alternative) LOS.

Requisition. A request for an asset, item, tool, or service. The requested entity can be procured from a vendor, or it can be acquired by an inter-departmental transfer.

Return on investment (ROI). A ratio that compares average annual savings to initial investment value.

Risk. The likelihood of a specific (usually unwanted) future outcome, combined with consequences that will follow in a particular context. Usually the outcome is defined as an unwanted event, but it could be an unintended benefit, which is an opportunity to add value.

Risk management. The systematic process of identifying and managing risks and opportunities for a project or business.

Risk profile. A description of risk exposure that includes how risk is evaluated and identifies at-risk assets.

Run-to-failure (RTF) approach. An approach in which no routine maintenance tasks are scheduled or performed on the asset.

S

Service life. The period of time after installation during which a building or its parts meets or exceeds the performance requirements.

Service life planning. The preparation of the brief and design for a building and its parts to achieve the desired design life; also called service life design.

Small hub airport. Commercial service airports that have more than 10,000 passenger boardings each year and have at least 0.05% but less than 0.25% of total passenger boardings within the United States in the most current calendar year ending before the start of the current fiscal year.

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Stakeholders. All persons or groups who have an interest in an organization's performance, success, or the impact of its activities. We are all stakeholders, and we may experience our stakeholder status in one or more ways (e.g., as members of the general public, on-site and off-site employees, travelers, and/or bystanders affected by the activities around us).

Strategic plan. A plan containing the long-term goals and strategies of an organization. Strategic plans have a strong external focus, cover major portions of the organization, and identify major targets, actions and resource allocations relating to the long-term survival, value, and growth of the organization.

Supervisory control and data acquisition (SCADA). A computerized system often used to collect real-time maintenance information for monitoring and control of assets.

Suppliers. Construction providers, which may include designers, consultants, building contractors, and manufacturers of components.

T

Triple bottom line accounting. An expansion of the traditional account reporting framework to take into account ecological and social factors in addition to financial factors.

U

UNIFORMAT II Code. A format for classifying building elements and related site work. Elements of the code are major building components that perform a given function, regardless of the design specification, construction method, or materials used.

Uptime. The time during which an asset is providing a service.

V

Value. The final output value that results from a business process such as manufacturing a product or delivering a service, or from a personal activity such as running a home. Value has three essential characteristics: (1) it can only be defined by the ultimate customer or user; (2) it is created by the provider; and (3) its creation is why providers exist (from the customer's point of view).

W

Warehouse. A storage facility where materials and spare parts are stocked.

Work order. An instruction with the accompanying details and information needed to perform a defined scope of work.

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APPENDIX A

Case Studies

Airports

Port Authority of New York and New Jersey

About PANYNJ's Aviation

The Aviation Department of the Port Authority of New York and New Jersey (PANYNJ) is considered a regional powerhouse that generates in excess of $2.5 billion per year. It employs 1,257 full-time and 259 temporary staff positions. Its Aviation Department helps safely, securely, and efficiently connect nearly 125 million domestic and international passengers to their work locales, leisure destinations, and families each year.

PANYNJ is an authority that is co-managed by the states of New York and New Jersey. PANYNJ's Aviation Department includes six airports: John F. Kennedy International Airport (JFK), Newark Liberty International Airport (EWR), LaGuardia Airport (LGA), Stewart International Airport (SWF), Teterboro Airport (TEB), and Atlantic City International Airport (ACY). Figure A-1 presents the Aviation Department's organization chart.

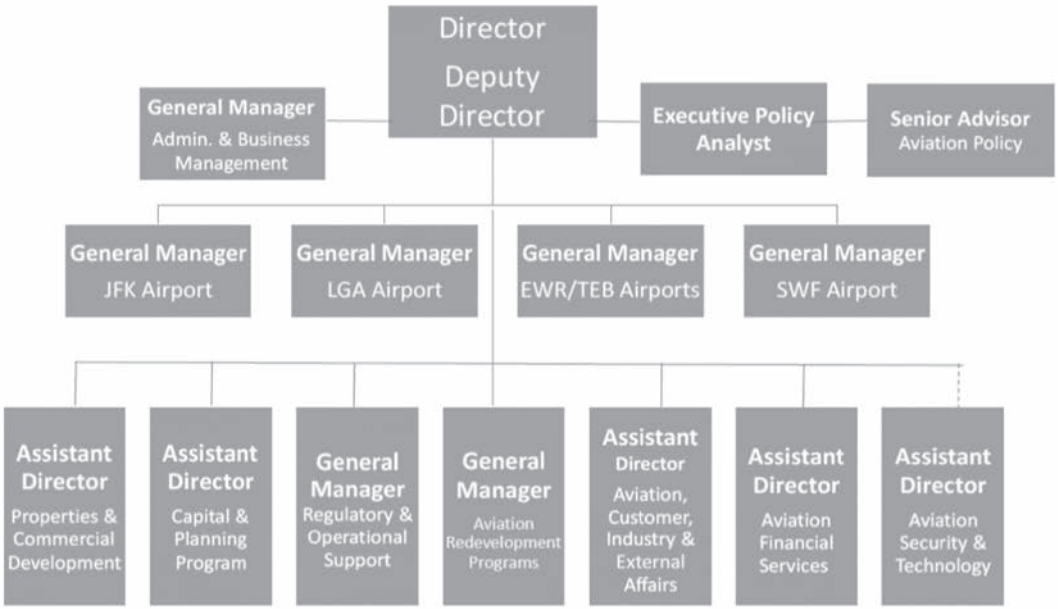


Figure A-1. PANYNJ Aviation Department organizational chart.

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JFK serves as the region's chief international gateway to the world (60 million passengers in 2016), whereas LGA serves as the premier short haul-facility (30 million passengers in 2016). SWF was the first major transportation facility to be newly acquired by PANYNJ in more than a generation. Unlike PANYNJ's other airports, which face significant growth constraints, SWF offers the potential to increase capacity and augment service already offered by the major airports in the New York–New Jersey region. EWR serves lucrative national and international markets (40 million passengers in 2016), whereas TEB serves general aviation clients and is a highly valued reliever airport to the three commercial facilities. In 2013, PANYNJ entered into a management services agreement with the South Jersey Transportation Authority to assist it in managing ACY.

The Aviation Department's 2017–2019 business goals and strategic initiatives are closely aligned with PANYNJ's strategic priorities and aim to deliver safe, secure, and efficient facilities while enhancing its competitiveness, investing in world-class infrastructure, and improving customer experience and business partner satisfaction, among other goals. Its strategic initiatives are designed to improve airport safety, develop its workforce, improve maintenance and development of core facility assets, and advance major redevelopment projects.

The vision statement for PANYNJ's Aviation Department is to be a "world-class integrated airport system that meets the demand for air transportation over the next 30 years."

The department's mission is to:

- Ensure the safety, security, and efficiency of our airport system to best serve all our customers
- Build infrastructure and provide services to meet the demand for air transportation of people, goods and services
- Facilitate job creation and economic growth to enhance the quality of life for those who live, work, and visit the New York/New Jersey region

While Aviation's initiatives are critical to meeting its near-term and long-term objectives, the Aviation Department at PANYNJ is also facing an unprecedented set of risks and challenges from within and beyond PANYNJ. These risks and challenges span the arenas of workforce development, federal aviation regulations, and infrastructure maintenance. Some key issues are described as follows.

- By 2019, 33% of represented staff will be eligible to retire and 40% of non-represented staff will be eligible to retire, many of whom currently fill mission-critical positions.
- Although the FAA recently lowered EWR from a Level 3, slot-controlled airport to a Level 2 airport, the agency continues to enforce slot restrictions at JFK and LGA, effectively limiting growth at those two airports.
- The failure of Congress to raise the Passenger Facility Charge rate above its current ceiling of $4.50 per passenger in the latest FAA reauthorization proceedings limits Aviation's ability to upgrade and enhance critical infrastructure.
- The ability of the airports to expand is very minimal because the properties are landlocked.
- Airports, especially in the New York and New Jersey area, face greater security needs in light of terrorist activity within and around major international airports, such as the March 2016 attacks on the Brussels Airport.
- Airports need to plan for adoption of a comprehensive Safety Management System to meet future FAA safety regulations.
- PANYNJ airports must adapt and respond to technological and business changes in the market for airport ground access, such as the rise of Lyft and Uber car-sharing services, which have led to declining short-term transactions at parking facilities.
- A growing list of aging assets will need replacement, and ongoing and increasing predictive and/or preventive maintenance (PM) routines are required to maintain existing assets.

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To address these challenges and begin taking advantage of potential opportunities, PANYNJ has put a business plan together that included high priority initiatives. Some of these initiatives include:

- Identification of current staff able to assume mission-critical positions over the next few years, and ongoing recruitment to fill mission-critical positions from within and outside the agency.
- Procurement of a state-of-the-art parking revenue management control system that will provide incremental revenue generation through dynamic pricing models based on both historical and real-time parking usage.
- A number of complex, long-term planning and redevelopment efforts aimed at modernizing the airport system, such as:
  - Development of a long-term strategic technology plan with a principle focus on fiber management, airside and landside networks, and connectivity
  - LGA redevelopment (development of LGA into a world-class, 21st century airport)
  - JFK redevelopment 2.0 (investments in infrastructure to accommodate forecasted growth, modernize aging facilities, and enhance security and service)
  - EWR Terminal A redevelopment (construction of a replacement terminal with room for expansion to meet future air passenger demand)

Maintenance Organization

Asset management and maintenance across PANYNJ is a major part of the strategic plan. The maintenance business units at each airport are working toward accomplishing the following goals.

- Meet current maintenance challenges
- Maintain Level A customer service standards
- Operate a safe and reliable airport system

The main challenges that are facing maintenance are the following:

- Finding and retaining qualified labor
- Maintaining existing infrastructure and expanding facilities
- Ensuring data collection accuracy and retention to assist in work planning and garnering accurate feedback from the field about assets and failure rates
- Managing budget constraints to allow for continued maintenance of additional assets without the addition of new staff, and balancing contract services with in-house staff to accomplish workload
- Investing in the upkeep of the existing maintenance management information system (MMIS) through training people and maintaining the system
- Implementing a succession plan to manage knowledge in the face of pending retirements

On the other hand, PANYNJ is actively working toward improving its asset management by:

- Involving the Chief Operations Officer (COO) in recognizing that MMIS needs to be changed to a modern system
- Initiating implementation of the CMMS for electrical assets serving the airfield
- Adhering to all regulatory compliance requirements to become a force of change
- Thinking about the future by building a robust CMMS program that would translate into a robust enterprise asset management (EAM) system in the future

The maintenance program at PANYNJ can be described as a mixture of proactive and reactive based on the asset class. The program is mainly driven by regulatory requirements. In general, it is divided between 60% PM activities and 40% CM activities.

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Asset Management Program at PANYNJ

The CMMS at PANYNJ was implemented in the early 1990s. A committee made up of staff from the different line department and business units took part in the evaluation and selection of software. The main drivers behind implementing the CMMS were easy access, analysis, and reporting of data; electronic documentation; standardized electronic management of assets airport-wide; and better management of available resources using schedulers and planners.

System Requirements

PANYNJ is transitioning their CMMS application. The system requirements discussion in this section covers both the existing and future systems. The chosen CMMS must have the ability to process work orders, manage materials and inventory, support PM, use cost controls, interface with other software, and produce required reports to make informed maintenance decisions.

The functional requirements of the CMMS selection criteria include:

- Track asset information such as cost center, department, location, etc.
- Allow for cost and repair information to be accessed at multiple asset levels
- Save work orders to the asset history as one line of information or full detail
- Access all information in the asset history by sorting or building ad hoc reports
- Provide a bill of materials for each asset
- Provide user-defined access screens for storing information relative to asset items/types, which also can be selected and sorted by reports

The PM requirements considered in selecting the CMMS include the ability to:

- Schedule PM actions (PMs) by calendar
- Allow for unlimited numbers of PMs per asset
- Schedule PMs for any specific date and/or day of the week
- Forecast labor resources, material requirements, or special tools for PMs due for any specific time period
- Allow for multiple crafts scheduled on a PM
- Combine all PMs that are due for an asset manually
- Generate PM work orders a multiple intervals including, but not limited to, daily, weekly, monthly, annually, and at user-defined intervals
- Provide system reports identifying overdue PMs, incomplete PMs, or results of PM inspections

The work order requirements that were originally considered when selecting the CMMS included the ability to:

- Track information at the work order task/step level for planned versus actuals, for labor, materials, and tools
- Produce a report of all work orders, which can be sorted by their current status in the work order workflow
- Track work order backlog by craft, crew, department, planner, supervisor, or other defined categories
- Update work order status either manually or automatically via workflow configurations
- Allow user to access stores, personnel, tools, contractors, etc., without leaving the work order module
- Create notifications and escalations and send these items to personnel internally or externally

The Maintenance, Repair, and Operating Supplies (MRO) inventory and procurement requirements that were originally considered in selecting a CMMS include the ability to:

- Integrate inventory system with the PANYNJ's own purchasing system
- Automatically track material and supply costs to the work order

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Additional system requirements considered for MRO include the ability to:

- Generate a spare parts reorder report when the quantity on hand drops below a minimum level
- Manage multiple warehouses
- Generate a purchase requisition once a reorder point is reached
- Notify the planner automatically when a part is received
- Automatically track materials and supply costs to the work order on an itemized basis

The following management reporting capabilities were originally considered when selecting a CMMS:

- Produces reports at a user-defined interval, e.g., daily, weekly, monthly, or yearly
- Produces reports – summary reports, lists of information, exception reports, or a combination
- Uses a report writer in addition to standard system reports
- Designs reports intended for use by system managers, maintenance managers/supervisors, and other maintenance personnel

Management reporting requirements that were additionally considered included:

- Has a maintenance budget reporting module
- Reports and forecasts asset downtime
- Tracks asset downtime costs
- Tracks asset failures, including mean time between failures

Implementation requirements that were considered in selecting the CMMS vendor included:

- The vendor performing software implementation, hardware installation, data gathering, data loading, and user training
- The vendor having fully documented installation plans
- The vendor having software experts, maintenance experts, training experts, and airport operations subject matter experts on staff
- The system needed some customization

Maintenance software analysis requirements that were additionally considered when selecting the CMMS included the ability to:

- Be operated by maintenance personnel
- Apply necessary modules to meet the organization's needs (PMs, work orders, inventory, cost tracking, scheduler, etc.)
- Provide single, concurrent user capabilities
- Closely integrate all system modules
- Archive files and retrieve files for reports spanning long time periods, and other legacy system functions
- Integrate its functions with: payroll, accounts payable, time keeping, etc.

The system components relevant to maintenance must be able to be updated periodically by the vendor, and the system must be able to display analytical information in a graphic format.

CMMS Implementation

When implementing its CMMS, PANYNJ followed an implementation flow consisting of the following steps:

- Establish steering committee for a CMMS
- Establish an implementation team

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- Establish site project teams for each airport
- Promote kick-off meeting for site and project teams
- Install the CMMS software (hardware was in-house)
- Initialize the CMMS software (create multiple system environments, including development, test, train, and production)
- Provide initial training for the site and project teams
- Define organizational procedures for the CMMS use
- Develop written the CMMS procedures
- Restructure the organization/business processes
- Use data formats to start the data collection process
- Begin data entry process
- Begin user training on the CMMS
- Begin using work order system
- Monitor all usage of the CMMS
- Audit the CMMS for 30 days
- Continue regulatory and criticality audits

As expansion of the implementation of the new system continues, the plan is to establish inventory management for materials and supplies, to expand the asset list to include those other than electrical airfield assets maintained by PANYNJ employees, and to also include assets that are owned by PANYNJ and maintained by others.

During the CMMS implementation, the team faced the following challenges:

- Properly defining and documenting the system user requirements
- Assessing vendor's qualifications
- Software testing
- Defining an implementation plan (at the agency level)
- Obtaining sufficient training and/or documentation
- Correctly estimating time needed to collect and load data

Most data entered into the CMMS with the electrical airfield asset implementation was available in MMIS. The CMMS currently tracks labor hours and costs, but in the new system additional data will be collected and input; specifically, PANYNJ assets maintained by others, and, potentially, PANYNJ tenant buildings.

The benefits of having a CMMS implemented at PANYNJ include better insight and management of:

- Regulatory compliance
- Asset criticality
- Capital project management
- Data quality
- Accountability
- Troubleshooting and trending
- Information granularity (down to the asset level)
- Identification of required resources
- Implementation of best management practices

Aviation utilizes a CMMS 100% of the time to allow the department to charge activities back to the airlines; however, contractors' staff only utilize the CMMS 30% of the time. The reason for this low percentage is the lack of linkage between the systems. The plan is to break down these silos in a new CMMS and utilize it more.

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Maintenance and inventory management is monitored but not managed through a CMMS. The plan is to improve this capability on the system once the new CMMS is in place.

Asset Classes in CMMS

The asset classes added to the CMMS at PANYNJ were based on the criticality of the assets, regulatory compliance, safety, and ensuring full transparency on each asset.

Table A-1 shows the assets that are managed through the CMMS. The future asset classes to be added to the CMMS are presented in Table A-2.

Table A-1. Asset classes in CMMS.

Asset Class	Asset Class	Asset Class
- Terminal buildings	- HVAC	- Fleet
- Runway lighting	- Electric	- Airfield signage
- Facilities	- Plumbing	- Utility – Treatment facilities
- Hangars	- Deicing runway chemicals	- Fire protection
- Parking garages	- Tide gates	- Airfield lighting
- Pavement (Airfield)	- Signage	- Environmental

Table A-2. Asset classes to be added to CMMS.

Asset Class	Asset Class	Asset Class
- Jet bridges	- Elevators	- Buried piping
- Escalators/moving walkways	- Baggage handling system	- Traffic signals

Data

Data points are collected based on asset class and stored on a server. In the current system, data is mainly used for labor reporting and asset failure frequency. In the new CMMS, parts availability and planned and scheduled maintenance work orders will be added to the collected data.

Data quality assurance is accomplished in several ways. On a quarterly basis, airfield electrical data is reviewed against regulatory requirements as they relate to airport certification by each airport's certification manager. Fire protection and life safety asset maintenance is completed in accordance with the National Fire Protection Association's (NFPA) code and is reviewed periodically by the PANYNJ's Risk Management Department to ensure compliance. Environmental regulatory work is reviewed regularly by each facility's environmental manager and is frequently inspected by state and federal regulatory agencies. Finally, all non-regulatory data is managed by unit supervisors and maintenance management staff at various frequencies.

Currently, minimal data analysis is being conducted. Except for airfield electrical assets, for which work order completion rates are being reported monthly, other asset types have simplistic, non-categorized, quarterly work order completion rates being reported. Once the new CMMS is in place, however, the plan is to employ the system's capability for more sophisticated, automated analysis for all asset and work order types.

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CMMS Integration

The CMMS at PANYNJ is only minimally integrated with other systems. As part of implementing the new CMMS, more systems will be integrated within the CMMS. Table A-3 shows the current and future systems integration with the CMMS. PANYNJ chose these particular systems because they were considered of high value.

The main operational drivers behind the systems integration are to capture actual cost, increase efficiency, have its CMMS ready for EAM, and promote easy visibility of data.

Table A-3. CMMS integration with other systems.

Current	Future
- Resources management	- Safety management
- Scheduling	- Wildlife hazard management plan
- Human resources	- Pavement management
- Airfield lighting	- Incident management
- Financial software	- Fleet management
- Time and labor	- Property management
	- Fueling
	- Storeroom
	- Geographic Information/GIS
	- Inspection application (Part 139)
	- Fire protection

The CMMS integration with other systems has proven to have an impact upon the different activities at PANYNJ by providing better planning of projects (capital and operating) and providing reliable data to inform budgets and maintenance decision making.

Performance Measures

PANYNJ utilizes performance measures to report and manage the condition of assets on behalf of the various business units. Maintenance managers and the Assistant Director for Capital developed the measures. Some of these measures are used internally within Maintenance, some are for regulatory reporting, and others are reported to the COO and Aviation.

These measures are reported in a report format on monthly basis for the business unit and on a quarterly basis for the COO. In the new CMMS, however, the data will be available in Star Center and in report format.

Some of the measures that are measured and reported utilizing the CMMS data are the following:

- PM work order performance
- Distribution of hours by work order type
- Facility PM work order performance by unit/trade (Figure A-2 illustrates the reporting for this measures in terms of actual PM hours and scheduled PM work orders for electrical, mechanical, structural, and environmental.)
- FAR 139 compliance for the applicable asset classifications identified in FAA advisory circular 150/5340-26C “Maintenance of Airport Visual Aid Facilities”

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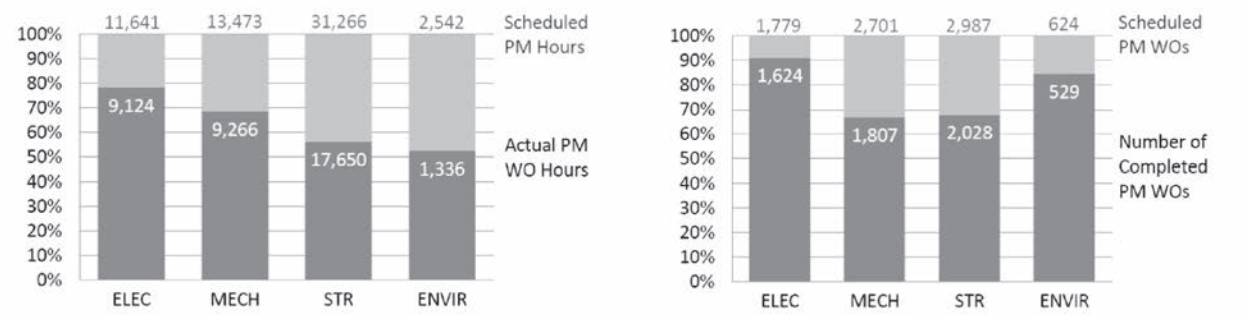


Figure A-2. Facility PM work order performance by unit/trade.

Decision Making

The CMMS data is not used on regular basis because of the quality and usefulness of existing data. As part of the CMMS upgrade, however, the plan is to fully utilize the generated data and information to make robust and informed maintenance decisions. Additionally, the data will be used to verify progressive return on investment per asset class.

The list of reports produced by the CMMS is not extensive. The list of current reports includes:

- Work order priority analysis
- Supervisor work order workflow performance
- Work order cost report (ad hoc basis)
- Completed work order performance
- Work order backlog summary
- Preventive maintenance forecast
- Preventive maintenance overdue

The future CMMS will add the following reports to the list above:

- Planner performance
- Scheduling activities
- Inventory
- Supervisor/skilled trade work order performance
- Asset repair history
- Asset maintenance costs history
- Asset maintenance cost exception (actual vs. planned)
- Safety work order backlog (regulatory/safety compliance)
- Stock item usage report
- Work order waiting report

The information produced by the CMMS is reported to leadership through:

- KPIs (high/low priority and regulatory compliance)
- Periodic reports
- Briefings

In the future, the plan is to add dashboards to the suite of reporting methods to leadership.

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Salt Lake City International Airport

About SLC

SLC is 15 minutes from the center of downtown Salt Lake City and serves more than 23 million passengers each year. SLC is the 25th busiest airport in North America and the 85th busiest airport in the world (2016).

Seven airlines and their affiliates operate 330 flights daily to more than 91 non-stop destinations, including international non-stop service to Canada, France, Mexico, and the Netherlands. SLC is a hub for Delta Air Lines, which operates flights that make up 72 percent of the airport's total market share.

SLC covers 7,697 acres and has four runways, including three carrier runways and one general aviation runway. The current passenger facilities consist of three passenger terminals with five airside concourses and 82 aircraft gates.

SLC is building a new terminal complex to better serve its passengers. The current airport terminal was built in 1961, and the airport has since grown into a hub airport with many flights that arrive and depart simultaneously. Its security needs have also changed significantly since 1961, as have earthquake safety standards.

The new SLC construction project broke ground in 2014, and the first phase of the $2.9 billion project is scheduled to be completed in 2020. The three-story terminal and concourse will eventually replace 29 outdated structures that are connected, including three separate terminals and five concourses.

Architectural firm HOK designed the 1.7 million square foot terminal to be convenient, inspiring, and sustainable. The new design celebrates Utah's natural beauty and reputation as an outdoor recreation hub. Floor-to-ceiling glass is used to provide expansive views to the airfield and mountains beyond and is intended to create an uplifting ambiance. The facility's centerpiece will be a soaring interior space called the Canyon, which will house security screening areas, shopping, and dining facilities.

The new facility is designed to achieve a Leadership in Energy and Environmental Design (LEED) Gold certification as well as to incorporate a sustainable design that minimizes the airport's environmental footprint. The efficient configuration of the terminal and gate locations will reduce fuel use and aircraft emissions, while efficient mechanical and lighting systems will provide additional energy savings. Plus, airlines plan to convert all of their airport ground equipment to alternatively fueled vehicles.

These changes and upgrades are part of the airport's strategic plan to improve the services provided to the passengers, airlines, freight services, concessions, and the city-at-large to become one of the best airports in the world. The planned upgrades will also improve operations and maintenance (O&M) activities to drive reliability, robustness, and resiliency to the airport assets.

The challenges that SLC is facing include:

- Continuing to operate the airport facilities while constructing the new airport terminal
- Dealing with an aging workforce and facilities
- Changes in leadership
- Budgetary constraints
- Acquiring and maintaining a professional workforce
- Talent planning and up-to-date training on new equipment
- Aligning the workforce's talent with airport's strategies and goals
- Meeting stakeholder expectations (travelers, airlines, concessions)
- Having the infrastructure in place to handle the increasing number of passengers

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Maintenance Organization

The Maintenance Business Unit at SLC is part of the Department of Airports in Salt Lake City (Figure A-3). The main challenges the SLC Maintenance Department faces are:

- Scheduling of projects
- Budgetary constraints
- Supporting new terminal construction and commissioning while maintaining the old
- Dealing with an aging CMMS
- Acquiring and maintaining a professional workforce
- Aligning the workforce's talent with airport's strategies and goals
- Training on new equipment and their controls

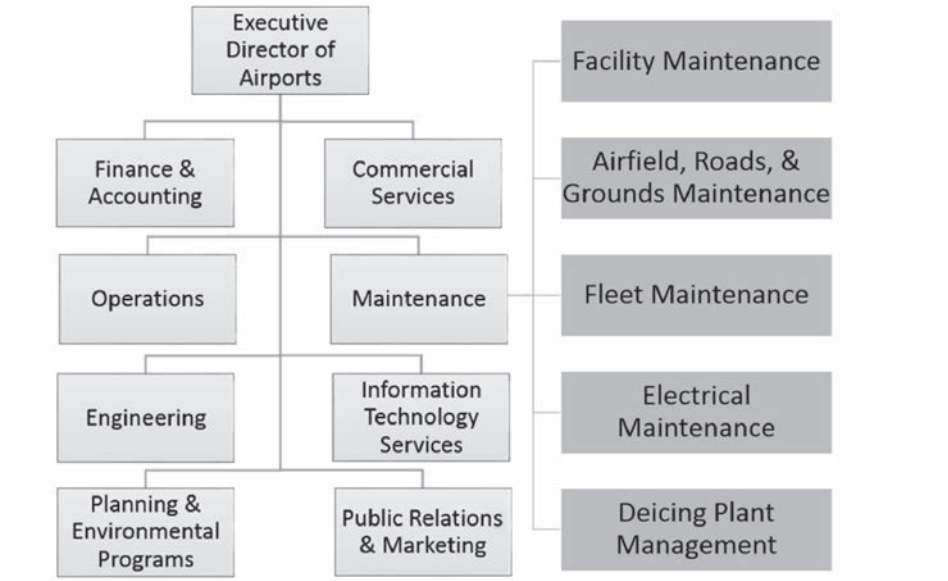


Figure A-3. SLC Organization with breakdown of maintenance structure.

The main concerns that Maintenance has with its existing maintenance management software include:

- Reports are limited (e.g., Maintenance cannot run reports for individual labor, and reports are bulky and cannot be formatted).
- The system is not user-friendly for work orders. It takes a long time to enter data, does not track user input and changes, and does not allow photographs to be attached to individual work orders.
- The system has limited security features.
- The system is not web-based and does not allow real-time updates.
- The system has limited options for customer feedback and tracking repeated maintenance issues.
- System upgrades and updates are time consuming and occur on an individual basis per workstation.
- The system is not GIS-centric (e.g., it does not allow individual assets to be mapped so users can easily find, update, or report work to be done).
- The system is not enterprise-wide. The existing system does not allow the airport to house work order information, warehouse inventory, certain financial aspects, etc. Having an enterprise-wide system would allow better tracking of true maintenance and operations cost.

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Maintenance is working under a strategic plan to meet the following goals:

- Implement maintenance best practices (e.g., add structure to existing asset system, develop business plans, and ISO 55000 standard)
- Tie inventory with the CMMS
- Implement Wi-Fi mobile solutions
- Integrate related asset programs
- Use a paperless system to initiate, develop, track, and report work orders status and results
- Improve on the reporting system
- Adopt clearer communication processes across less organizational layers

The maintenance program at SLC can be described as 65% PM and 35% CM. It should be noted that this ratio is dependent on the types of assets. For example, Fleet Services performs a high amount of PM, whereas Plumbing performs a high amount of CM.

Asset Management Program at SLC

The main driver behind implementing a CMMS was to address the airport's aging facilities, workforce, and retirement issues by using the system to better track and retain institutional knowledge. Its workforce challenges have provided the impetus to improve knowledge management by documenting and automating processes, tracking maintenance, and instituting a reporting system to aid the budget process tracking older equipment.

The CMMS was implemented to achieve many related benefits, such as improving access to critical information; improving quality of information; streamlining business and operational processes; implementing maintenance best practices; and gaining efficiencies in overall asset management.

Implementing the CMMS at SLC was a phased process, initiated with a low cost interim solution until the overall needs were gathered and funding secured, at which time a fully expanded CMMS was ultimately rolled out.

System Requirements

The chosen CMMS has the ability to manage work order processing, PM, cost controls, and required reports to inform maintenance decisions. The specific functional requirements of the CMMS included the ability to:

- Track asset information such as cost center, department, location, etc.
- Allow for cost and repair information to be accessed at multiple asset levels
- Save work orders to the asset history as one line of information or full detail
- Access all information in the asset history by sorting or building ad hoc reports

The PM requirements that were considered in selecting the CMMS included:

- Schedule PMs by calendar, meter readings, operational parameters (real time)
- Allow for unlimited numbers of PMs per asset
- Schedule PMs for any specific date and/or day of the week
- Schedule PMs for assets or facilities
- Generate the PM work orders; daily, weekly, monthly, or at user-defined intervals
- Provide system reports; overdue PMs by number, incomplete PMs, or results of PM inspections mobile/paperless

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The work order requirements that were considered when selecting the CMMS included the ability to:

- Track information at the work order task/step level for planned versus actuals, for labor, materials, tools, contractors, and safety
- Produce a report of all work orders which can be sorted by their current status in the work order workflow
- Track work order backlog by craft, crew, department, planner, supervisor, or other defined categories
- Update work order status either manually or automatically via workflow configurations
- Allow user to access stores, personnel, tools, contractors, etc., w/without leaving the work order module

The following management reporting statements were considered in selecting a CMMS:

- The system produces reports; daily, weekly, monthly, yearly, or any user-defined interval
- Reports produced by the system are summary reports, just lists of information, or exception reports
- Reports were designed to be used by system managers, managers, or maintenance personnel

The implementation requirements that were considered in selecting the CMMS included:

- The vendor performing full turn-key implementation, software implementation, data loading, system training, and user training
- The system requiring some customization

The maintenance software analysis requirements that were considered when selecting the CMMS included the system's ability to:

- Be operated by maintenance personnel
- Apply necessary modules to meet the organization's needs (PMs, work orders)
- Provide single, semi-multi, or multi-user capabilities
- Closely integrate all system modules
- Archive files and retrieve files for reports spanning long time periods
- Update software on a regular basis

CMMS Implementation

When implementing its CMMS, SLC followed an implementation flow consisting of the following steps:

- Establish steering committee for the CMMS (only for end users)
- Establish the site implementation team
- Establish site project teams
- Promote kick-off meeting for site and project teams
- Install the CMMS software (hardware was in-house)
- Initialize the CMMS software (create multiple system environments, including development, test, train, and production)
- Provide initial training for the site and project teams
- Begin data entry process
- Begin user training on the CMMS
- Begin using work order system

During the CMMS implementation, the team faced the following challenges:

- Assessing current and future needs
- Properly documenting the system user requirements or "get user input"

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- Conducting a good search in the CMMS marketplace
- Assessing vendor's qualifications
- Software testing
- Putting an implementation plan together
- Obtaining sufficient training and/or documentation
- Correctly estimating time needed to collect and load data

The Maintenance Department utilizes the CMMS 95% of the time for the planning and scheduling of its maintenance activities and 30% of the time for its maintenance inventory and purchasing.

All SLC assets are incorporated into its CMMS. By implementing the CMMS, the airport has seen benefits in:

- Planning and scheduling maintenance activities
- Safety
- Asset longevity
- Productivity
- Assigning accountability
- Budget planning

Asset Classes in CMMS

SLC added asset classes to its CMMS based on their criticality, their impact on the operational services provided to customers (airlines, travelers, and concessions), regulatory compliance, and safety. Table A-4 shows the assets that are managed through the CMMS, and future asset groups to be added to the CMMS are presented in Table A-5. The asset hierarchy at SLC is a flat one, as shown in Figure A-4.

Table A-4. Asset classes in CMMS.

Asset Class	Asset Class	Asset Class
- Terminal buildings	- HVAC	- Fleet
- Runway lighting	- Electric	- Airfield signage
- Facilities	- Plumbing	- Utility – Treatment facilities (sewage lift)
- Hangars	- Power plant (central utility plant)	- Stormwater infrastructure (pump stations)
- Parking garages	- Baggage handling system	- Fire protection
- Pavement (Airfield)	- Jet bridges	- Deicing system (only structure)
- Airfield lighting	- Fueling system (airlines and vehicles)	

Table A-5. Asset classes to be added to CMMS.

Asset Class	Asset Class	Asset Class
- Glycol recovery	- Elevators	- Security system
- Escalators/Moving walkways	- Chemicals expensing	

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Adding assets to the CMMS is an informal process. Representatives from Maintenance are consulted when a new asset is in the planning phase. They provide their feedback and input during the design, construction, and commissioning phases. The Maintenance Supervisor accepts and approves the commissioning of the asset(s), based on recommendations and information provided from the group. A key element of the process is assigning an Asset ID to the assets listed in the plans.

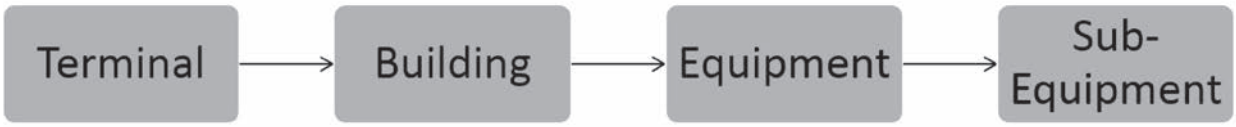


Figure A-4. Horizontal asset hierarchy at SLC.

Data

There are many data points collected through the CMMS. These data points include PM, corrective maintenance, labor, parts, repair history, and meter readers. This data is used to inform decision makers about:

- Labor hours
- Repeat history
- Failures
- Incidents
- Crew size
- Availability of resources
- Condition and maintenance history
- Budgets

The collected data are saved on an SQL server. This data, before being finalized, is randomly checked for any errors. Maintenance does not follow a specific methodology to analyze the data to produce reports; rather, reports are produced on an as-needed basis.

Fleet Services can provide a specific example of how the CMMS data triggers a PM. When a driver fuels a vehicle, mileage is entered into the fuel meter. This entered data will trigger a PM for the vehicle, based upon miles driven.

CMMS Integration

The CMMS is not integrated with other systems at SLC. Systems integration is, however, a part of the new strategic plan and new airport terminal construction. Such integration will be discussed and documented throughout the different business units at the airport. Figure A-5 provides a high-level view of how the CMMS can be integrated into SLC's maintenance activities. SLC's plan is to have the following systems integrated:

- Resource management
- Financial software
- Safety management
- Wildlife hazard management
- Pavement management
- Fleet management
- Time and labor
- Property management

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- Fueling system for vehicles and generators
- Airfield lighting
- Storeroom
- GIS
- Inspection application (Part 139)
- Attic stock tracking

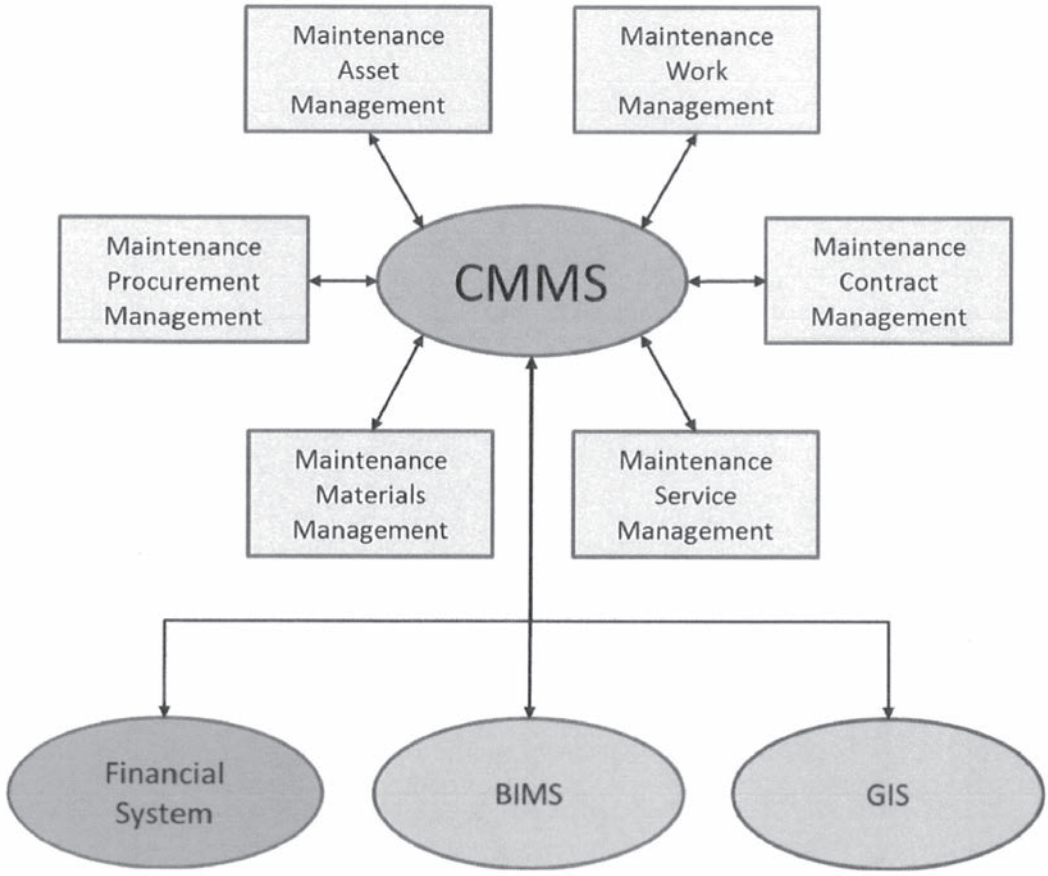


Figure A-5. CMMS integration into SLC's maintenance activities.

Performance Measures

SLC uses performance measures to manage and report the condition of assets across the different business units. Maintenance and other superintendents developed these measures, but Maintenance is the only business unit that tracks and uses the measures to understand its performance.

These measures are reported and communicated to the leadership via e-mails and meetings on a quarterly basis or whenever there is a need.

As an example, Fleet Maintenance has a total of 966 vehicles and pieces of equipment with an asset value of $61 million. Its operating budget is $4.3 million, and the acquisition budget is $4.8 million (FY 2016). Fleet Maintenance maintains diverse vehicles and pieces of equipment, ranging from line lasers and weed whips, to fire trucks and runway snow removal equipment. In addition, Fleet Maintenance maintains the airport's garbage trucks, glycol recovery vehicles, paint strippers, airfield sweepers, deicer trucks, stationary and portable generators, passenger ramps, backhoes, and graders.

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SLC adopts new measures by reviewing industry standards and learning from other airports. An example of measures for Fleet Services is presented in Table A-6.

Table A-6. Measures for Fleet Services.

Measure	Recorded Value	Industry Standard
Vehicle In-Commission (Degree to which fleet units are available for fleet customers)	92.7%	>90%
24-hour Turnaround Time (How quickly vehicles are returned to service)	85%	>70%
Shop Return Rate (Fleet customer returns to shop with same issue without specific time period)	<1%	<1%
Billable Hours (Productivity of technicians in regards to direct vs. indirect labor)	70%	70%
PM Performance Rate (Percentage of PM done on or before scheduled date)	96.3%	--

Decision Making

Maintenance uses the CMMS on regular basis to make robust and informed decisions for its maintenance practices. Reports from the CMMS assist in increasing wrench time and reducing downtime for maintenance activities, which will positively impact the assets operations and improve on customer service. These reports include:

- Work order costs report
- Completed work order performance
- Work order backlog summary
- Work order waiting reports
- Asset repair history
- Asset maintenance costs repair
- Preventive maintenance overdue report

The information produced by the CMMS is reported to leadership through:

- KPIs
- Periodic reports
- Briefings

Seattle-Tacoma International

About SEA

SEA is the largest airport in the Pacific Northwest Region of the United States. It is the fifth fastest growing airport in the world. In 2016, SEA served over 45 million passengers, which makes it the country's ninth largest airport. In 2014, Delta established a hub at SEA for its Trans-Pacific services, which was one of the key contributors to its recent growth.

As part of its "Century Agenda" goals, SEA plans to become the West Coast "Gateway of Choice" for international travel, double the number of its international flights and destinations, and meet the region's

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air transportation needs for the next 25 years through cost-effective expansion of domestic and international passenger and cargo services.

SEA is one of the pioneer airports in pursuing and adopting sustainable practices. It is committed to sustainable growth that balances environmental, economic, and social outcomes. SEA is also utilizing sustainability goals and objectives as the basis for identifying airport development alternatives and developing criteria to evaluate such alternatives. This is driving SEA's efforts to evaluate and construct new and renovate existing buildings and facilities in a sustainable manner.

SEA is the first airport to implement an avian radar system that provides around-the-clock monitoring of wildlife activity across the airfield. This system was designed and implemented to reduce and protect wildlife activity on and near the field.

The challenges that SEA is navigating to achieve its strategic plan are:

- Meeting stakeholders' growth demands
- Capacity constraints at terminals and facilities
- Managing the passenger volume at each gate, since SEA has some of the highest volume in the nation

Maintenance Organization

The Aviation Maintenance Department at SEA is one of the few such units at airports with vision and mission statements and guiding principles. The unit's vision statement is as follows:

"As a strategic aviation business partner, we strive to minimize the long-term total cost of ownership of airport facilities and systems. We are skillful at continuous change, and value collaborating with our internal and external customers.

We focus on maintainability, safety, flexibility, and innovation to operate and maintain our world-class international airport."

Its mission statement is:

"Ensure that all airport facilities, equipment and systems function properly 24 hours a day, 365 days a year."

Aviation Maintenance operates in accordance with the following guiding principles, intended to help them drive maintenance toward having a safe operating environment, available and reliable assets, and minimized cost:

- The development of people, optimal use of equipment and inventory, and our ability to leverage technology
- Reduced equipment and system failures through PM practices
- The utilization of well-defined and understood operating procedures
- The use of KPIs based on industry best practices

Aviation Maintenance recognizes that there are challenges they face as they strive for excellence, including:

- Over-utilization of assets
- Making assets available for maintenance work
- Resource constraints in terms of staffing and availability of highly skilled and specialized technicians
- A preponderance of aging facilities, when funds are normally directed toward new assets
- Obtaining information about new commissioned assets

Despite these challenges, the Aviation Maintenance Department continues to make progress on achieving its goals. Based on recent numbers, the maintenance program spends 83% of its efforts on PM

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and 17% of its efforts on corrective maintenance (CM). Its PM to CM ratio places SEA at the top of comparative airport performance in the nation.

CMMS Selection at SEA

In 1996, SEA decided to improve on its asset management practices by implementing a CMMS. The main driver behind the CMMS implementation was the desire to:

- Identify airport facilities, assets, and systems
- Document maintenance personnel activities
- Better documentation on asset life cycle
- Map and manage the assets
- Identify resources utilized to maintain each asset

System Requirements

The chosen CMMS can manage: work order processes, materials and inventory management, PMs, cost controls at the asset level, integrations with other software applications, and required reports for different organizational levels to facilitate informed decisions. The specific functional requirements of the CMMS included its ability to:

- Track asset information such as cost center, department, location, etc.
- Allow for cost and repair information to be accessed at multiple asset levels
- Save work orders to the asset history as one line of information or full detail
- Access all information in the asset history by sorting or building ad hoc reports
- Provide a bill of materials for each asset
- Provide user-defined screens for storing information relative to asset items/types, which also can be selected and sorted by reports

The PM requirements that were considered in selecting the CMMS included the ability to:

- Schedule PMs by calendar, meter readings, operational parameters (real time)
- Allow for unlimited numbers of PMs per asset
- Allow for multiple crafts scheduled on a PM
- Schedule PMs for any specific date and/or day of the week
- Forecast (labor resources, material requirements, or special tools) for PMs due for any specified time period
- Schedule PMs for assets or facilities
- Combine all PMs that are due for an asset automatically or manually, or not at all
- Generate the PM work orders daily, weekly, monthly, or at user-defined intervals
- Provide system reports (e.g., overdue PMs by number, incomplete PMs, or results of PM inspections)

The work order requirements that were considered when selecting the CMMS included:

- Track information at the work order task/step level for planned versus actuals and for labor, materials, tools, contractors and safety
- Produce a report of all work orders that can be sorted by their current status in the work order workflow
- Track work order backlog by craft, crew, department, planner, supervisor, or other defined categories
- Update work order status either manually or automatically via workflow configurations
- Allow the user to access information on stores, personnel, tools, contractors, etc., without leaving the work order module

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A-20 Guidebook for Advanced Computerized Maintenance Management System Integration at Airports

- Create notifications, escalations and send these items to personnel internally or externally
- Allow for work process flows and assignments based on the end-user role

The MRO inventory and procurement requirements that were considered when selecting the CMMS included:

- Generate a spares reorder report when the quantity on hand drops below a minimum level
- Manage multiple warehouses
- Generate a purchase requisition if a reorder point is reached
- Notify the planner or requestor, when a part is received, which work orders can now be filled
- Produce performance reports for purchasing section that include overdue purchase orders (POs), inactive parts, or inventory valuation
- Automatically track cost to the work order on an itemized basis

The following management reporting statements were considered in selecting a CMMS:

- The system produces reports daily, weekly, monthly, yearly, or at any user-defined interval
- Reports produced by the system are summary reports, just lists of information, or exception reports
- The system has a report writer in addition to standard system reports
- Reports are designed to be used by system managers, managers, or maintenance personnel
- The system reports and forecasts asset downtime
- The system tracks asset downtime costs

The implementation requirements that were considered in selecting the CMMS included:

- The vendor performing full turn-key implementation, software implementation, data loading, system training, and user training
- Vendor has fully documented installation plans
- Vendor having software experts, maintenance experts, and training experts
- The system requiring no customization

The maintenance software analysis requirements that were considered when selecting the CMMS included the system's ability to:

- Be operated by maintenance personnel
- Apply necessary modules to meet the organization's needs (PMs, work orders, inventory, integration, etc.)
- Provide single, semi-multi, or multi-user capabilities
- Closely integrate all system modules and external software applications (Planning/Scheduling tool, Mobile tool, etc.)
- Archive files and retrieve files for reports spanning long time periods
- Update software on a regular basis
- Display analytical information in a graphic format
- Integrate its functions with: payroll, accounts payable, general ledger, time keeping, etc.

CMMS Implementation

When implementing the CMMS, SEA followed an implementation flow consisting of the following steps.

- Establish steering committee for the CMMS
- Establish the site implementation team
- Establish site project teams

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- Promote kick-off meeting for site and project teams
- Install hardware and the CMMS software
- Initialize the CMMS software (create multiple system environments, including development, test, train, and production)
- Provide initial training for the site and project teams
- Define organizational procedures for the CMMS use utilizing training manuals
- Develop written the CMMS procedures within the training manuals
- Restructure the organization/business processes (Note: one of the main organizational changes that was considered during the CMMS implementation was to move storerooms from under Finance to become part of Maintenance)
- Begin developing various data collection formats (templates)
- Use the formats to start data collection process (asset, stores, PMs, personnel, purchasing, accounting)
- Begin data entry process
- Begin user training on the CMMS
- Establish stores stock levels
- Begin using work order system
- Monitor all usage of the CMMS
- Audit the CMMS for 30 days
- Consider having a periodic an audit program of the CMMS (site and enterprise levels)

The CMMS implementation at SEA did not face any particular challenges and proved to be very beneficial to the Aviation Maintenance Department as well as the airport at large. It has allowed Aviation Maintenance to effectively manage the assets and increase their availability, reliability, and uptime. Additionally, it provides meaningful information in support of managing staffing and resources needs. The Aviation Maintenance Department utilizes the CMMS for asset management, work management, materials management, and reporting capabilities.

Asset Classes in CMMS

The asset classes added to the CMMS at SEA were chosen to be part of the system because they are critical to the different operational activities at the airport and because maintenance resources are involved in their maintenance and uptime. The asset classes that are part of the CMMS are presented in Table A-7. The plan is to add the new terminal connecting bridge to the list of assets in the CMMS.

Table A-7. Asset classes in CMMS.

Asset Class	Asset Class	Asset Class
- Terminal buildings	- HVAC	- Security system
- Runway lighting	- Electric	- Signage
- Facilities	- Plumbing	- Utility – treatment facilities (sewage lift)
- Power center	- Elevators	- Stormwater infrastructure
- Parking garages	- Escalators/Moving walkways	- Fire protection
- Pavement	- Jet bridges	- IT
- Baggage handling system	- Rolling stock	- Custodial

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SEA has an excel spreadsheet called "CMMS Data Input Form" that is populated with asset information and then is automatically uploaded to the CMMS. The CMMS has a function that reads data sheets and uploads them to the system. Four groups are responsible for entering information into the spreadsheet: architect; engineer (at 90% design); contractor, and the maintenance shops.

Data

Information about all assets maintained at SEA is collected, input, and managed in the CMMS, then saved on the SQL server. There is a formal process and in place for data collection and documentation. The CMMS Data Input Form collects new asset information, which includes some of the following attributes:

- Equipment ID number
- Equipment description
- Geolocation
- Model number
- Install or purchase date
- End of equipment warranty
- Estimated equipment value
- Expected equipment lifespan
- Classification
- Asset type
- Work group

The assets are organized by a geolocation functional hierarchy. In the future, the asset hierarchy will be updated and individual assets tracked using a UniFormat classification code.

SEA employs a quality assurance (QA) process through which all the collected data and analysis is reviewed by the CMMS Business Analysis team and then validated by the various department leadership teams. The process covers every stage of the data life cycle: collection, retention, analysis, and reporting. SEA decision makers say the most beneficial CMMS information is that related to:

- Asset management
- Work management
- Inventory management

CMMS Integration

The CMMS at SEA is integrated with the following systems: resource management, payroll, planning and scheduling, a mobile CMMS, fueling, baggage handling system, elevators and escalators, and lightweight directory access protocol.

The main drivers behind integrating these systems are to eliminate redundant work, ensure accuracy of information, and increase data visibility. The impact of this integration has been:

- Increased productivity
- Transparent decision making
- Better budget allocations and savings
- Improved maintenance activities

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Performance Measures

SEA senior leadership and maintenance managers are responsible for developing performance measures to measure and evaluate the overall performance of facilities and systems at the airport. These measures are incorporated within their CMMS and updated daily.

Introducing a new performance measure starts by senior leadership and the Maintenance Review Board (Business Process Systems Change Board) identifying a need to measure and track a particular area of performance. Once the measure is created, measuring criteria is developed, and data availability or measurement is identified. The new measure is then put into practice.

Aviation Maintenance is the department that employs most of these measures. The intention, however, is to distribute the information to additional business units' supervisors, managers, and leadership. Table A-8 presents some of the KPIs that Aviation Maintenance utilizes and reports on.

Some of the financial and environmental measures are as follows:

- Estimated 20-year (O&M) Cost ($/square foot)
- Total cost of ownership ($/square foot)
- Age of infrastructure
- Age relative to expected life
- Total energy consumption (kilowatt hours, gallons)
- Energy consumed per passenger or area

Decision Making

The CMMS data is used on regular basis by Aviation Maintenance and leadership at the airport to make robust and informed decisions. Data is presented in reports format. These reports include:

- Scheduling activities
- Inventory
- Procurement
- Work order costs report
- Completed work order performance
- Work order backlog summary
- Asset repair history
- Asset maintenance costs repair
- Stock item usage report
- Work order waiting report
- Preventive maintenance
- Preventive maintenance overdue report

The information produced by the CMMS is reported to leadership through:

- KPIs
- Periodic reports
- Dashboards
- Briefings
- Scorecards

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Table A-8. Maintenance KPIs.

KPI Description	Calculation Type
All completed/closed PM count [Year To Date] (YTD)	Decimal
All canceled/incomplete PM count (YTD)	Decimal
On-time PM completion rate w/target completion on date (YTD)	Percent
Total “tool time” labor hours	Decimal
Total “non-tool time” labor hours (Admin, Leave, Training)	Decimal
Total “non-tool time” leave labor hours	Decimal
Total count of assets	Decimal
Total assets with spare parts – All storerooms	Decimal
Total assets with no spare parts – All storerooms	Decimal
Total count stock keeping unit (SKU) associated to equipment – All storerooms	Decimal
Total assets with no PM or route	Decimal
Total assets with no work orders associated	Decimal
Total assets with no replacement date	Decimal
Distribution Center (DC) inventory valuation	Decimal
Craft storeroom inventory valuation	Decimal
All storerooms inventory valuation	Decimal
Total inventory records	Decimal
Total SKUs on hand	Decimal

Toronto-Pearson International Airport (YYZ)

About YYZ

As a global hub for approximately 440,000 flights a year, YYZ is the largest and busiest airport in Canada. Its network of both regional and international destinations is among the most extensive of any airport, and it is considered the second largest access point into North America, second only to JFK International Airport. By the number of flights, it is the 13th busiest airport in the world; by total passenger traffic, it is the 33rd busiest airport in the world; by international passenger traffic, it is the 23rd busiest airport in the world. It has recently experienced an unprecedented annual compounded growth rate of 8%. In 2016, it served more than 44 million passengers and by 2033, it is expected to surpass 64 million passengers and 570,000 flights annually, as detailed in the 20-year strategic plan.

An independent, non-profit entity called the Greater Toronto Airports Authority (GTAA), manages the airport. The GTAA operates within the south-central region of Ontario on a commercial basis to set airport user fees, which are used to develop, improve, and maintain the facilities. Approximately 1,500 people are employed directly by the GTAA. Approximately 380 of those are employed by the *Airport Planning and Technical Services* (APTS) branch, which provides engineering, maintenance, project management, and technical performance services.

In providing its planning and technical services, the GTAA considers the airport’s long-term growth strategy, as well as the unique structural demands of the airport and airline industries. Key considerations include the infrastructure planning and large-scale development efforts that are required to accommodate future traffic demand, improve passenger flow, and create dwell-time so transit passengers have an opportunity to enjoy the travel experience.

The focus at YYZ is on managing its substantial growth in a sustainable manner, while honoring passenger needs in every initiative it undertakes with the overarching goal of becoming one of the best

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airports in the world for customer service. The vision is to be "The best airport in the world; Making a difference; Connecting the world" within the next 20-years. Three equal principles underscore its three-part vision: customer experience, financial sustainability, and operational excellence.

The clear challenges that YYZ must address to achieve its vision and strategic plan are to:

- Meet stakeholder expectations (travelers, airlines, concessions)
- Have the right amount/type of infrastructure in place to handle the increasing number of passengers

Maintenance Organization

YYZ is a progressive organization that is seeking excellence in its operations and maintenance activities. It is on the path to meet the Reliability-Centered Maintenance (RCM) requirements, comply with ISO 55000 standard requirements, and apply appropriate Internet of Things (IoT) technology to achieve real-time asset health.

The maintenance business unit at YYZ is part of the APTS. Figure A-6 presents the organizational chart of APTS.

Maintenance is a key element to the success of the YYZ 20-year strategic plan. Figure A-7 illustrates the 20-year strategic framework summary. Figure A-8 presents the hierarchy of needs at the airport to achieve operational excellence.



Figure A-6. YYZ planning and technical services leadership organization chart.

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Figure A-7. YYZ's vision, goals, values, and behaviors.

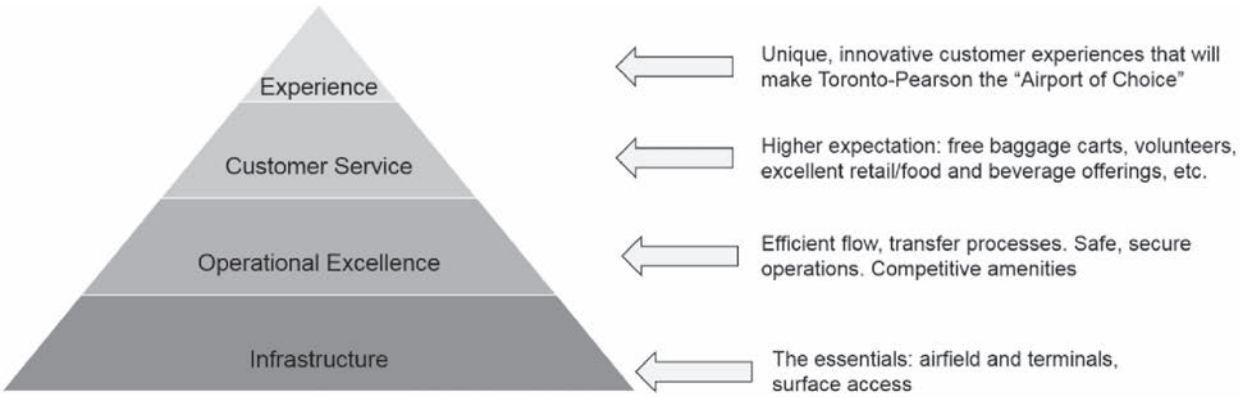


Figure A-8. Hierarchy of needs at the airport.

APTS 5-year strategic focus is to:

- Plan, design, and construct facilities on time and on budget that support aviation growth, customer requirements, and financial sustainability, as defined by YYZ's strategic framework
- Improve asset management system and technical performance to achieve customer service and reliability at the lowest total LCC

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- Build a high performance team that drives business results through s focus of leadership, business acumen and technical competence
- Be recognized as a world leader in environmental stewardship

There are, however, a few challenges that the maintenance business unit is facing while on this journey to excellence. These challenges include:

- Achieving and maintaining high customer service expectations
- Maintaining asset condition, performance, and availability to meet increasing demands
- Increasing asset uptime and reliability, including the availability of assets to meet peak hour volume (passengers, luggage, planes)
- Maintaining and improving operational processes to achieve operational excellence, despite competing capital priorities and the lack of trained and specialized technicians and supervisors and other resources

The maintenance program activities are 30%-40% PM and 60%-70% CM. There are, however, many resources being expended on the jet bridges system. If the jet bridges work is excluded, the PM to CM ratio is equally split. The goal is to achieve the industry standard 80-to-20 PM to CM ratio.

Asset Management Program at YYZ

In the 1970s, Transport Canada decided to computerize its systems. An airport maintenance management system (AMMS) was developed and deployed in all airports in Canada at that time. When Canadian airports were subsequently transferred to private management, they were given AMMS for their own internal management.

From 1996 until recently, this AMMS has been in place and has been managed and developed internally by YYZ. In 2013, due to the limitations and challenges of its AMMS, YYZ began the process of selecting a CMMS (described in the following section) and it is now fully in place. The assets are being phased into the selected CMMS, with the phase-in expected to be completed in 2017.

The desired outcomes from implementing the CMMS are:

- Reliable work order management
- Reduced maintenance costs via accurate predictive maintenance
- Availability of FMEA [Failure modes and effects analysis] and root cause analysis
- Ability to eliminate basic frequency based maintenance
- Better data collection
- Capability to perform real-time asset health analytics

CMMS Selection Process

Before selecting CMMS software, YYZ completed a detailed and comprehensive vetting process meant to ensure that their selection would support operational excellence through improved customer satisfaction, increased asset reliability, and reduced wrench time.

This process began in 2013, when the airport started working with a consultant to assist in mapping-out the journey and developing a comprehensive request for proposals that set forth required functionalities of the system (described in the following section).

The airport involved representatives from business units across the airport in vetting four vendors. The vendors were required to walk the attendants through the entire process and show how the system would function and how it would phase-out the AMMS currently in place. Because of this process, the airport selected its new CMMS because it incorporated best practices and the ability to support the vision of the airport's leadership.

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The selected vendor conducted 50 workshops with the different business units to discuss organizational and process changes to be impacted and how these changes would work to improve processes at the airport. One example of an organizational change was for inventory; after the CMMS implementation, Inventory was moved from Finance to be incorporated within the Asset Management functions.

System Requirements

The chosen CMMS can perform work order processing; materials and inventory management; PM; cost controls at the asset level; and to develop the required reports needed at different organizational levels to support informed decision making. The specific functional requirements of the CMMS included the ability to:

- Track asset information such as cost center, department, location, etc.
- Allow for cost and repair information to be accessed at multiple asset levels
- Save work orders to the asset history as one line of information or full detail
- Access all information in the asset history by sorting or building ad hoc reports
- Provide a bill of materials for each asset
- Provide user-defined screens for storing information relative to asset items/types, which also can be selected and sorted by reports

The PM requirements that were considered in selecting the CMMS included the ability to:

- Schedule PMs by calendar, meter readings, operational parameters (real time)
- Allow for unlimited numbers of PMs per asset
- Allow for multiple crafts to be scheduled on a PM action
- Schedule PMs for any specific date and/or day of the week
- Forecast labor resources, material requirements, or special tools for PMs due for any specified time period
- Schedule PMs for assets or facilities
- Combine all PMs that are due for an asset automatically, manually, or not at all
- Generate the PM work orders daily, weekly, monthly, or at user-defined intervals
- Provide system reports on overdue PMs by number, incomplete PMs, or results of PM inspections

The work order requirements that were considered when selecting the CMMS included the ability to:

- Track information at the work order task/step level for planned versus actuals, for labor, materials, tools, contractors and safety
- Produce a report of all work orders which can be sorted by their current status in the work order workflow
- Track work order backlog by craft, crew, department, planner, supervisor, or other defined categories
- Update work order status either manually or automatically via workflow configurations
- Allow user to access stores, personnel, tools, contractors, etc., without leaving the work order module
- Create notifications and escalations, and send these items to personnel internally or externally

The MRO inventory and procurement requirements that were considered when selecting the CMMS included the ability to:

- Generate a Spares Reorder Report when the quantity on hand drops below a minimum level
- Manage multiple warehouses
- Integrate the inventory system with the vendors' own purchasing system
- Generate a purchase requisition if a reorder point is reached
- Notify the planner, when the part is received, which work orders can now be filled

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- Produce performance reports for purchasing section that include; overdue purchase orders, inactive parts, or inventory valuation
- Automatically track cost to the work order on an itemized basis

The following management reporting capabilities were considered in selecting a CMMS:

- The system produces reports daily, weekly, monthly, yearly, and at any user-defined interval
- The system produces summary reports, just lists of information, or exception reports
- The system has a report writer in addition to standard system reports
- The system produces reports to be used by system managers, managers, or maintenance personnel
- The system has a maintenance budget reporting module
- The system reports and forecasts asset downtime
- The system tracks asset downtime costs

The implementation requirements that were considered in selecting the CMMS included:

- The vendor performing full turn-key implementation, software implementation, data loading, system training, and user training
- The vendor having software experts, maintenance experts, training experts, and airport operations subject matter experts on staff
- The system requiring no customization

The maintenance software analysis requirements that were considered when selecting the CMMS included the system's ability to:

- Be operated by maintenance personnel
- Apply necessary modules to meet the organization's needs (PMs, work orders, inventory, integration, etc.)
- Provide single, semi-multi, or multi-user capabilities
- Closely integrate all system modules
- Archive files and retrieve files for reports spanning long time periods
- Update software on a regular basis
- Integrate its functions with: payroll, accounts payable, general ledger, time keeping, etc.

CMMS Implementation

When implementing its CMMS, YYZ followed an implementation flow consisting of the following steps:

- Establish a steering committee for the CMMS
- Establish the site implementation team
- Establish site project teams
- Promote a kick-off meeting for site and project teams
- Install hardware and the CMMS software
- Initialize the CMMS software (create multiple system environments, including development, test, train, and production)
- Provide initial training for the site and project teams
- Define organizational procedures for the CMMS utilizing training manuals
- Develop written CMMS procedures within the training manuals
- Restructure the organization/business processes (Note: one of the main organizational changes considered during the CMMS implementation was to move storerooms from under Finance to become part of Maintenance)

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A-30 Guidebook for Advanced Computerized Maintenance Management System Integration at Airports

- Begin developing various data collection formats (templates)
- Use the formats to start the data collection process (asset, stores, PMs, personnel, purchasing, accounting)
- Begin the data entry process
- Begin user training on the CMMS
- Establish stores stock levels
- Begin using the work order system
- Monitor all usage of the CMMS
- Audit the CMMS for 30 days

During the CMMS implementation, the team faced the following challenges:

- Assessing current and future needs
- Properly documenting the system user requirements or "get user input"
- Conducting a good search in the CMMS market place
- Assessing vendor's qualifications
- Software testing
- Putting an implementation plan together
- Obtaining sufficient training and/or documentation

The CMMS implementation at YYZ was completed in phases, as groups of assets were entered into the CMMS. The first group consisted of assets related to jet bridges, the baggage handling system, and the fleet, and made up approximately 40% of total assets. After this initial phase was tested, more asset groups were added, representing 80% of total asset groups, followed by the remaining to reach 100% of asset groups over the next couple of years. From this point, an asset group was input into the CMMS, and maintenance activities for that group were planned and scheduled through the system.

Inventory functions are now 100% conducted in the CMMS. Purchase orders are generated in the CMMS and then passed for financial processing.

Since the CMMS implementation has just been initiated, its benefits have not yet been measured. There are, however, some early wins. For example, the visibility of work orders has greatly increased, and process improvement opportunities have been identified.

One of the implementation challenges has been that all maintenance groups use the CMMS, but some vendors who conduct maintenance on certain asset groups do not. The goal is to use the CMMS to conduct all asset work and programs by 2018.

Asset Classes in CMMS

The asset classes added to the CMMS at YYZ were based on the asset hierarchy that was established many years ago in the initial system. Once the CMMS vendor was selected, the cross-functional implementation team held many sessions with the different business groups to identify business requirements of the system. Insights from these sessions were combined with best practices that the vendor offered the project.

The asset classes that were first entered into the system where chosen because they are critical to the different operational activities at the airport and are frequently maintained. The asset classes that are part of the CMMS are presented in Table A-9, and the ones to be added in the future are in Table A-10.

The migration of existing assets from the AMMS to the CMMS application was completed in phases. The first group to be input was composed of assets that are independent from other assets, are easy to manage, are close in proximity, and do not interfere with passenger flow. Fleet assets are such a class. After these assets were migrated, the Asset Management team moved to the next asset class and so on.

For new capital improvement projects, the vendor is required to supply the airport with information about the assets, such as the bill of materials, O&M manual(s), and asset tags (bar coding). These

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requirements are detailed in a Testing, Commissioning, Acceptance, and Turnover document issued by YYZ. This document is part of every project to make sure the vendor understands the project and its related deliverables.

The asset classes are broken down into three hierarchies: (1) Geographic, (2) Functional (asset perspective), and (3) Electrical.

Table A-9. Asset classes in CMMS.

Asset Class	Asset Class	Asset Class
- Terminal buildings	- HVAC	- Fleet
- Runway lighting	- Electric	- Signage
- Facilities	- Plumbing	- Utility – treatment facilities (sewage lift)
- Hangars	- Elevators	- Stormwater infrastructure
- Parking garages	- Escalators/Moving walkways	- Fire protection
- Pavement (airfield)	- Jet bridges	- Baggage handling system

Table A-10. Asset classes to be added to CMMS.

Asset Class	Asset Class	Asset Class
- Power plant	- Fueling system	- Security system
- Deicing facility		

Data

There are many data points that YYZ maintenance staff collect through the CMMS. These data benefit system users and decision makers by providing:

- Asset performance information
- Asset functional location and history
- Effectiveness in planning, scheduling, and execution of workflow
- Asset condition information, which is relayed to technicians and vests decision power upon them
- A platform for planning and execution excellence (parts, tools, equipment, and skills)
- Improved oversight of maintenance activities by providing information at the job site

The collected data goes through a QA/QC process, which starts when the data is collected. The technician makes sure that this data is collected and registered correctly in the system. The functional team reviews it and then the data becomes available for reporting and further analysis, including an annual audit and review. Data is stored on site in two database management systems. Both systems have the capability to store, analyze, and retrieve data efficiently. There is a plan to migrate the data into a Cloud-based system.

CMMS Integration

Many systems are active at YYZ. These systems can stand alone or be integrated in order to inform other systems. On a system-by-system basis, required touchpoints are identified, allowing information to flow in a single direction or bi-directionally. The CMMS collects data and informs other systems in place

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and uses data from other systems (bi-directional). Figure A-9 illustrates the different systems at YYZ that are integrated with the CMMS and how the data flows between them.

There are many benefits that arise from integrating systems, one being reducing the need for users to learn, use, and sign-in to multiple systems. This translates to a reduction in the time required to complete a job, with an increase in system efficiency and data yield.

At YYZ, the CMMS is integrated with the following systems.

- Financial system – brings people information to the CMMS
- Time management system
- Incident management system
- Fuel system
- Building management – mobile product interface for alarms from building management system
- Asset health program

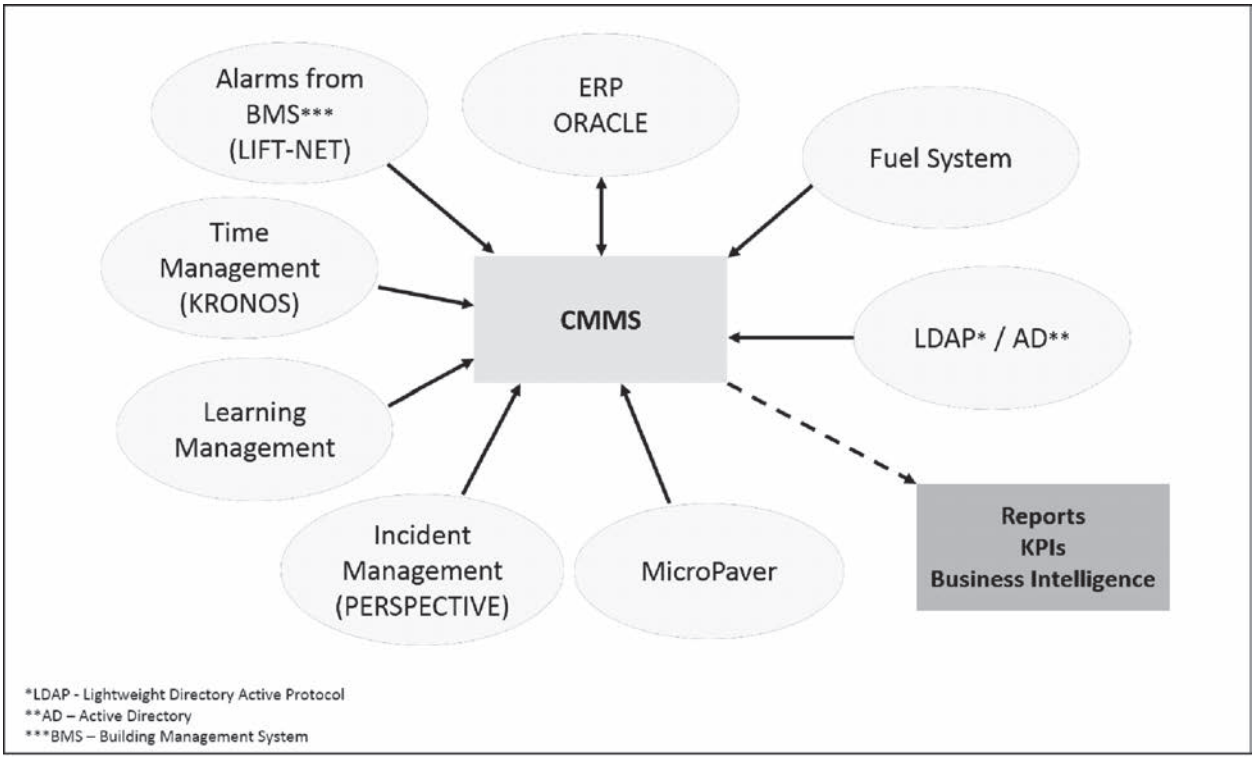


Figure A-9. CMMS integration.

Performance Measurement

YYZ leadership and management have developed performance measures to help them manage and report the condition of their assets to various business units. The information is stored and then shared and presented through enterprise software.

Depending on the business unit and its needs for the measures' information, the reports are produced on a daily, weekly, and monthly basis. Usually, the daily and weekly editions are reserved for technicians and supervisors, whereas the monthly reports are intended for leadership.

Once the measures are reported and reviewed, actions are planned during the weekly status meetings. Actions taken can include performing:

- Deeper dive analysis
- Historical trending

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- Continued measurement
- Engineering and technical intervention
- Process changes (if needed)

Figures A-10 through A-14 present various performance measures that are measured at YYZ related to baggage handling systems, people moving systems, elevators, escalators, and passenger boarding bridges (jet bridges). Figure A-15 shows the calculations equations for these metrics.

Performance Metric

	Target	Dec 2015	YTD
Baggage Service Availability	99.5%	99.8%	99.9%
Degraded Mode % normalized for construction	<4.6%	1.1%	1.5%
Jams/1000 (over 1 min) *	<3.6	3.2	2.9
PBB – Service Availability **	97.4%	98.6%	99.1%
Elevators	96.0%	96.6%	95.9%
Escalators	96.0%	96.0%	95.5%
Moving Walkways	96.0%	97.5%	95.7%
HSW - FEM11 ***	95.0%	98.0%	89.8%
HSW - FEM12	95.0%	95.3%	91.0%
HSW - FEM12 Normalized for GTAA Holds and Approved Outages		98.9%	93.9%
Automated People Mover	99.5%	99.7%	99.3%

* Jams: Baggage unable to move due to being stuck.
** PBB: Passenger Boarding Bridges; ***HSW: High Speed Walk; FEM: Alias of specific unit.

Figure A-10. Key performance metrics during December 2015.

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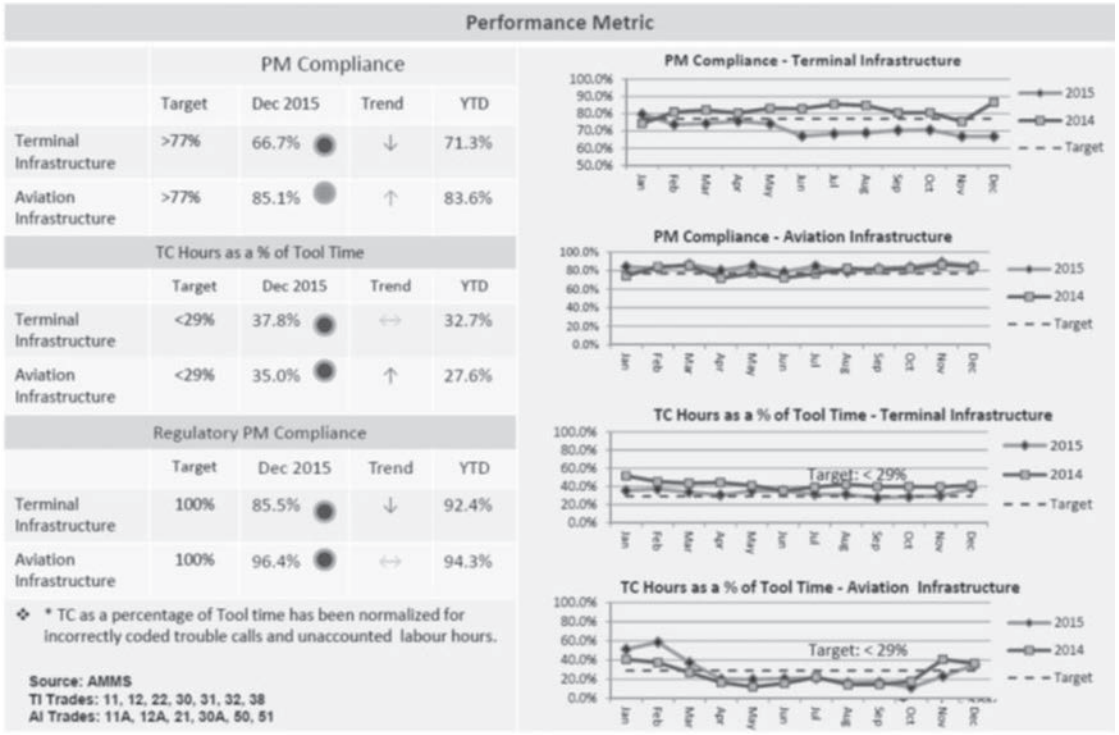
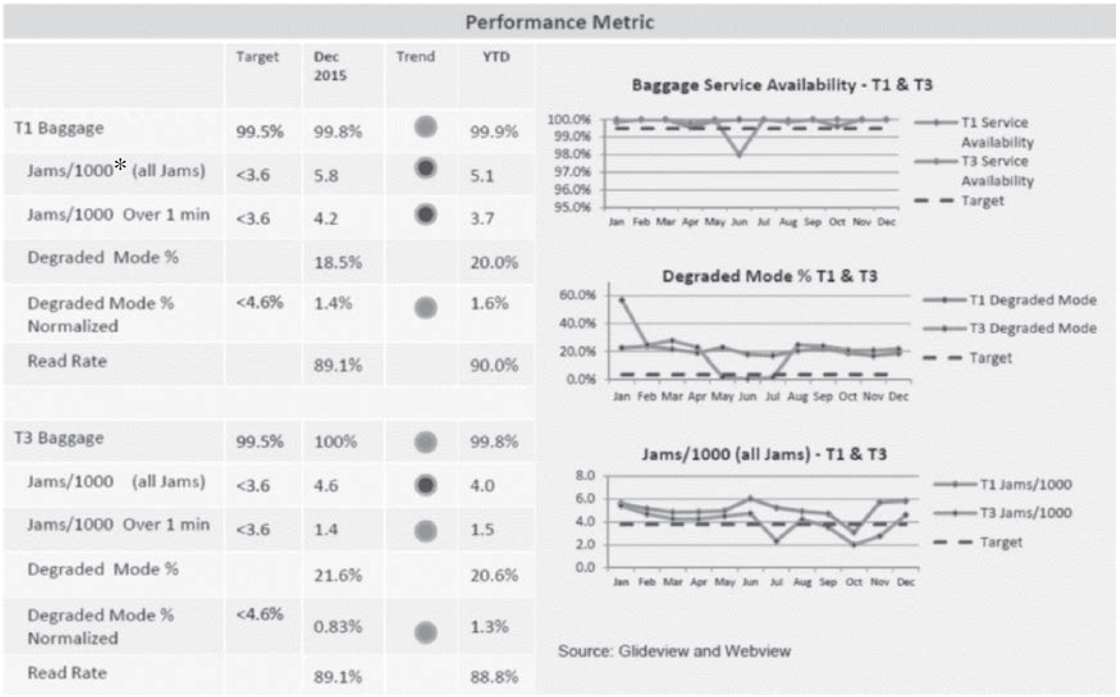


Figure A-11. PM performance metrics during December 2015.



* Jams: Baggage unable to move due to being stuck.

Figure A-12. Baggage performance metrics during December 2015.

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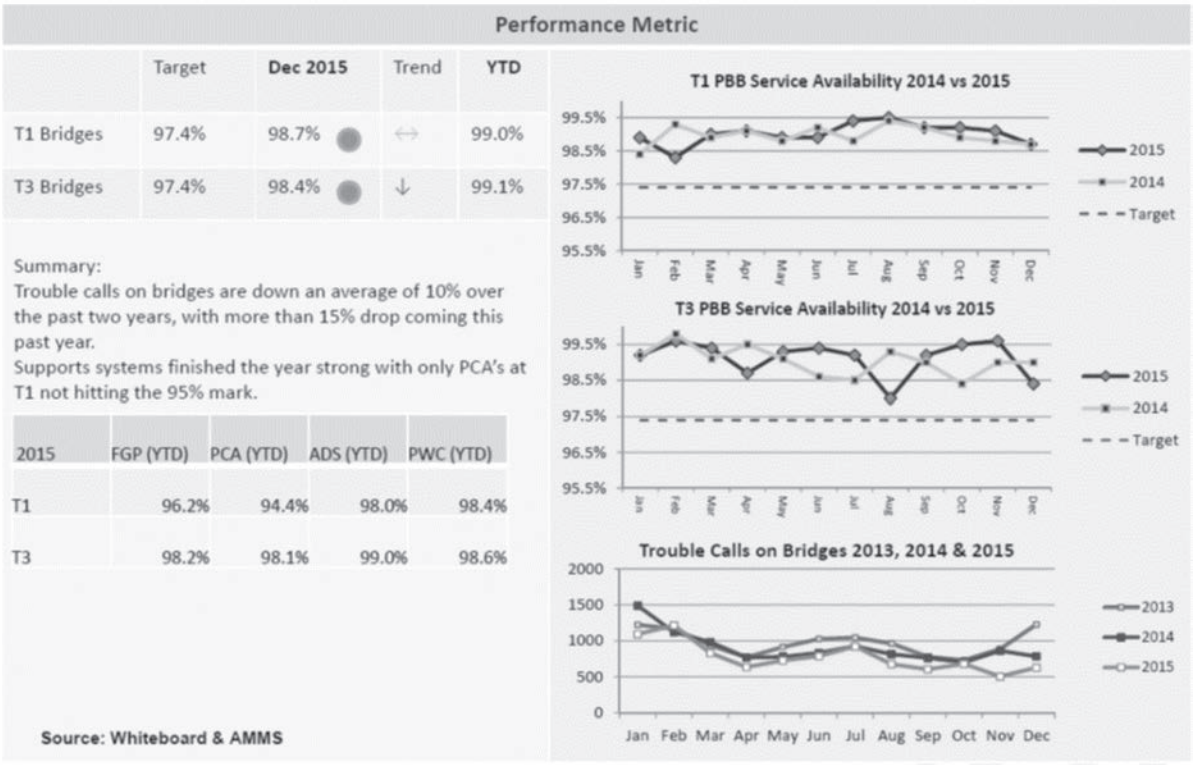


Figure A-13. Passenger boarding bridges performance metrics during December 2015.

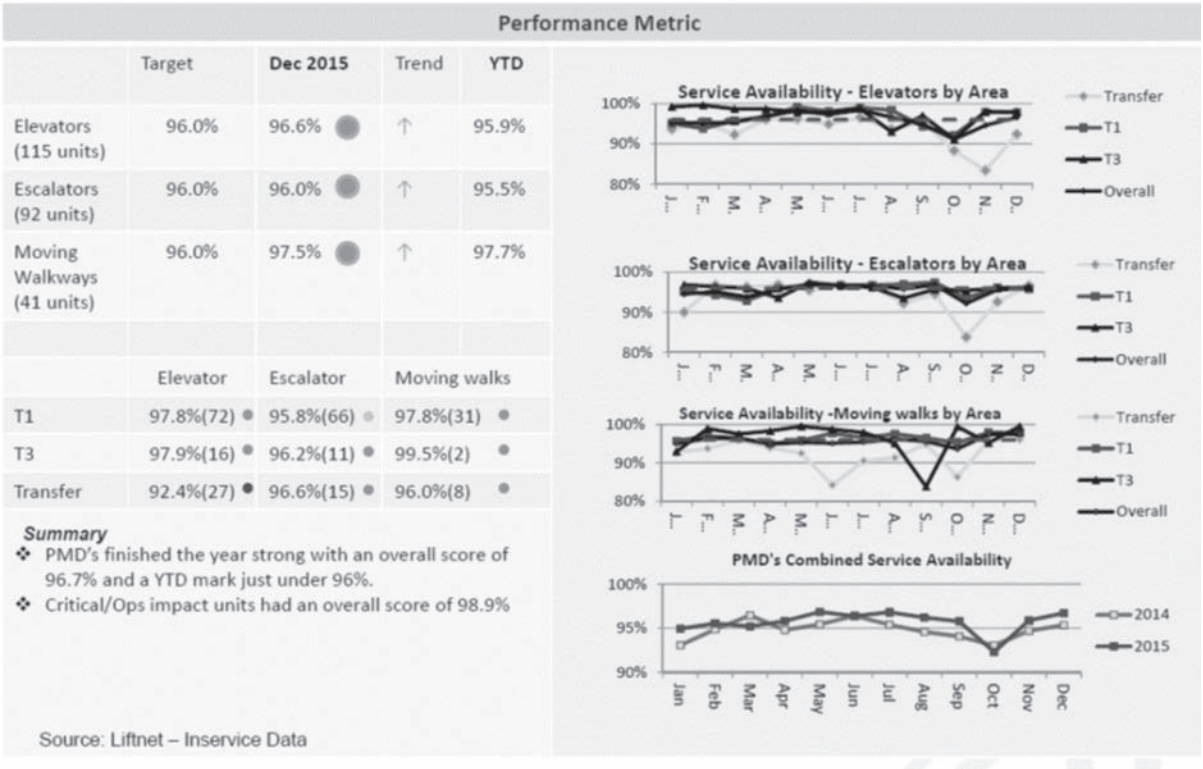


Figure A-14. People moving systems performance metrics during December 2015.

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PBB Calculations

$$SA_{Bridges} = \frac{\#\ of\ Bridges\ x\ Total\ Operating\ Time - Total\ Out\ Of\ Service\ Time}{\#\ of\ Bridges\ x\ Total\ Operating\ Time} = \frac{(86\ bridges\ x\ \#\ of\ hours\ in\ Month - Total\ Out\ Of\ Service\ Time(PBB,ALV,CPY)\ (hrs)}{86\ bridges\ x\ \#\ of\ hours\ in\ Month}$$

PMD's

$$SA_{elevators} = \frac{\#\ of\ Units\ x\ Total\ Operating\ Time - Total\ Out\ Of\ Service\ Time}{\#\ of\ Units\ x\ Total\ Operating\ Time} = \frac{(115\ x\ \#\ of\ hours\ in\ Month - Total\ Out\ Of\ Service\ Time\ (hrs)}{115\ x\ \#\ of\ hours\ in\ Month}$$

$$SA_{escalators} = \frac{\#\ of\ Units\ x\ Total\ Operating\ Time - Total\ Out\ Of\ Service\ Time}{\#\ of\ Units\ x\ Total\ Operating\ Time} = \frac{(92\ x\ \#\ of\ hours\ in\ Month - Total\ Out\ Of\ Service\ Time\ (hrs)}{92\ x\ \#\ of\ hours\ in\ Month}$$

$$SA_{moving\ walks} = \frac{\#\ of\ Units\ x\ Total\ Operating\ Time - Total\ Out\ Of\ Service\ Time}{\#\ of\ Units\ x\ Total\ Operating\ Time} = \frac{(41\ x\ \#\ of\ hours\ in\ Month - Total\ Out\ Of\ Service\ Time\ (hrs)}{41\ x\ \#\ of\ hours\ in\ Month}$$

$$SA_{High\ speeds} = \frac{Total\ Operating\ Time - Total\ Out\ Of\ Service\ Time}{Total\ Operating\ Time} = \frac{(18\ hrs\ x\ \#\ of\ days\ in\ month - Total\ Out\ Of\ Service\ Time\ during\ 0500{-}2300\ (hrs)}{18\ hrs\ x\ \#\ of\ days\ in\ month}$$

Baggage

$$SA_{Baggage} = \frac{Total\ Operational\ Hours - Total\ time\ of\ all\ Operational\ Impact\ Events}{Total\ Operational\ Hours}$$

Figure A-15. Performance metrics calculation equations.

Decision Making

YYZ employs a CMMS to make robust and informed decisions. The reports that the CMMS produces are being utilized by multiple levels across the organization. These reports assist in increasing appropriate wrench time and reducing downtime for maintenance activities, which will, over time, have a positive impact on asset operations and improve customer service. Reports driving these benefits include:

- Inventory – stores catalogue, counting inventory, and stock out
- Stock item usage report
- Completed work order performance
- Work order backlog summary
- Work order waiting reports
- Preventive maintenance
- Preventive maintenance overdue report

The information produced by the CMMS is reported to leadership through:

- KPIs
- Periodic reports
- Dashboards
- Briefings
- Scorecards

General Mitchell International Airport (MKE)

About MKE

MKE is a county-owned, joint civil-military public airport located five miles south of the central business district of Milwaukee, Wisconsin. It is named after United States Army Air Service General Billy Mitchell, who was raised in Milwaukee and is often regarded as the father of the United States Air

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Force. Along with being the primary airport for Milwaukee, MKE is sometimes described as Chicago's third airport, as many travelers in the suburbs north of Chicago use it as an alternative to Chicago's O'Hare Airport and Chicago's Midway Airport. It is also used by travelers throughout Wisconsin and Northern Illinois. The airport also hosts the General Mitchell Air National Guard Base on the eastern area of the airport property. MKE covers an area of 2,180 acres that contains five runways ranging in length from 4,183 to 9,690 ft.

MKE maintains utilities, pavement, lights, signs, leased properties, and other assets. MKE also manages a host of airport-specific features such as navigational aids, security systems, alerts, weather feeds, airline information, and others.

The airport began using a CMMS for work orders in the late 1980s. That software was used until 2007, when there was a catastrophic system hardware failure. Attempts to restore the system and its data failed. Fortunately, the airport had already developed requirements for a replacement system, evaluated software to meet those requirements, planned an implementation, and purchased the new CMMS. The airport used hand-written work orders until the new system could be implemented. That process took less than four days.

Before the new CMMS, MKE struggled with a series of systems ranging from computer applications to an established set of forms. Their work order system was an admittedly underutilized legacy program that could not be deployed to its full potential. After many years of struggling with this program, the Maintenance Department embarked on a plan to replace the system.

One of the requirements for the replacement system was that the CMMS would integrate with the airport's existing GIS. The GIS was already a mature system (first implementation started in 1998), and the GIS database already contained many of the airport's assets, including general and airfield-specific lights, signs, pavement, buildings, fences, gates, navigational features, security items, and other assets.

The new CMMS was installed in early 2011 and went live on December 11, 2011. The drivers behind the CMMS implementation include that fact that MKE had an existing antiquated system, predominantly paper based, without ties or links to operational events. With the CMMS, MKE now foresees the replacement and elimination of aging and inefficient applications, along with improvements to project management, data sharing, and planning capabilities.

Some of the challenges MKE faces include:

- Staffing shortages
- Talent planning
- Moving from reactive to proactive maintenance
- Budget constraints

In the future, MKE plans to introduce more work order processes such as labor entry, statistics, and inventor, and add the ability to track work orders in the GIS web application. MKE aims to develop a variety of reports to further assist airport management with the data analysis.

Organization Background

MKE has four shops that work both the airside and landside:

- South Shop – general maintenance
- Electricians
- HVAC-plumbers
- Custodians

The primary user of the CMMS at MKE is Airside Operations. Airside Operations maintains airfield and public spaces, including the terminals. The airport's GIS has mapped building interior floor plans and

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room-related assets. These maps can be accessed through the CMMS and used to call up an asset by location and initiate or follow up any work orders related to the mapped asset.

The system is used inside the buildings by Landside Operations and shop supervisors (maintenance, electrical, HVAC, plumbing, custodians) to manage equipment, materials, and labor statistics. Security and safety personnel use the system to manage specific work orders, including Part 139 work orders by date. IT manages IT assets and tracks IT-related service requests and work orders.

The system is also used to track airport rental space through the Property Management system integration with GIS. Users can determine who is renting a space, how to contact the lessee, and how to access leased space in an emergency. The parking management contractor and the county's fire department also use the system.

CMMS Program at MKE

The CMMS is used to manage hundreds of assets and an array of maintenance procedures throughout the airport. The airport has compiled a comprehensive GIS database that includes general and airfield-specific lights, signs, pavement, buildings, fences, gates, navigational features, security items, and other assets – all managed with the CMMS.

Airport operations care for both airfield and public spaces, including the actual terminal buildings. These features are all accessible within the CMMS, including many data layers that are linked to the airport's Property Management software. Though the attributes associated with these features are very detailed, the airport continues to add more refined details within interior spaces such as lights, electrical outlets, carpet, wall coverings, and others.

The CMMS has streamlined work. In the past, work orders would go to the wrong location, and there was no way to adjust the recipient. Work orders would then have to be canceled and re-created; paper copies were routed around the airport, and often a work order would be closed prior to final inspection. With the CMMS, the time it takes to complete work orders has been shortened, and routing of work has improved dramatically with virtually no errors. The staff is happy to find the right work order at the appropriate maintenance shop.

Each day of the year, airports perform a variety of FAA-mandated inspections. With the ability to customize the CMMS, MKE has developed very specific work order templates. Their "Master Inspection Plan" guides staff through the FAA-required airfield inspections three times a day, as well as when an issue may require a runway inspection. This specific inspection (FAA 14 CFR Part 139) involves the examination of airfield assets including lights, signs, pavement, navigational aids, security items, and other assets. To accommodate multiple assets in an inspection, MKE customized a work order template with multiple tasks to act as an inspection form. Work orders resulting from this inspection are linked to the inspection as child work orders. This allows users to track work tied to a specific inspection while managing open work orders produced from subsequent inspections. The CMMS's flexible search capabilities and unique map interface only make the process easier.

At MKE, two offices oversee daily operations: Airside Operations and Landside Operations. Each of these uses logbooks to capture a veritable diary or "life" summary. Everything that occurs on the airfield is entered in the logbook – terminal incidents, weather, staffing, airline issues, and so on. MKE customized the CMMS Service Requests to facilitate the capture of events that occur at the airport. Staff enters these events into custom fields, customized templates, contact information, etc. Work orders and LOGBOOK items are trackable and searchable with the CMMS, which has eliminated a great deal of paperwork. Entries contain in one place all the information that would otherwise be recorded on multiple forms and by different offices.

An example of how the CMMS manages a LOGBOOK incident might be when a person falls in the airport terminal and sustains an injury. Before the CMMS, an airline employee would call the Operations Control Center (OCC), who would then contact police, emergency medical technicians, and Landside Operations. The Landside Coordinator would respond to the scene, complete a lengthy form detailing the

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incident, replicate the information in a paper logbook, and make copies of the form to provide to the OCC. The OCC Operator would enter the same information into another logbook and make copies for the Security, Safety, and Properties offices. The Property Manager would then enter the information into a database, make copies, and forward those copies to Risk Management and the airport insurance provider.

With the CMMS, the OCC Coordinator enters incidents directly into the airport's LOGBOOK, which automatically dispatches the Landside Coordinator and alerts police and rescue. Using a tablet PC, the Landside Coordinator adds detailed incident information to the same record in the LOGBOOK, including a pinpoint map of where the incident occurred, photos, and witness statements. With all the information in the LOGBOOK, the Landside Coordinator closes the record, which is then immediately available to Security, Safety, Properties, Risk Management, and insurance providers. All personnel are now immediately aware of what occurred and have access to the same information – and more information than in the past.

MKE now has a way to track events, categorize issues, and easily view the event status at multiple airport departments. Through a series of reports, information is automatically disseminated to departments, airlines, and others in a much more efficient manner, better communicating everything from outages to day-to-day issues. The CMMS has helped the airport virtually eliminate the trails of paper that once piled up, resulting in a considerable savings in resources and time. The CMMS has streamlined work processes, simplified user tasks, and created a better working environment. From a data management perspective, MKE can make changes to one map service immediately available to all users in both the CMMS and GIS systems. This eliminates the necessity to constantly update maps for multiple departments every time there is even the slightest change. The CMMS has also helped the airport better justify staffing needs, have a firm handle on labor and work statistics, and have a much more accurate picture of the time, effort, and efficiency of their maintenance efforts.

The CMMS's flexibility has proven invaluable to MKE. Airport-specific requirements demanded unique customizations to the service request and work order forms. The software was configured to add an airport-specific panel to the service request form while non-applicable fields were removed. MKE used universal custom fields and custom field templates to ensure the collection of key information.

System Requirements

MKE used a complete list of maintenance criteria when defining the system requirements prior to selecting the CMMS. This list included:

- Preventive Maintenance Criteria
  - Schedule PMs by calendar, meter readings, operational parameters (real time)
  - System allows for unlimited numbers of PMs per asset
  - System allows for multiple crafts scheduled on a PM
  - The ability to schedule PMs for any specific date and/or day of the week
  - System will schedule PMs for assets or facilities
  - System will combine all PMs that are due for an asset automatically, manually, or not at all
  - System generates the PM work orders daily, weekly, monthly, or at user-defined intervals
  - System reports overdue PMs by number, incomplete PMs, or results of PM inspections
- Work Order Criteria
  - System tracks information at the work order task/step level for planned versus actuals, for labor, materials, tools, contractors, and safety
  - System produces a report of all work orders that can be sorted by their status in the work order workflow
  - System is capable of tracking work order backlog by craft, crew, department, planner, supervisor, or other defined categories

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  - System updates work order status either manually or automatically via workflow configurations
  - When planning a work order, the user can access stores, personnel, tools, contractors, etc., without leaving the work order module
  - System provides the ability to create notifications and escalations and send these items to personnel internally or externally

- MRO Inventory and Procurement Criteria
  - System generates a Spares Reorder Report when the quantity on hand drops below a minimum level
  - System can manage multiple warehouses
  - System produces performance reports for purchasing section that include overdue POs, inactive parts, or inventory valuation
  - System automatically tracks cost to the work order on an itemized basis

- Management Reporting Criteria
  - System produces reports daily, weekly, monthly, yearly, or at any user-defined interval
  - Reports produced by the system are summary reports, lists of information, or exception reports
  - System has a report writer in addition to standard system reports
  - Reports were designed to be used by system managers, managers, or maintenance personnel
  - System has a maintenance budget reporting module

CMMS Implementation

The MKE CMMS selection team was made up of Maintenance, Operations, and leadership staff. As part of the selection and implementation, the following items were considered:

- The system can track asset information such as cost center, department, location, etc.
- The system allows for cost and repair information to be accessed at multiple asset levels
- Work orders can be saved to the asset history as one line of information or full detail
- All information in the asset history can be accessed by sorting or building ad hoc reports
- The system provides a bill of materials for each asset
- The system provides user-defined screens for storing information relative to asset items/types, which also can be selected and sorted by reports

During the CMMS implementation, the selected vendor was required to perform a full turn-key implementation, including software implementation, hardware installation, data gathering, data loading, system training, and user training. The vendor also provided a fully documented installation plan, and their staff was made up of software experts, maintenance experts, training experts, and airport operations subject matter experts.

As part of the initial CMMS implementation, the following activities were considered and implemented:

- Work order processing
- Materials management/inventory management
- Preventive maintenance
- Cost controls
- Reporting

MKE established and then followed the following process flows:

- Establish a steering committee for CMMS system
- Establish the site implementation team and site project teams

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- Establish kick-off meetings for site and project teams
- Install hardware and the CMMS software
- Initialize CMMS software (create multiple system environments, including development, test, train, and production)
- Provide initial training for the site and project teams
- Define organizational procedures for CMMS use
- Develop written CMMS procedures
- Restructure the organization/business processes as needed
- Begin developing various data collection formats
- Use the formats to start data collection process (asset, stores, PMs, personnel, purchasing, accounting)
- Begin data entry process
- Begin user training on CMMS system
- Begin using work order system
- Monitor all usage of CMMS system
- Audit CMMS system for 30 days (180 days)
- Continue 30-day audits for the 1st six months

During the implementation, MKE experienced and addressed several issues that included assessing current and future needs, properly documenting the system user requirements, and correctly estimating time needed to collect and load data.

The MKE system is capable of being Cloud based, but during the implementation it was decided not to put the system in the Cloud, mainly due to the additional data that the MKE system handles. MKE's implementation choice was to use virtual machines in a controlled and secured local environment.

Assets in CMMS

MKE reviewed the short-term and long-term asset classes and the associated asset data that would be required to achieve full benefit from the system. Most of the asset classes were already located in MKE's GIS system, and staff was already familiar with it. Soon after, MKE expanded the data collection and added assets to meet the needs of other departments.

Additional assets and data were added by requests and staff discussions about how the assets benefited the CMMS. MKE works with the departments, collects the spatial locations of assets, collects the associated attributes, populates them into a mapping service, and publishes to the CMMS.

Some of the other drivers in selecting the asset classes included the ability to meet the FAA GIS data requirements and meet the needs of the Maintenance and Operations staff. The following is the list of asset classes included:

- Terminal buildings
- Runway lighting facilities
- Hangars
- Parking garages
- Pavement
- HVAC
- Electric
- Power plant
- Plumbing
- Elevators
- Escalators/moving walkways
- Baggage handling system
- Bridges

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- Rolling stock (to be implemented as part of Storeroom/Inventory function in spring of 2018)
- Utility/treatment facilities on site
- Fueling system
- Security system (partial)
- Signage
- Buried piping (water, sewer, gas, electric, etc.)
- Storm water infrastructure (culverts, swales, inlets, basins, etc.)
- Deicing system
- Fire protection

Data

At MKE, the data structure/system hierarchy was organized around the FAA design. That design is not always the most useful structure for use within a CMMS. Two examples of this are Pavement and Utilities. The FAA hierarchy has nearly 12 layers of data that are all pavement. MKE built a pavement layer, which is a consolidation of the 12 layers into one layer for ease of use within the CMMS. Utilities are developed in-house at a greater level of detail than is required by the FAA. The structures of those data sets do not match the general structure of the FAA requirements, so MKE built a script to transfer between detailed datasets into the general FAA data set.

MKE collects different types of data, which include data such as date and times (initiation of work, mitigation, in and out of pending, completions, closed), labor, equipment used, materials used, tasks involved, and links to associated work orders or service requests, links to operational log entries (incidents that trigger work), and PM dates and times.

MKE has found that some of the most beneficial data for decision making includes work order timing. The times are developed through a series of calculations to determine average and unique times to mitigate issues, complete work, and times from initiation to close of work, and then those times are used as comparisons and in calculations to determine performance measures and KPIs.

MKE has an ongoing data QA/QC process in place. Data is reviewed by departmental supervisors for incorrect data, and the data is scrubbed as part of developing the KPIs. Data with issues outside the parameters of expected results are flagged and reviewed to ensure data was entered properly. MKE uses an SQL database structure that is part of the CMMS database.

MKE regularly analyzes all system data. Calculations within the database provide the values of times for mitigation, pending, completion, and close. Those values are compared against standard operating procedures (SOPs) within the dashboard KPI application to determine averages, percentages, etc., and values of success/failure are determined. Management has established goals for performance dependent on categories and type of work.

Data points are reported to all levels of management through KPI dashboards and published documents. Data is automatically communicated to staff. The data is established automatically based on date/time entries within the work order. The data points that are communicated are supported by the following metadata:

- Work order initiated date, time, and initiated by
- Submitted to
- In to pending (date/time), and pending reason
- Out of pending
- Completion date, time, and completed by
- Closed date, time, and closed by
- Time to mitigation
- Time in pending

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- Time to completion
- Time to close
- Labor hours, crew hours (personnel list)
- Associated tasks, coinciding with certain work orders
- Associated LOGBOOK entries, service requests, and inspections

Data is also integrated with data from other system via the KPI dashboard. Data is merged from multiple database tables and integrated for analysis. Analysis and updates are live. New work orders in the system constantly update the performance measures.

CMMS Integration

MKE's CMMS is GIS-centric, embedded within the application. It is also the operational LOGBOOK, the reporting mechanism for Part 139 reporting, Wildlife Management, and Incident Management, and is linked to Property Management data. It will be used as the inventory (storeroom) function. Additional integrations include:

- Security
- Safety management
- Wildlife Hazard Management Plan
- Pavement management
- Scheduling
- Electronic logbooks
- Incident management
- Property management
- Airfield lighting
- Storerooms
- Geographic information
- Enterprise asset management (EAM)
- Inspection application (Part 139)

For the most part, the system integration is self-contained within the CMMS. Some integrations are connected via application programming interfaces within the application tied to database views and links. The overall drivers in developing these integrations were to eliminate duplicate applications and streamline data processing and management. MKE has obtained significant benefits from the development and implementation of system integrations, namely the ability to eliminate duplicate applications and centralization of data and data processing (reporting, analysis).

Performance Measurement

Primarily, MKE measures system performance through the use of KPIs and reporting. Various reports are used to measure overall productivity and gauge the system's ability to meet MKE's CMMS needs. KPIs and reports were developed by management and maintenance leaders, are used within the Maintenance Department, and are seen by executives at the Milwaukee County Department of Transportation. MKE does quarterly reviews of performance measures.

For the most part, new KPI development, relative to performance measurement, is instigated by management and reviewed by CMMS and GIS staff to ensure the availability of the data required to calculate the KPI. Performance measures are reviewed by management and senior leadership. Any gaps determined are closed, which could result in a workflow change, change to an SOP, and or change in personnel responsible for the assets and work involved.

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Decision Making

MKE's CMMS is used on a regular basis to make cost-effective management decisions for capital, maintenance, and operations decisions.

As part of the decision-making process, the following reports are used:

- Performance measures – times to completion, percentage on time, reasons for not hitting measure
- Work order priority analysis
- Inventory
- Supervisor work order performance
- Supervisor/skill work order performance
- Work order costs report
- Completed work order performance
- Asset repair history
- Asset maintenance costs repair
- Safety work order backlog
- Stock item usage report
- Work order waiting report
- Preventive maintenance
- Preventive maintenance overdue report

Non-Airports

One non-airport organization was included among the case studies. The findings with this organization are relevant in that the utility's maintenance functions much as an airport in terms assets and facilities.

Gwinnett County Department of Water Resources (GCDWR)

About GCDWR

Gwinnett County is in the state of Georgia, approximately 30 miles northeast of Atlanta. It comprises 16 municipalities, has an area of 437 square miles, and has a population of 859,304 (as of the 2013 Census).

GCDWR is a ratepayer-funded utility that serves the county with water, wastewater, and stormwater services. GCDWR's water and wastewater facilities consist of two water treatment plants with a total water production capacity of 248 million gallons per day (MGD), three wastewater treatment plants with a capacity of 98 MGD, 217 wastewater pump stations, 10 tanks, and nine booster stations. In addition, it has 8,000 miles of pipes and 32,000 facility assets that are worth $9 billion. GCDWR has 614 positions, 242,000 active water accounts, and 168,000 active sewer accounts.

GCDWR has a strategic plan, developed in 2012, that is being updated. The utility's mission is "*To provide superior water service at an excellent value,*" and its vision is "*to be widely recognized as a leader in the water industry.*" The utility is dedicated to protecting public health and strives to continuously improve through personnel commitment, innovation, efficiency, and education.

Two main challenges that GCDWR faces are:

- Keeping, developing, and harnessing institutional knowledge
- Recruiting qualified technicians to the department

GCDWR is working extensively on these two challenges by developing a platform to encourage its younger staff members to work and interact with seasoned employees and to increase their involvement and responsibilities. Positional cross training is available and encouraged for those employees who want

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to learn more about plant operations or maintenance. The county is changing its development program for maintenance technicians to include Qualification Cards to support employee retention and advancement. An example of a new qualification that supports job advancement is to pass the Certified Maintenance Reliability Technician or Certified Maintenance Reliability Professional exam.

Organizational Background

Organizationally, the CMMS is part of the Asset Reliability Section under the Facility Operations Business Unit. The Facility Operations team's vision is to achieve "*Maintenance, Operations, and Reliability Excellence (MORE).*" In addition to the CMMS, the asset reliability section includes predictive maintenance, precision alignment, maintenance planning and scheduling, and reliability engineering. Additional responsibilities include asset criticality ranking, root cause failure investigations, RCM2 [Reliability-Centered Maintenance Mindset], asset management best practices implementation, and maintenance and KPI reporting. Figure A-16 presents the organizational structure for Facility Operations.

Asset Reliability deals with various assets as a system from a criticality perspective and strives to ensure asset availability and reliability while having minimal impact and ensuring non-interruption of water operations and services.

In 2014, GCDWR conducted a study, survey, and assessment about the condition of asset management best practices and its current gaps. Based on the results, a steering committee developed an asset management best practices roadmap and began implementing the changes it envisioned. The committee was assisted by five focus-area teams from different sections.

Initially, maintenance activities were proactive in thinking, but reactive in nature. Currently, they are proactive both in thought and in action. Prior to 2014, anyone would have been able to approve and start a work order. Now it has shifted to become the responsibility of plant operations to review, prioritize, and approve the work orders prior to being planned and scheduled. The most recent statistics show that 92% of the work orders across all facilities are planned and scheduled.

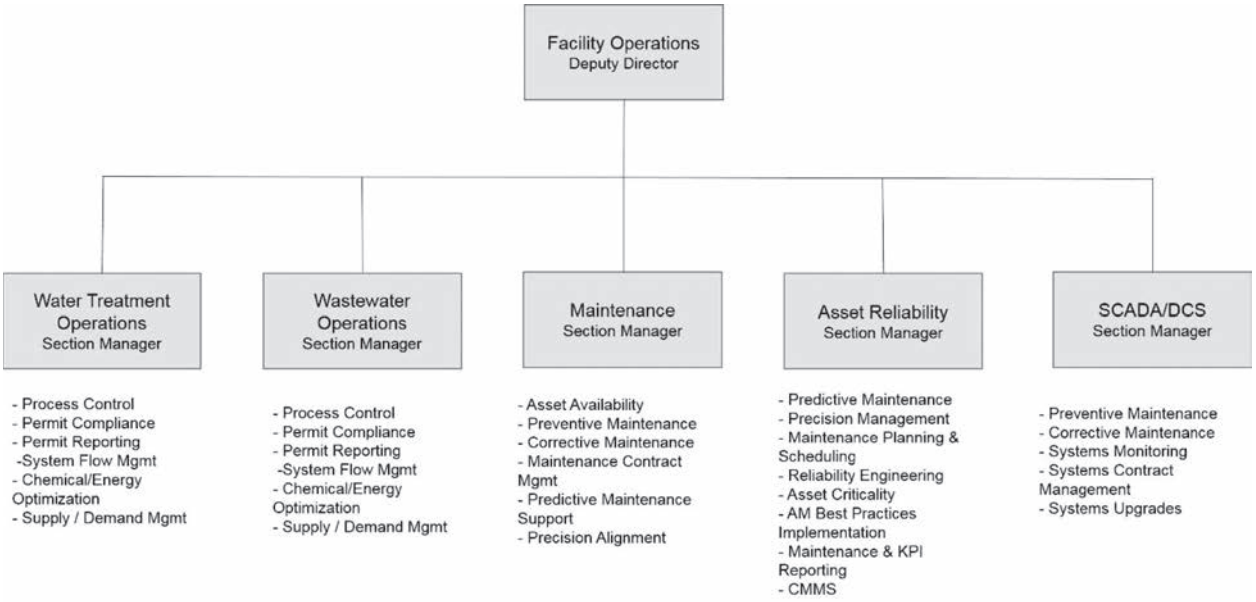


Figure A-16. Facility operations organizational chart.

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CMMS Selection

The implementation of the CMMS started in 2002/2003, driven by maintenance and financial considerations. Leadership and managers were involved in the selection and implementation of the CMMS. The CMMS was under consideration because GCDWR needed to document what assets it owned and how they were being maintained, to classify the assets based on their condition and to track them in terms of cost and functionality. This documentation allowed the maintenance team to have more real-time asset information and communicate with leadership and other sections about maintenance performance and its impact on operations, thereby being able to continue providing superior quality of services to the citizens of the county.

GCDWR uses two types of the CMMS: one for vertical assets, and one for linear assets. The vertical asset CMMS is the system that is used within Facility Operations and it is the focus of the discussion in this case study. The CMMS is mainly maintained by the CMMS team that is under the Asset Reliability section of Facility Operations. The team is responsible for all aspects of the CMMS and utilizes vendors as needed for upgrades and enhancements.

The current CMMS covers work order processing, materials management, inventory management, PM, predictive maintenance, reactive maintenance, contractor maintenance, personnel, crafts, and reports development. For reports, Facility Operations prefers to use reports customized to their data rather than the system's built-in reports. It should be noted that more than 90% of the data for maintenance and reliability KPIs comes from the CMMS.

System Requirements

The functional requirements of the CMMS selection criteria included the ability to:

- Track asset information such as cost, department, location, etc.
- Allow for cost and repair information to be accessed at multiple asset levels
- Save work orders to the asset history as one line of information or in full detail
- Access all information in the asset history by sorting or building ad hoc reports
- Provide a bill of materials for each asset
- Provide user-defined screens for storing information relative to asset items, which also can be selected and sorted by reports

The PM requirements that were considered in selecting the CMMS included the ability to:

- Schedule PMs by calendar, readings, condition, operational parameters, etc.
- Allow for an unlimited number of PMs per asset
- Allow for multiple crafts scheduled on a PM
- Schedule PMs for any specific date and/or day of the week
- Forecast labor resources, material requirements, or special tools; for PMs due for any specified time period
- Schedule PMs for assets or locations
- Combine all PMs that due for an asset manually or by sequencing
- Generate the PM work orders manually or allow the system to auto-generate them daily, weekly, or monthly

The work order requirements that were considered when selecting the CMMS included:

- Tracking information at the work order task level for planned versus actuals, for labor, materials, tools, contractors, and safety
- Producing a report of all work orders which can be sorted by their current status in the work order workflow

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- Tracking work order backlog by craft, crew, department, planner, supervisor, or other categories
- Updating work order status either manually or automatically vie workflow configurations or list
- Providing user-defined screens for storing information relative to asset items/types, which also can be selected and sorted by reports

The MRO inventory and procurement requirements that were considered when selecting the CMMS included the ability to:

- Generate a spares reorder report when the quantity on hand drops below a minimum level
- Manage multiple warehouses
- Transfer items between storerooms and warehouses
- Integrate inventory system with the vendors' own purchasing system
- Generate a purchase requisition if a reorder point is reached
- Produce performance reports for purchasing section that include overdue purchase orders, inactive parts, or inventory valuation
- Automatically track cost to the work order on an itemized basis

The maintenance software analysis requirements that were considered when selecting the CMMS included the ability to:

- Be operated by operations and maintenance personnel
- Apply necessary modules to meet the organization's needs (PMs, assets, work orders, purchasing, inventory, integration, etc.)
- Provide single, semi-multi, or multi-user capabilities
- Closely integrate all system modules
- Archive files and retrieve files for reports spanning long time periods

The implementation requirements that were considered in selecting the CMMS included the following:

- The vendor performing full turn-key implementation, software implementation, data loading, system training, and user training
- The vendor having software experts, maintenance experts, training experts, and airport operations subject matter experts on staff
- The system requiring no customization
- The ability to update software on a regular basis

The preferred training method at GCDWR is to have the training conducted by the CMMS team and in-house users while having the vendor present for questions or clarifications. From experience, GCDWR does not prefer that the vendor conduct the training for the system because it is too rigid and does not meet the users' specific needs or other implemented processes at GCDWR.

The management reporting requirement was not part of the CMMS selection. It was not considered a key selection criterion because the system's data capture and reporting meets GCDWR needs without customization.

CMMS Implementation

When implementing the CMMS, GCDWR followed an implementation flow consisting of the following steps:

- Establish steering committee for the CMMS
- Establish the site implementation team
- Establish site project teams
- Promote kick-off meeting for site and project teams

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- Install hardware and the CMMS software
- Initialize the CMMS software (create multiple system environments, including development, test, train, and production)
- Provide initial training for the site and project teams
- Define organizational procedures for the CMMS use
- Develop written CMMS procedures
- Begin developing various data collection formats
- Use the formats to start data collection process (asset, stores, PMs, personnel, purchasing, accounting)
- Begin data entry process
- Establish stores stock levels
- Begin user training on the CMMS
- Begin using work order system
- Monitor all usage of the CMMS
- Conduct audits periodically, ad hoc, and at least once a year

Data contained in the CMMS is mainly for asset condition and performance and does not include financial, budgeting, human resources, and time keeping for payroll. The data in the database covers more than 95% of the assets and within 12 months it contains 100% of GCDWR's assets. Currently, 92% of all maintenance activities are formally planned and scheduled. Further, nearly 100% of the maintenance inventory and purchasing functions are documented in the system.

During the CMMS implementation, the team faced the following challenges:

- Assessing future current and needs (The team did not conduct this type of assessment efficiently, especially in communicating with employees about the system's importance and how it will benefit their work and improve on it.)
- Properly documenting the system user requirements or "getting user input"
- Conducting a good search in the CMMS marketplace (The implementation team wished that it had done more research in the CMMS marketplace.)
- Obtaining sufficient training and/or documentation from the vendor
- Correctly estimating the time needed to collect and load data (At the beginning of the implementation, it took the team seven months to collect and document 2,200 assets in the database.)

Assets in CMMS

GCDWR defines an asset as any item that has a value greater than $5,000.00 and a useful life greater than 1 year. From a CMMS perspective, Facility Operations defines an asset more broadly and includes many assets that do not meet the department's definition. The majority of the assets at GCDWR are managed through the CMMS. A CMMS is used to manage all Facility Operations assets. GCDWR has plans to add more assets, such as the central office facility, into a CMMS.

These assets were chosen to be added and be part of a CMMS to:

- Inventory the assets that GCDWR owns, operates, and maintains
- Learn more about the assets in terms of operation and maintenance resource requirements
- Develop the real maintenance cost
- Analyze equipment reliability
- Properly conduct root cause failure investigations on critical asset failures

The process of documenting and adding these assets to the CMMS database started manually. The main questions that the team asked when documenting the asset were, "How do we respond when it fails? Do we repair or replace?"

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A New Asset Form template was developed for new assets. The template was designed to be generic so as to apply to a variety of asset types. It includes 173 attributes, as shown in Appendix B.

As part of the submittal for new capital projects, the contractor completes and submits a New Asset Form to the project manager, who then forwards the form to the CMMS Administration Group to add to the CMMS. After Administration reviews the form for accuracy and data completeness, the asset IDs are assigned and added to the form, which is then sent to Finance so the CMMS data can match the Finance fixed asset registry. This asset documentation process by the Asset Management Group provides assets' useful life estimates to Finance.

Data

Asset data that are collected by GCDWR cover 173 attributes (Appendix B). These attributes include the information for any type of asset. The collected data is mainly used to support the performance measures that are used as input into decision making.

Collected data, prior to uploading and using it, is checked and cleansed through a QA/QC process:

- Individual collecting and providing the data reviews it.
- The CMMS group reviews the data.
- Planners/schedulers review and audit the data, after it is uploaded.
- Data is formally audited for accuracy.

The CMMS uses three servers: (1) Database; (2) Production; and (3) Development. Data are backed up nightly and three days of back-ups are consistently maintained. Additionally, the asset manuals are saved on a three-terabyte storage array. As part of the advancement in data storage, GCDWR is considering a Cloud solution through a technology group and IT. The CMMS analysis team is responsible for loading, and cleaning the data, as well as providing quality control.

Data from the CMMS is regularly queried, may be combined with other systems data, and used to develop actionable information. This type of information is produced in the periodic reports and updates.

CMMS Integration

At GCDWR, the CMMS is the main system used to manage maintenance activities. The data, information, and reports produced and managed via the CMMS are robust and reliable, and they encompass maintenance-specific activities and needs. At this time, GCDWR chooses not to integrate the CMMS with its other systems due to limited value added to the organization.

Performance Measurement

Facility Operations utilizes performance measures to manage and report the performance of the assets within GCDWR. Working with outside experts, GCDWR deputy directors, section managers, trades managers, and plant managers developed these performance measures, based on the utility's gap analysis and improvement plan that formed the basis for its journey to excellence.

A list of 18 performance measures are regularly measured and reported to leadership, then shared across facilities to all staff. These metrics illustrate how their performance is progressing, which metrics need more attention, and how these metrics can be improved. The list of these performance measures is presented in Table A-11. Appendix B presents figures of the historical data of the different performance measures.

The majority of the data (>90%) coming out of the CMMS is reported to and used by Facility Operations Leadership. The CMMS data file (in Microsoft Excel) is shared monthly via e-mail, and regular meetings are held to update section heads and leadership about the work and progress. Additionally, the CMMS data and spreadsheets are available on a share drive, soon to be moved to a BI

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platform. Additionally, the performance measures are communicated in a variety of reports: (1) monthly performance measures; (2) daily progress; (3) weekly number of assets down; (4) reactive maintenance; and (5) Red/Green.

The Red/Green report is a weekly report that is produced every Monday, highlighting the following information:

- Percentage of available labor schedule
- Scheduled work orders
- Incomplete work orders
- Scheduled success
- Percentage of break in work orders (mainly priorities 1, 2)

Table A-11. Performance measures.

Performance Measure	Performance Target	Leading/Lagging
Asset Availability %	Increasing Trend – 97%	Lagging
Mean Time Between Failure (MTBF) in Hours/Minutes	Increasing Trend	Leading/Lagging
Schedule Success % (Monthly Average)	80%	Leading
Ready to Schedule Backlog—Weeks	2–4 Weeks	Leading
% Proactive Labor Hours vs. Total Labor Hours (Proactive + Reactive)	80%	Leading
% of Predictive Maintenance Hours	15%	Leading
Average Mean Time to Repair in Hours	Decreasing Trend	Leading/Lagging
Planned and Scheduled Maintenance Hours (%)	90%	Leading
Planned and Scheduled Maintenance Work Orders (%)	95%	Leading
% of Availability Hours Scheduled (Monthly Average)	100%	Leading
Order Cycle Time in Days	< 1	Leading
PM Compliance %	95%	Leading/Lagging
Facility Operations Craft Utilization (%)	75%	Leading
Stock Outs	0	Leading
Maintenance Cost as a % of RAV	2-3.5%	Lagging
% of Emergency & Urgent Labor Hours vs. Total Labor Hours (All Work Types)	< 7.5%	Leading
Total Deferred Maintenance Work Orders	Decreasing Trend or (0)	Leading
Total Number Reported Maintenance Related Safety Incidents Last 12 Months	No Incidents in Last Month	Leading/Lagging

Decision Making

The CMMS is an integral part of the decision-making process at GCDWR. It is used, on regular basis, to make cost-effective management, capital, maintenance, and operational decisions. More specifically, the data from the CMMS assists GCDWR in identifying asset failure modes and maintenance activities. Ultimately, information from the CMMS drives decision making on the basis of performance, as measured by the various performance measures.

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The types of reports that the CMMS produces, are the following:

- Work order priority analysis
- Planner performance
- Scheduling activities
- Inventory
- Completed work order performance
- Work order backlog summary
- Asset repair history
- Asset maintenance repair costs
- Safety work order backlog
- Stock item usage report
- Work order waiting report
- PM
- PM overdue
- Predictive maintenance

This information is reported to the leadership through performance reports, periodic reports, the dashboard, and regular briefings.

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APPENDIX B

Gwinnett County Department of Water Resources List of Asset Attributes, CMMS Workflow, Work Order Prioritization, and KPIs

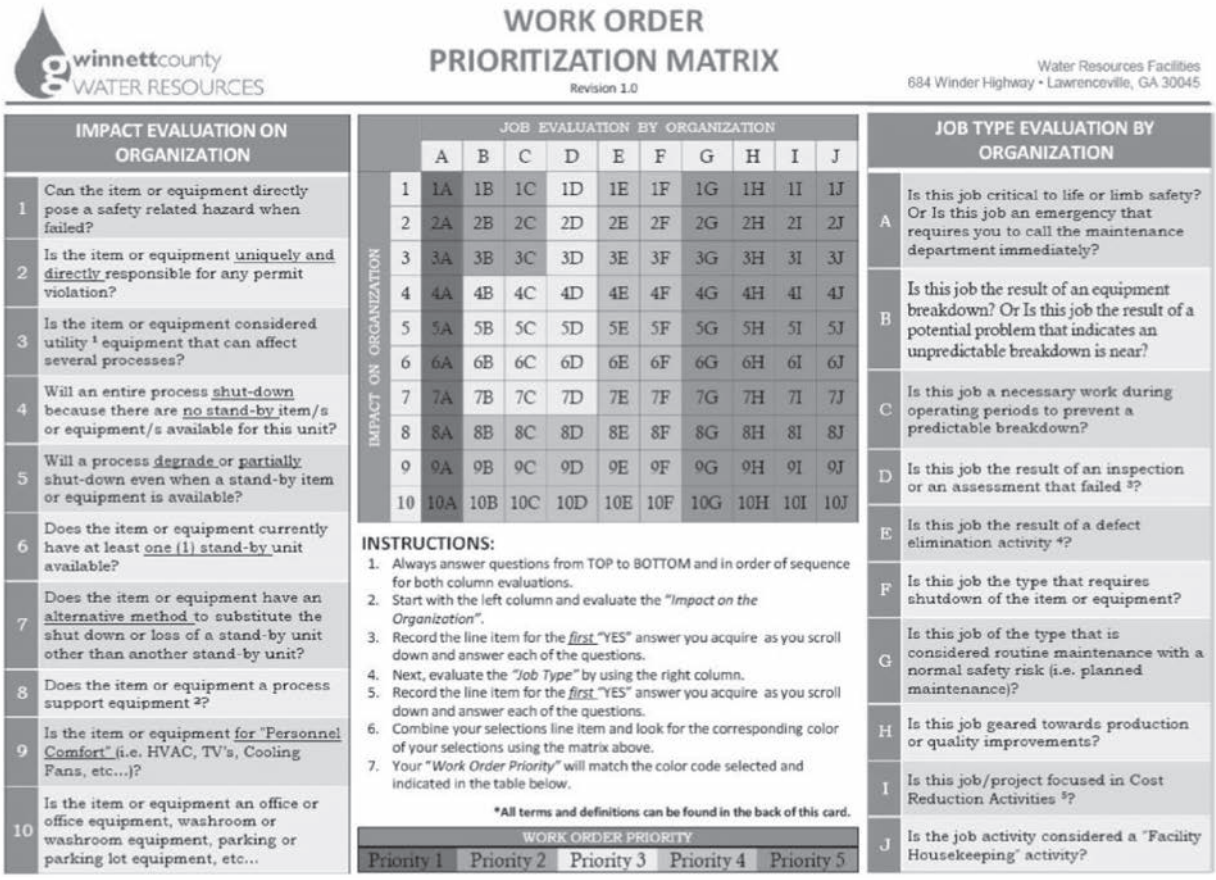
Table B-1. List of new asset attributes.

Attributes	Attributes	Attributes
DCS TAG_NO	Gear Reducer Manufacturer	Motor2 Manufacturer
New Asset? Y or N	Gear2Reducer Manufacturer	Motor3 Manufacturer
Structure Title	Gear Model	Motor4 Manufacturer
Equipment Description	Gear2 Serial	Motor2 Type
Manufacturer	Gear Reducer Type	Motor3 Type
Vendor	Gear2ReducerType	Motor4 Type
Model Number	Gear Reducer Ratio	Motor2 Horsepower
Serial Number	Gear2Reducer Ratio	Motor3 Horsepower
Purchase Price	Gear Input-Output	Motor4 Horsepower
Installed Date (Substantial Completion)	Gear2 Input-Output	Motor2 Insulation Class
Warranty Expiration date	Gear Torque	Motor3 Insulation Class
Pro Rate/ Extend Warranty	Gear2Torque	Motor4 Insulation Class
Useful Life (Years)	Gear Speed	Motor2 Enclosure
Impeller Size	Gear2 Speed	Motor3 Enclosure
Suction Line Size	Gear HP	Motor4 Enclosure
Discharge Line Size	Gear2 HP	Motor2 Frame
E_HP	Gear Weight	Motor3 Frame
E_VOLT	Gear2 Weight	Motor4 Frame
E_PHASE	Diameter	Motor2 Power Required
E_AMPs	Length	Motor3 Power Required
E_RPM	Width	Motor4 Power Required
GPM	Height	Motor2 Amps
MGD	Check Valve Size	Motor3 Amps
TDH	Check Valve Type	Motor4 Amps
Type	Base Elevation	Motor2 Service Factor
E_CAPACITY	Source	Motor3 Service Factor
Hose/Insert Type	Actuator Manufacturer	Motor4 Service Factor
Hose/Tube Size	Actuator Model	Motor2 Speed

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Attributes	Attributes	Attributes
Rated Pressure	Actuator Size	Motor3 Speed
Service	Actuator Serial No	Motor4 Speed
Capacity Per Revolution	Actuator HP	Motor2 FullLoad
Body Style	Actuator Base	Motor3FullLoad
Cast And Rotor	Actuator Coupling	Motor4FullLoad
E_Weight	Actuator Speed	Motor2NoiseLevel
Motor Manufacturer	Actuator Torque	Motor3NoiseLevel
Motor Serial	Actuator Temp Code	Motor4NoiseLevel
Motor Model	Lubrication	Motor2DesignTemp
Motor Type	Actuator Motor Rating	Motor3DesignTemp
Motor Horsepower	Actuator Elect Class	Motor4DesignTemp
Motor Insulation	Actuator Weight	Motor2DriveType
Motor Insulation Class	NEMA Enclosure	Motor3DriveType
Motor Enclosure	Construction Material	Motor4DriveType
Motor Frame	Actuator	Motor2Volts
Motor Power Req	Gear Box	Motor3Volts
Motor Amps	Valve Size	Motor4Volts
Motor Service Factor	Screw Diameter	Motor2Phase
Motor Speed	Screw Length	Motor3Phase
Motor Full Load	Flight Size	Motor4Phase
Motor Noise Level	Screen Width	Motor2Model
Motor Design Temp	E_Speed	Motor3Model
Motor Drive Type	E1Type	Motor4Model
Motor Weight	E2Type	Motor2Serial
Asset Location ID	E1Manufacturer	Motor3Serial
Inventory Number	E1Manufacturer	Motor4Serial
Asset Class	E2Manufacturer	Motor2Weight
Sub Number	E1ModelNo	Motor3Weight
	E2ModelNo	Motor4Weight
	E1SerialNo	
	E2SerialNo	

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gwinnettcounty WATER RESOURCES

WORK ORDER PRIORITIZATION MATRIX

Revision 1.0

Water Resources Facilities
684 Winder Highway • Lawrenceville, GA 30045

IMPACT EVALUATION ON ORGANIZATION

1	Can the item or equipment directly pose a safety related hazard when failed?
2	Is the item or equipment uniquely and directly responsible for any permit violation?
3	Is the item or equipment considered utility [1] equipment that can affect several processes?
4	Will an entire process shut-down because there are no stand-by item/s or equipment/s available for this unit?
5	Will a process degrade or partially shut-down even when a stand-by item or equipment is available?
6	Does the item or equipment currently have at least one (1) stand-by unit available?
7	Does the item or equipment have an alternative method to substitute the shut down or loss of a stand-by unit other than another stand-by unit?
8	Does the item or equipment a process support equipment [2]?
9	Is the item or equipment for "Personnel Comfort" (i.e. HVAC, TV's, Cooling Fans, etc...)?
10	Is the item or equipment an office or office equipment, washroom or washroom equipment, parking or parking lot equipment, etc...

JOB EVALUATION BY ORGANIZATION

IMPACT ON ORGANIZATION

	A	B	C	D	E	F	G	H	I	J
1	1A	1B	1C	1D	1E	1F	1G	1H	1I	1J
2	2A	2B	2C	2D	2E	2F	2G	2H	2I	2J
3	3A	3B	3C	3D	3E	3F	3G	3H	3I	3J
4	4A	4B	4C	4D	4E	4F	4G	4H	4I	4J
5	5A	5B	5C	5D	5E	5F	5G	5H	5I	5J
6	6A	6B	6C	6D	6E	6F	6G	6H	6I	6J
7	7A	7B	7C	7D	7E	7F	7G	7H	7I	7J
8	8A	8B	8C	8D	8E	8F	8G	8H	8I	8J
9	9A	9B	9C	9D	9E	9F	9G	9H	9I	9J
10	10A	10B	10C	10D	10E	10F	10G	10H	10I	10J

INSTRUCTIONS:

1. Always answer questions from TOP to BOTTOM and in order of sequence for both column evaluations.
2. Start with the left column and evaluate the *"Impact on the Organization"*.
3. Record the line item for the *first* "YES" answer you acquire as you scroll down and answer each of the questions.
4. Next, evaluate the *"Job Type"* by using the right column.
5. Record the line item for the *first* "YES" answer you acquire as you scroll down and answer each of the questions.
6. Combine your selections line item and look for the corresponding color of your selections using the matrix above.
7. Your *"Work Order Priority"* will match the color code selected and indicated in the table below.

***All terms and definitions can be found in the back of this card.**

WORK ORDER PRIORITY				
Priority 1	Priority 2	Priority 3	Priority 4	Priority 5

JOB TYPE EVALUATION BY ORGANIZATION

A	Is this job critical to life or limb safety? Or Is this job an emergency that requires you to call the maintenance department immediately?
B	Is this job the result of an equipment breakdown? Or Is this job the result of a potential problem that indicates an unpredictable breakdown is near?
C	Is this job a necessary work during operating periods to prevent a predictable breakdown?
D	Is this job the result of an inspection or an assessment that failed [3]?
E	Is this job the result of a defect elimination activity [4]?
F	Is this job the type that requires shutdown of the item or equipment?
G	Is this job of the type that is considered routine maintenance with a normal safety risk (i.e. planned maintenance)?
H	Is this job geared towards production or quality improvements?
I	Is this job/project focused in Cost Reduction Activities [5]?
J	Is the job activity considered a "Facility Housekeeping" activity?

Figure B-1. Work order prioritization matrix.

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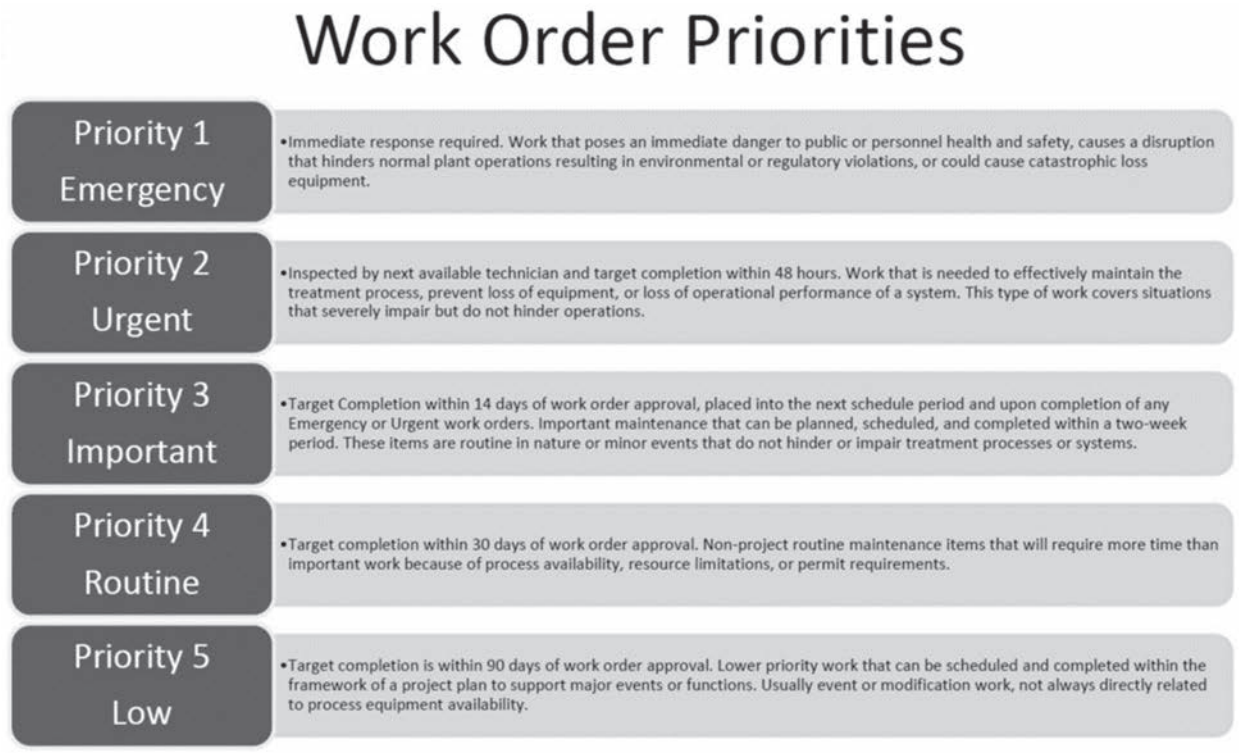


Figure B-2. Work order priorities.

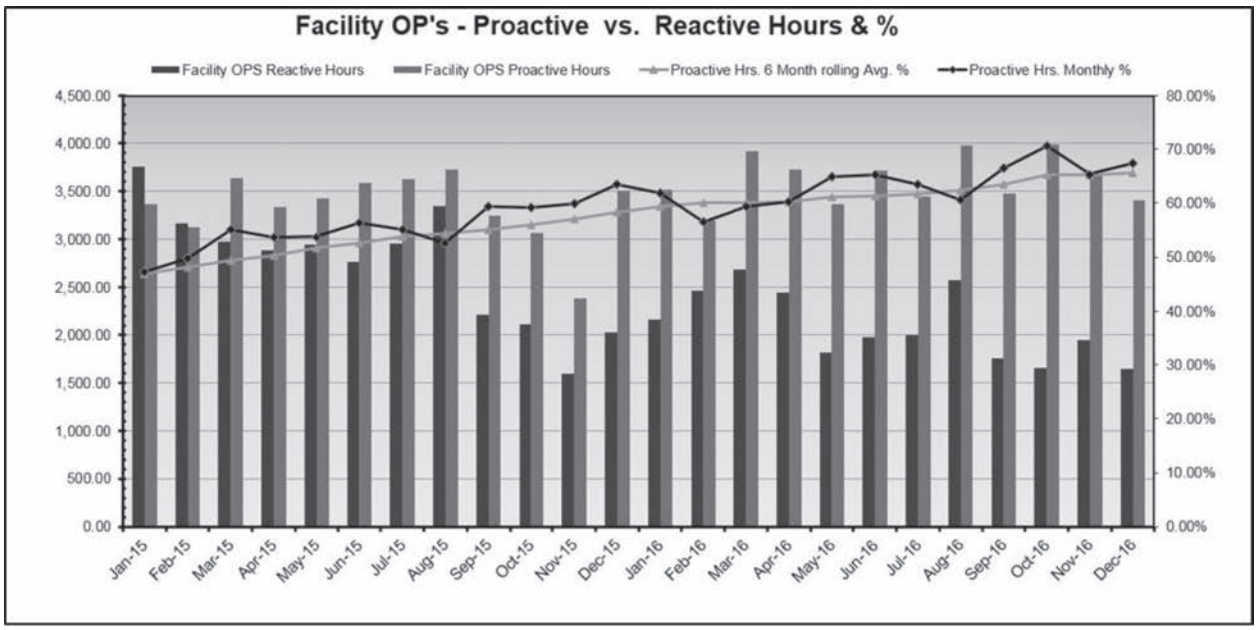


Figure B-3. KPI—proactive vs. reactive.

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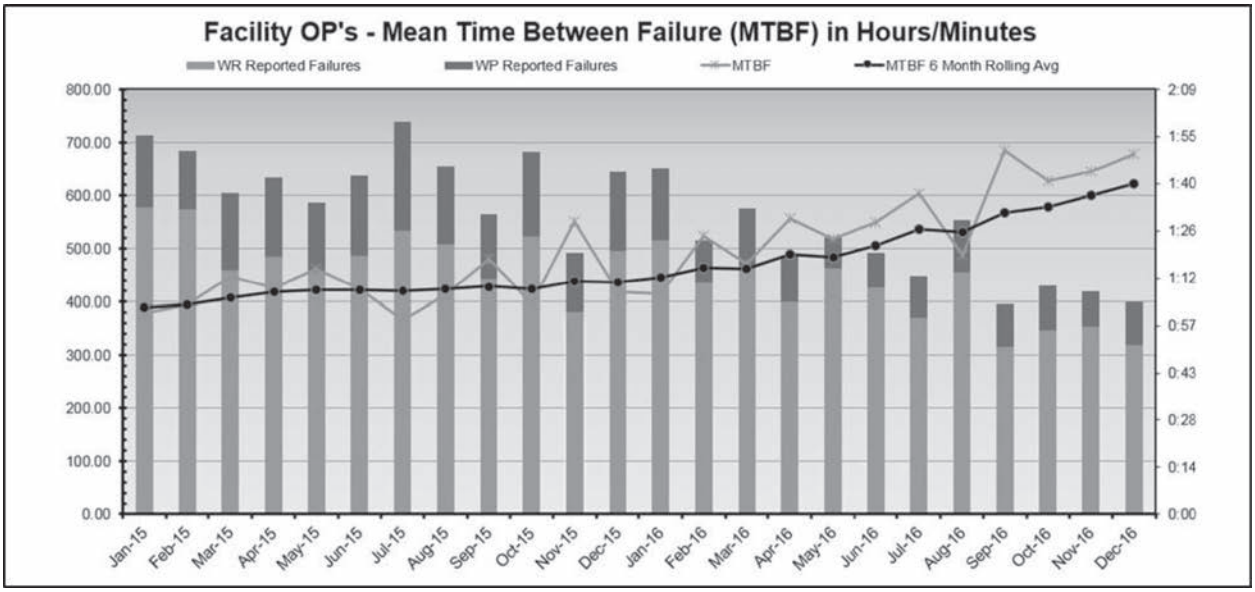


Figure B-4. KPI—mean time between failures.

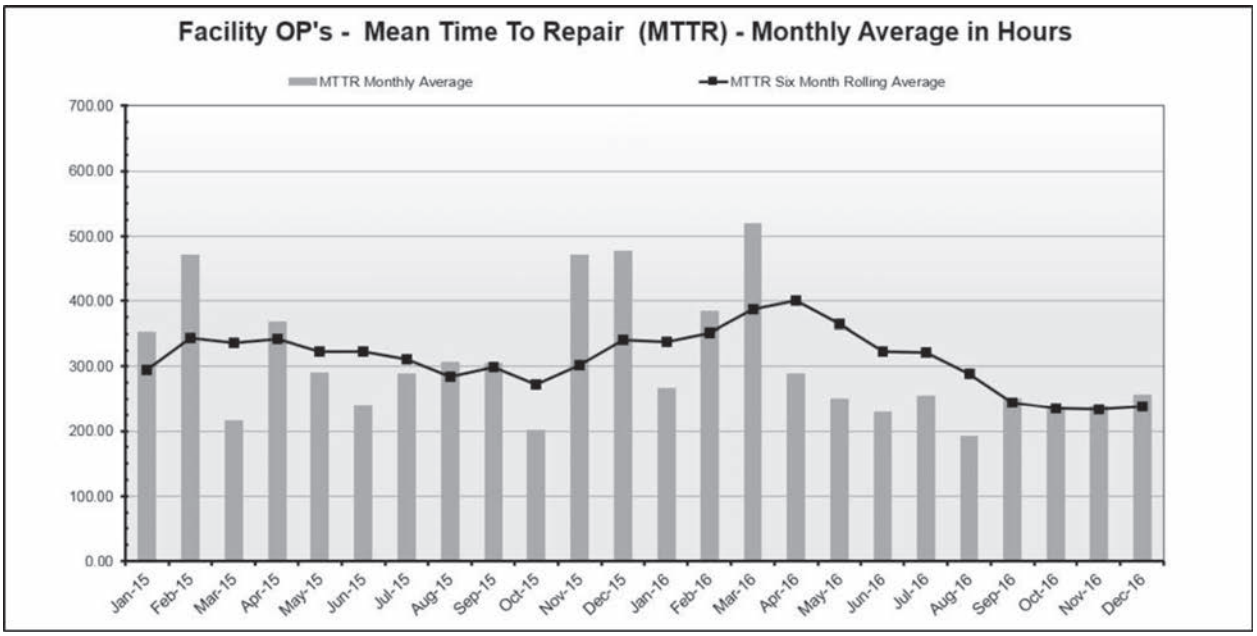


Figure B-5. KPI—mean time to repair.

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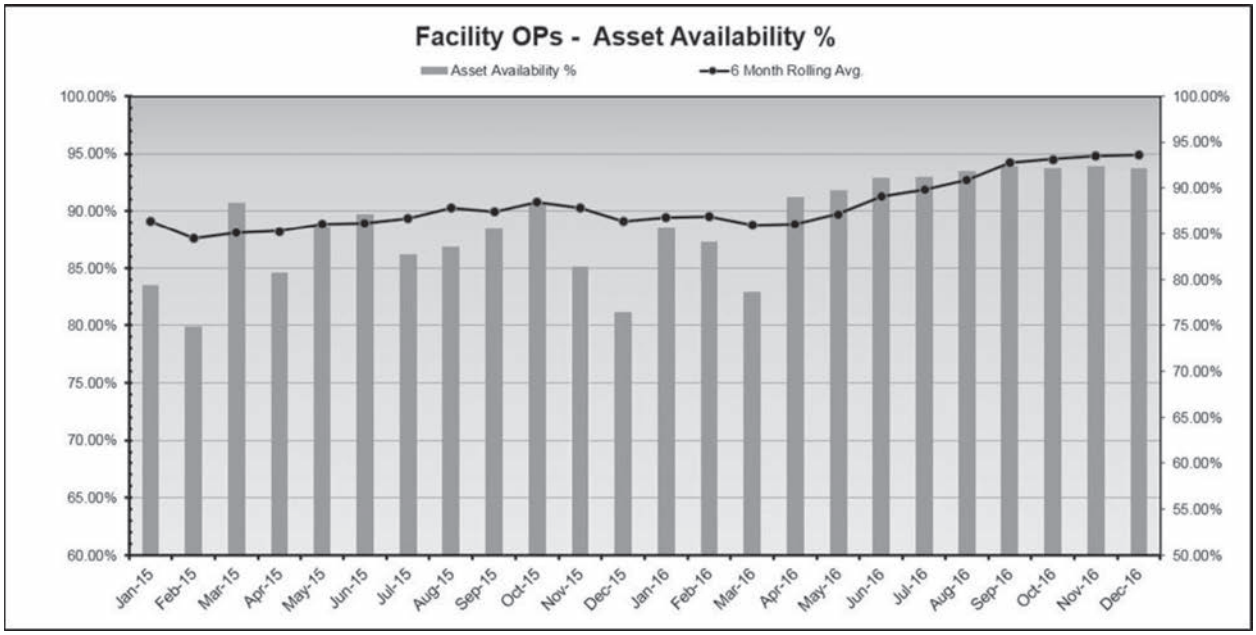


Figure B-6. KPI—asset availability.

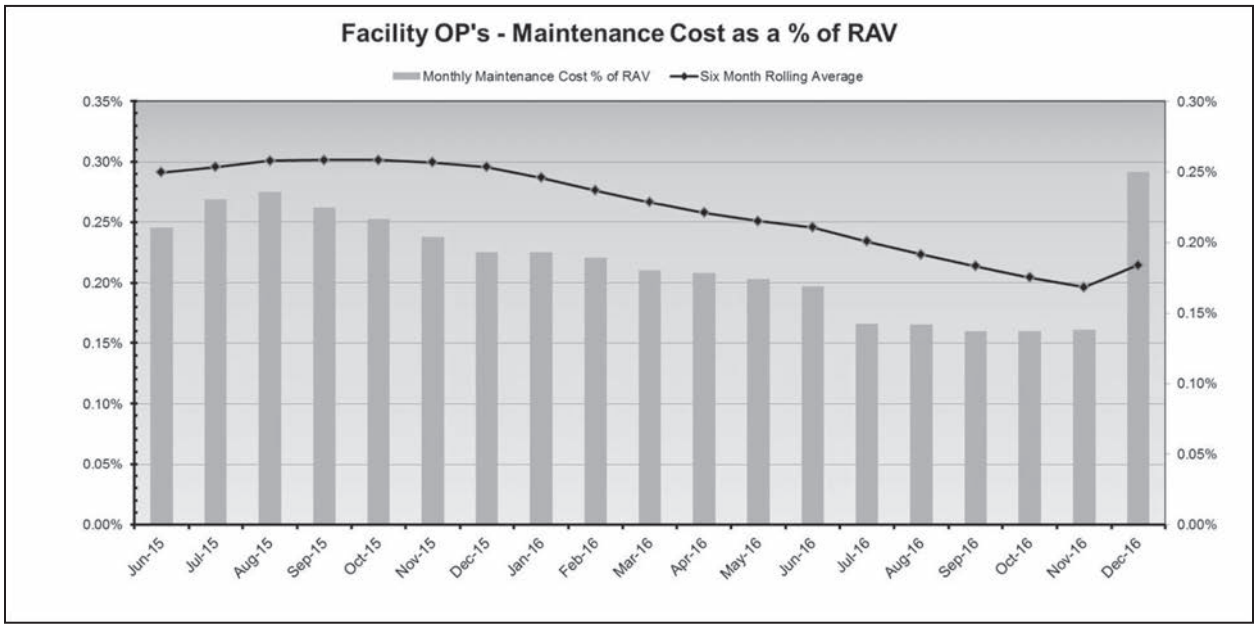


Figure B-7. KPI—asset availability.

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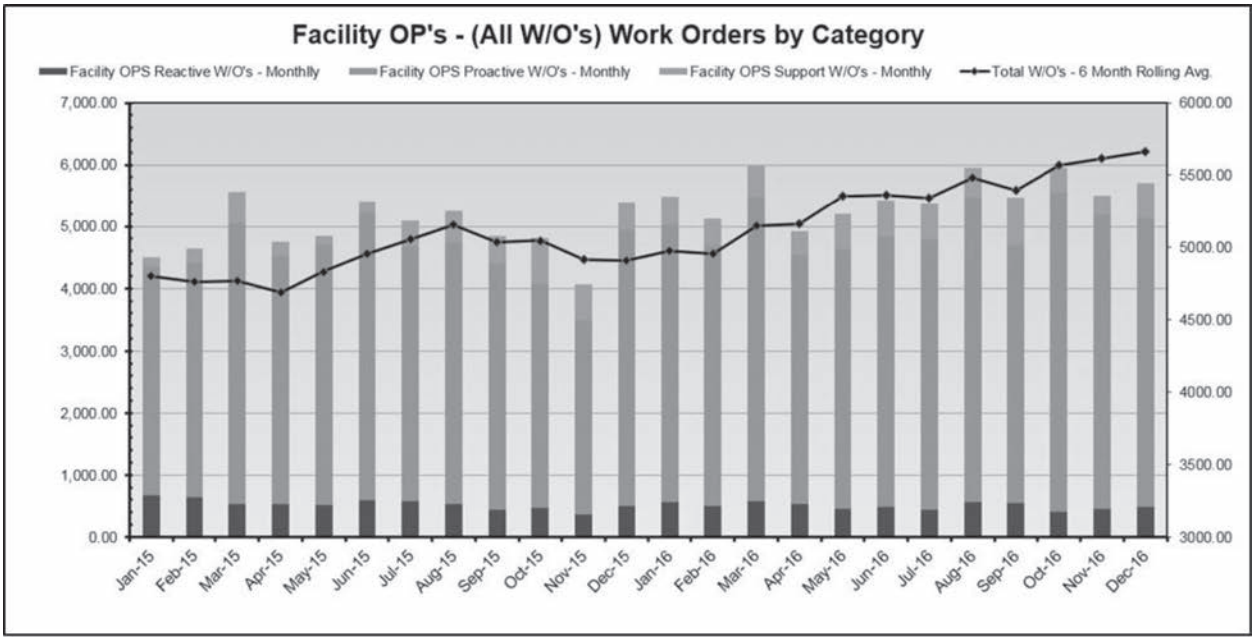


Figure B-8. KPI—work orders by category.

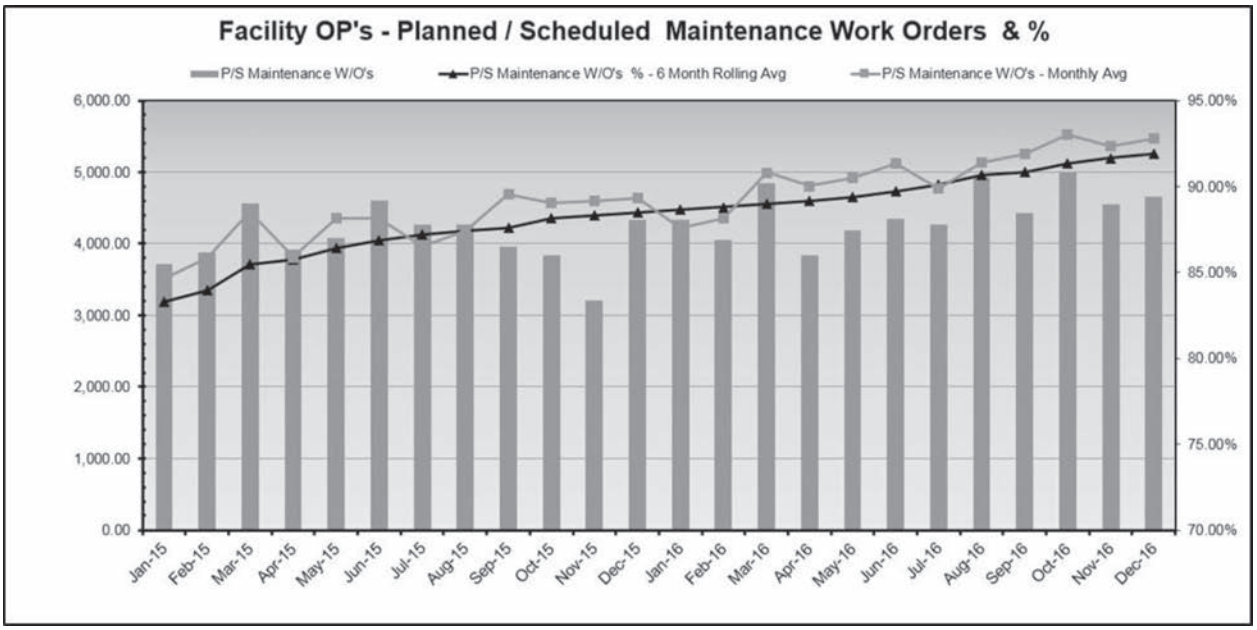


Figure B-9. KPI—planned/scheduled maintenance work orders.

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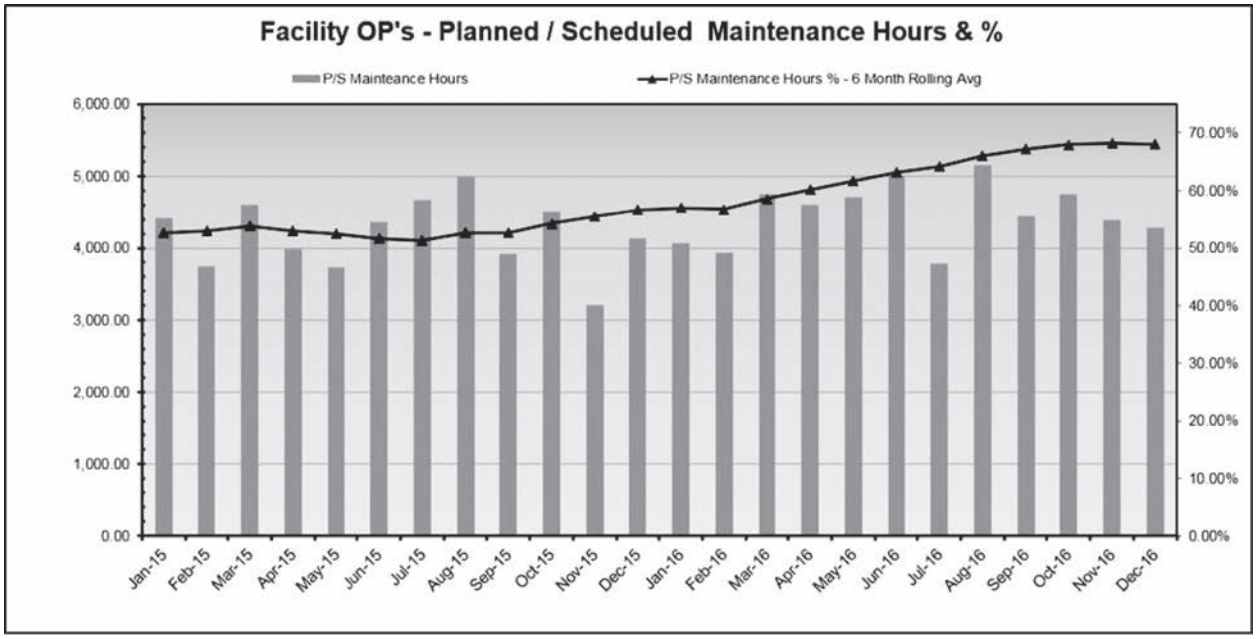


Figure B-10. KPI—planned/scheduled maintenance hours.

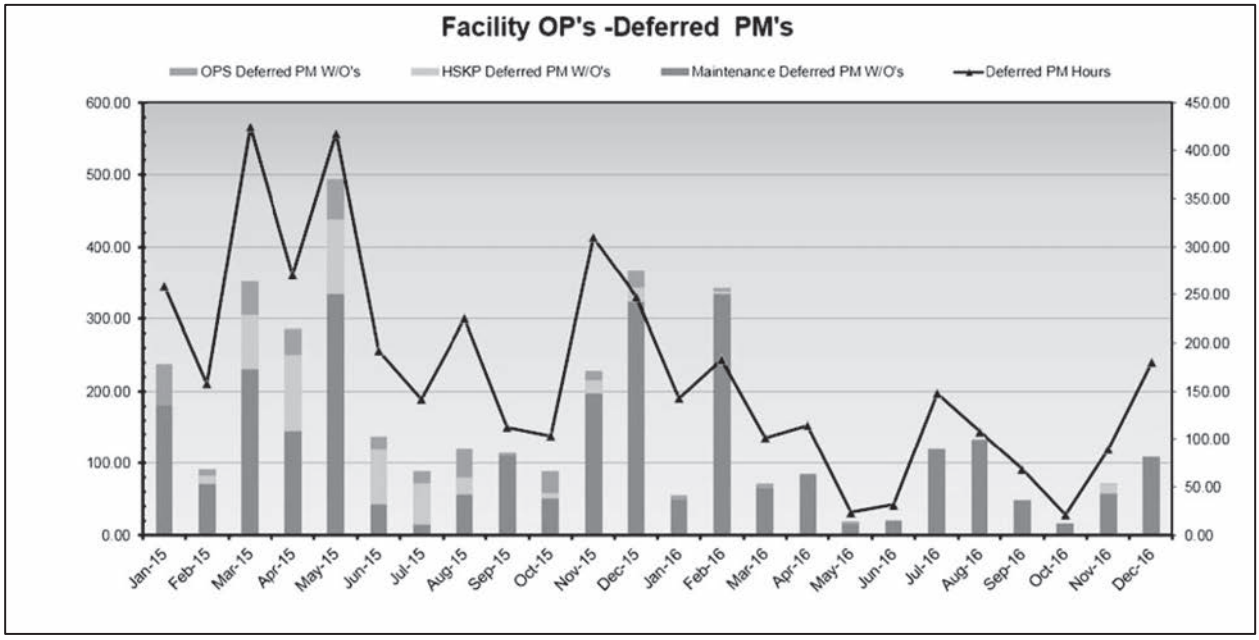


Figure B-11. KPI—deferred PMs.

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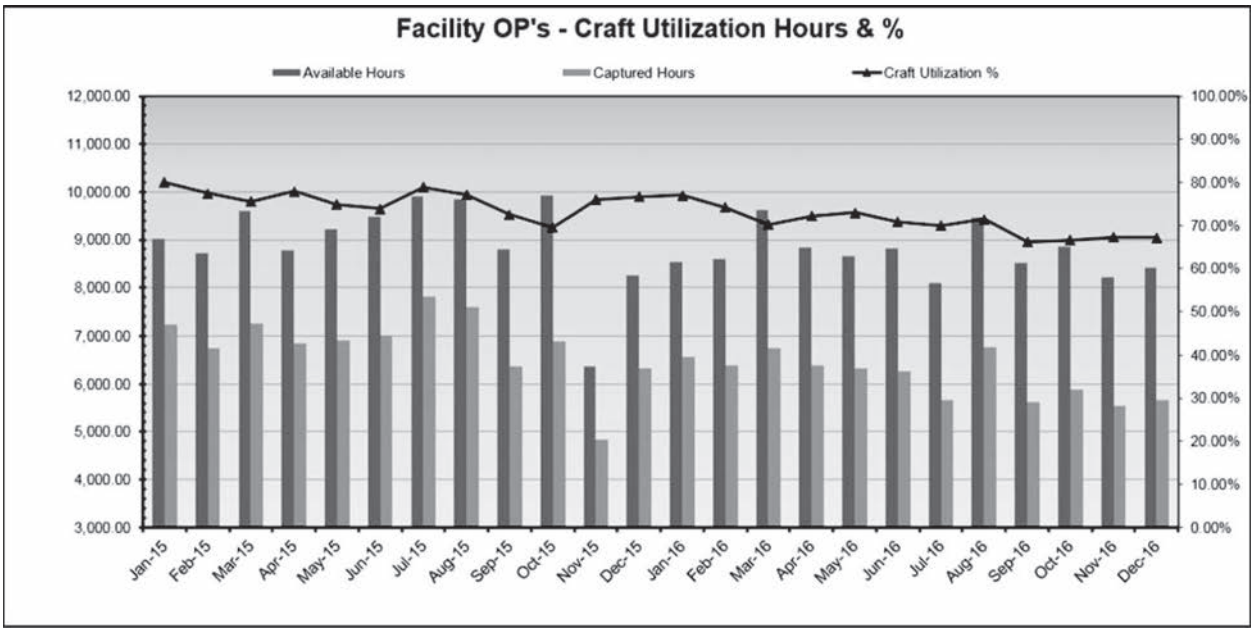


Figure B-12. KPI—craft utilization hours.

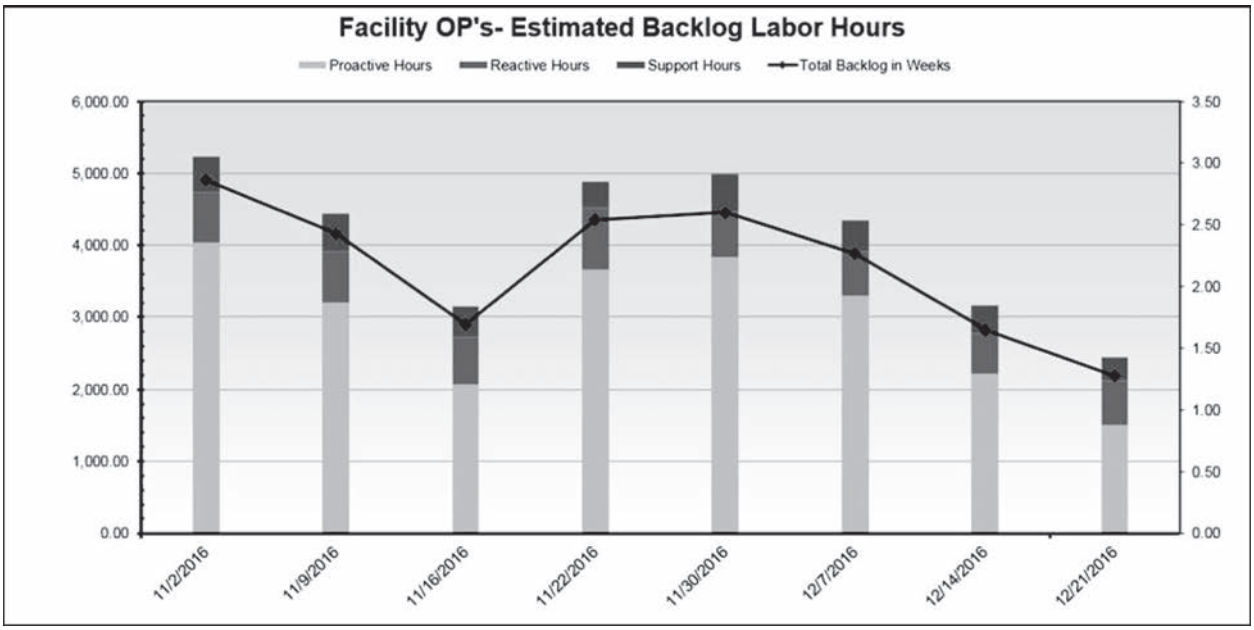


Figure B-13. KPI—estimated backlog labor hours.

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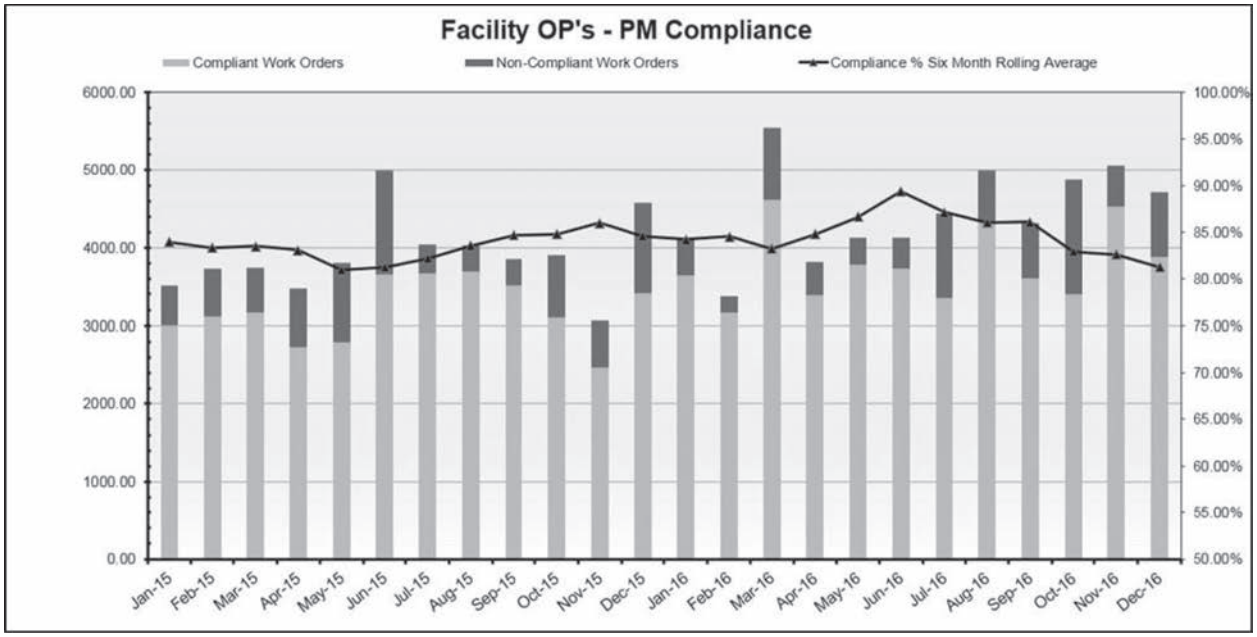


Figure B-14. KPI—PM compliance.

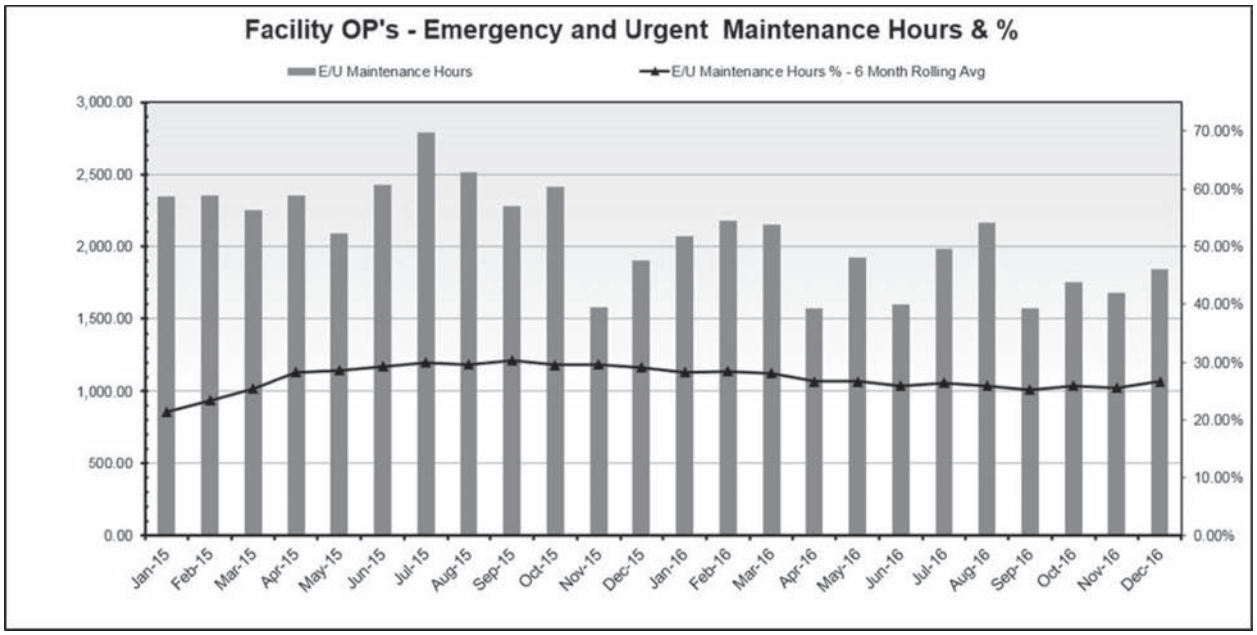


Figure B-15. KPI—emergency and urgent maintenance hours.

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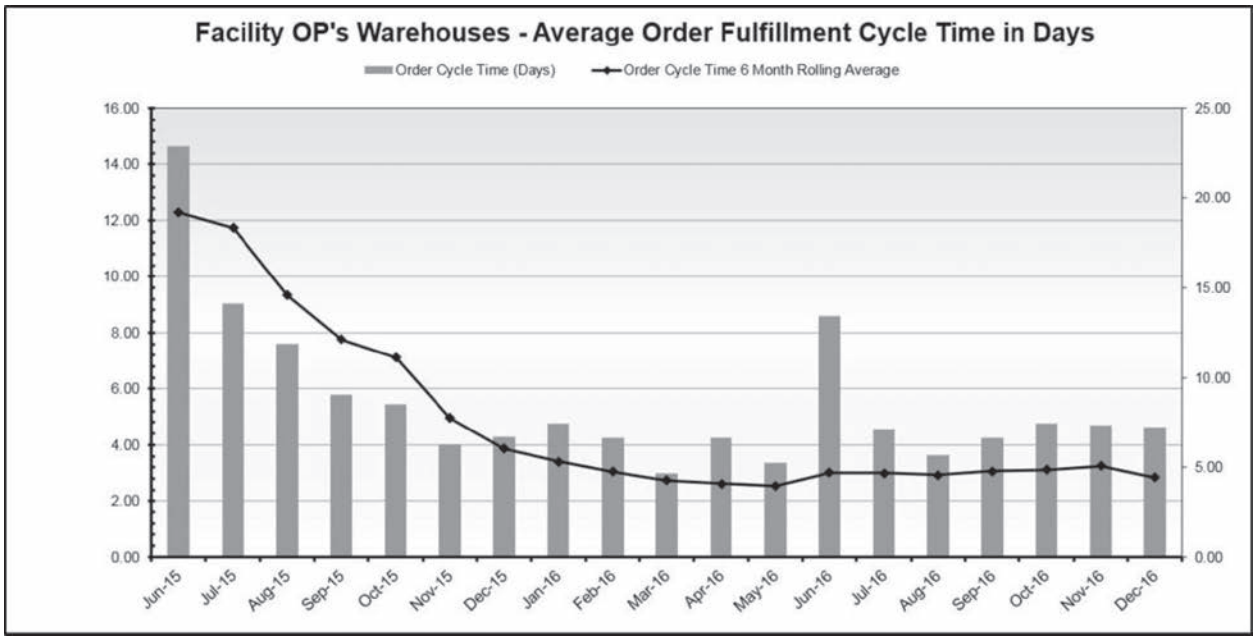


Figure B-16. KPI—average order fulfillment cycle time.

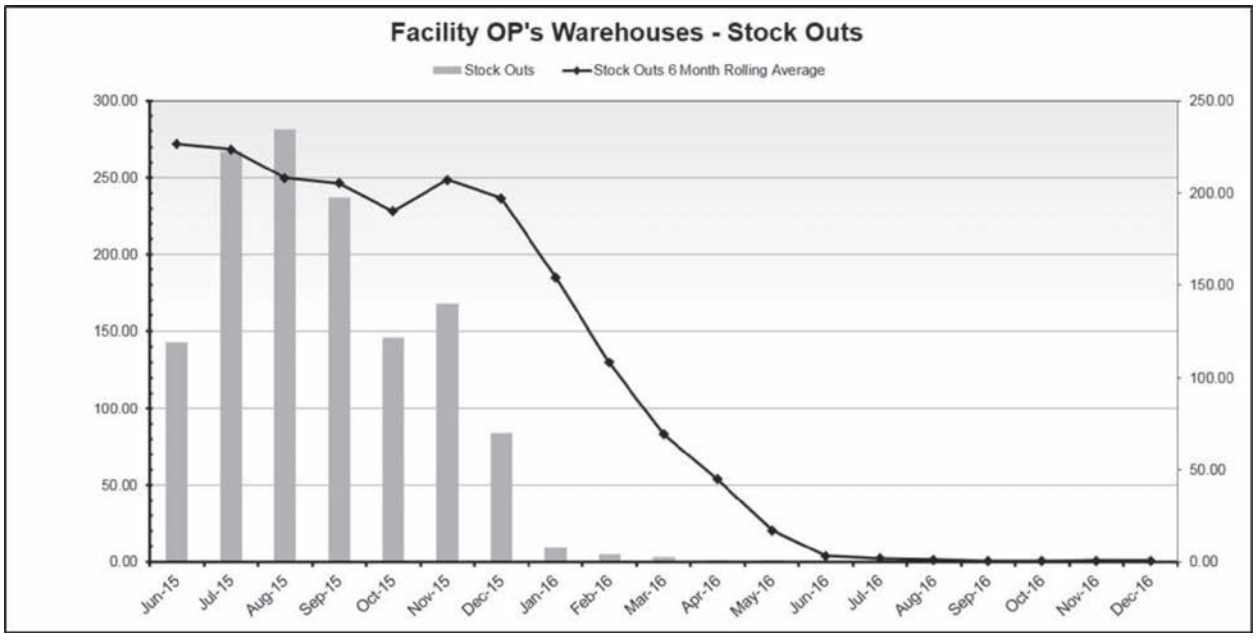


Figure B-17. KPI—stock outs.

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Abbreviations and acronyms used without definitions in TRB publications:

A4A	Airlines for America
AAAE	American Association of Airport Executives
AASHO	American Association of State Highway Officials
AASHTO	American Association of State Highway and Transportation Officials
ACI–NA	Airports Council International–North America
ACRP	Airport Cooperative Research Program
ADA	Americans with Disabilities Act
APTA	American Public Transportation Association
ASCE	American Society of Civil Engineers
ASME	American Society of Mechanical Engineers
ASTM	American Society for Testing and Materials
ATA	American Trucking Associations
CTAA	Community Transportation Association of America
CTBSSP	Commercial Truck and Bus Safety Synthesis Program
DHS	Department of Homeland Security
DOE	Department of Energy
EPA	Environmental Protection Agency
FAA	Federal Aviation Administration
FAST	Fixing America's Surface Transportation Act (2015)
FHWA	Federal Highway Administration
FMCSA	Federal Motor Carrier Safety Administration
FRA	Federal Railroad Administration
FTA	Federal Transit Administration
HMCRP	Hazardous Materials Cooperative Research Program
IEEE	Institute of Electrical and Electronics Engineers
ISTEA	Intermodal Surface Transportation Efficiency Act of 1991
ITE	Institute of Transportation Engineers
MAP-21	Moving Ahead for Progress in the 21st Century Act (2012)
NASA	National Aeronautics and Space Administration
NASAO	National Association of State Aviation Officials
NCFRP	National Cooperative Freight Research Program
NCHRP	National Cooperative Highway Research Program
NHTSA	National Highway Traffic Safety Administration
NTSB	National Transportation Safety Board
PHMSA	Pipeline and Hazardous Materials Safety Administration
RITA	Research and Innovative Technology Administration
SAE	Society of Automotive Engineers
SAFETEA-LU	Safe, Accountable, Flexible, Efficient Transportation Equity Act: A Legacy for Users (2005)
TCRP	Transit Cooperative Research Program
TDC	Transit Development Corporation
TEA-21	Transportation Equity Act for the 21st Century (1998)
TRB	Transportation Research Board
TSA	Transportation Security Administration
U.S.DOT	United States Department of Transportation

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